

Self-Supervised Learning

Joseph Bakarji

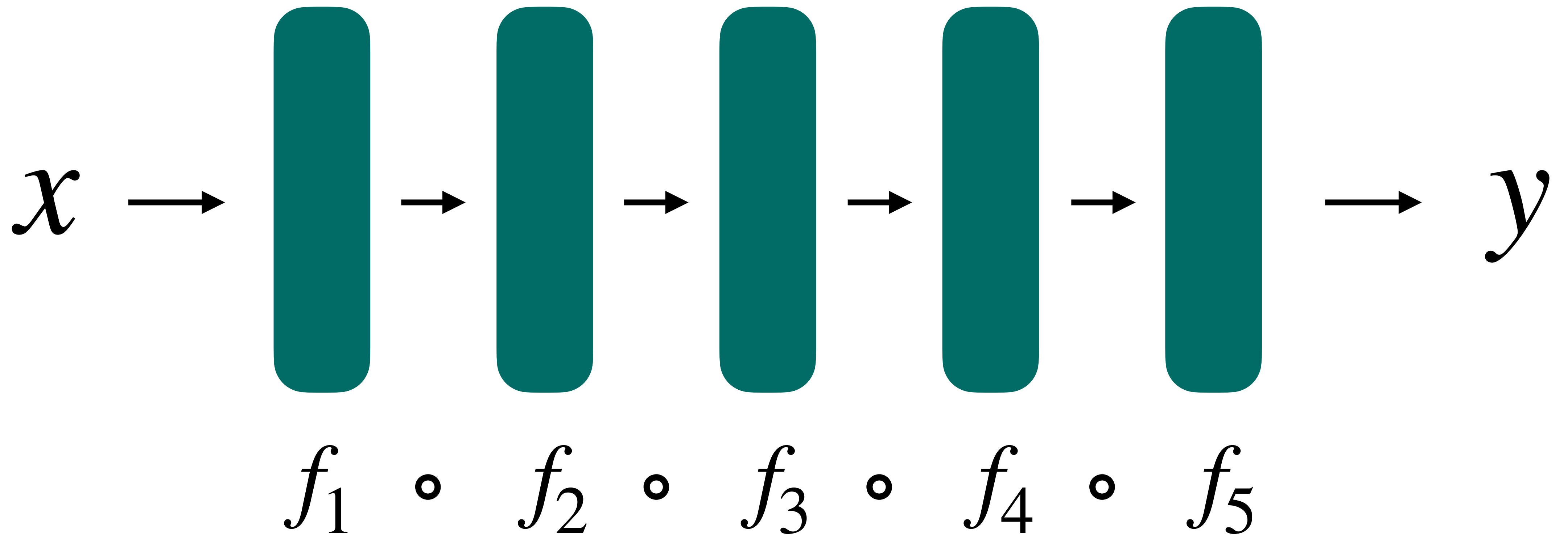
Supervised Learning

Supervised Learning (Training)



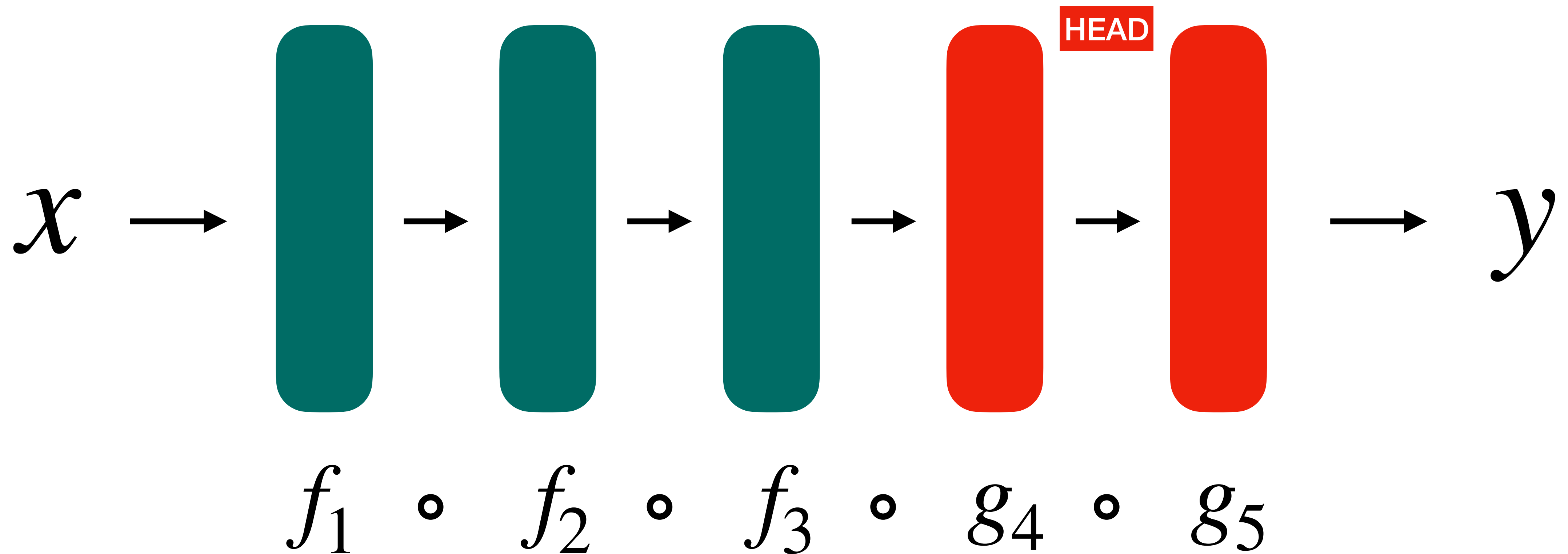
Transfer Learning

Train Model on Big Dataset of Images

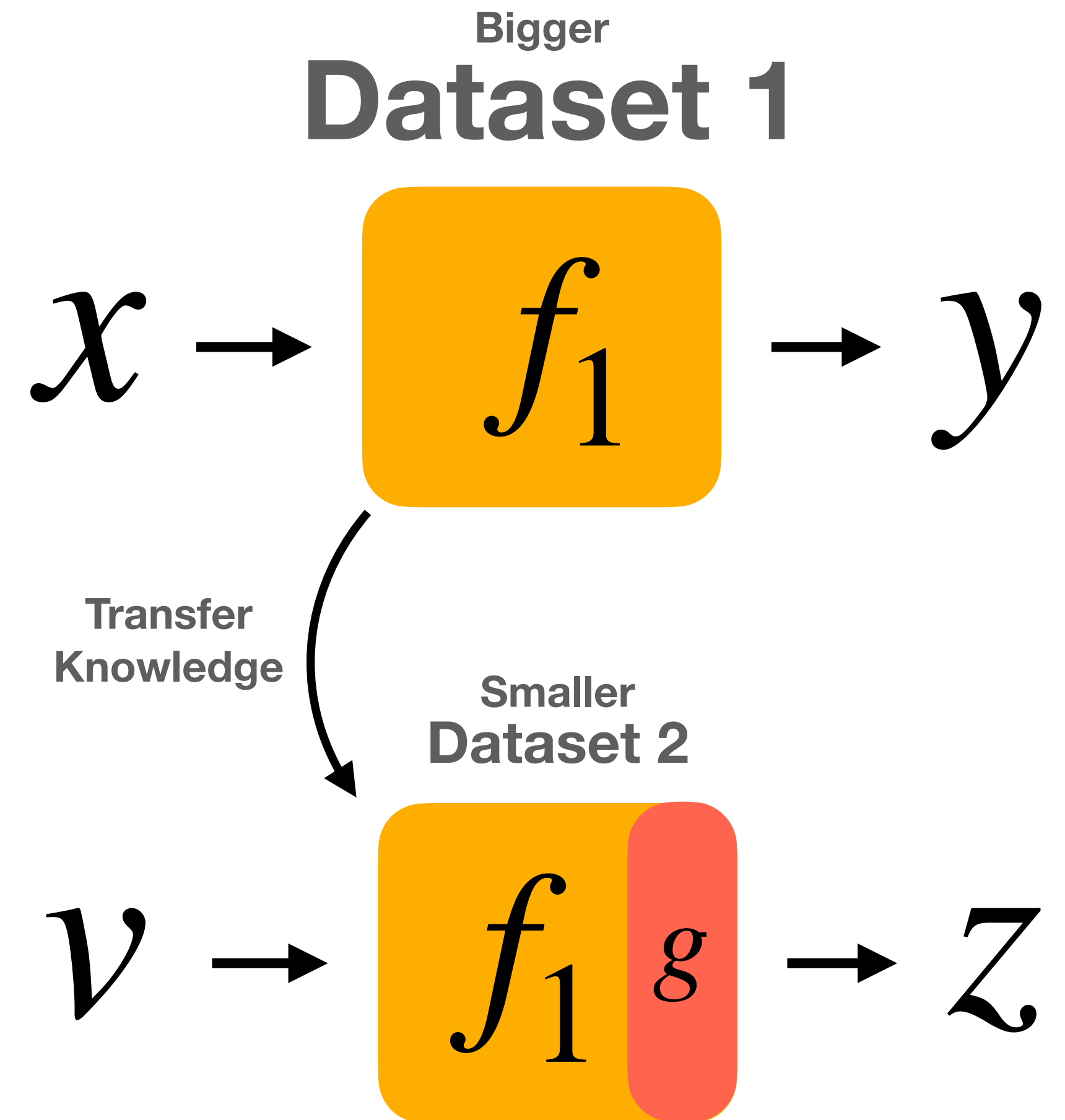
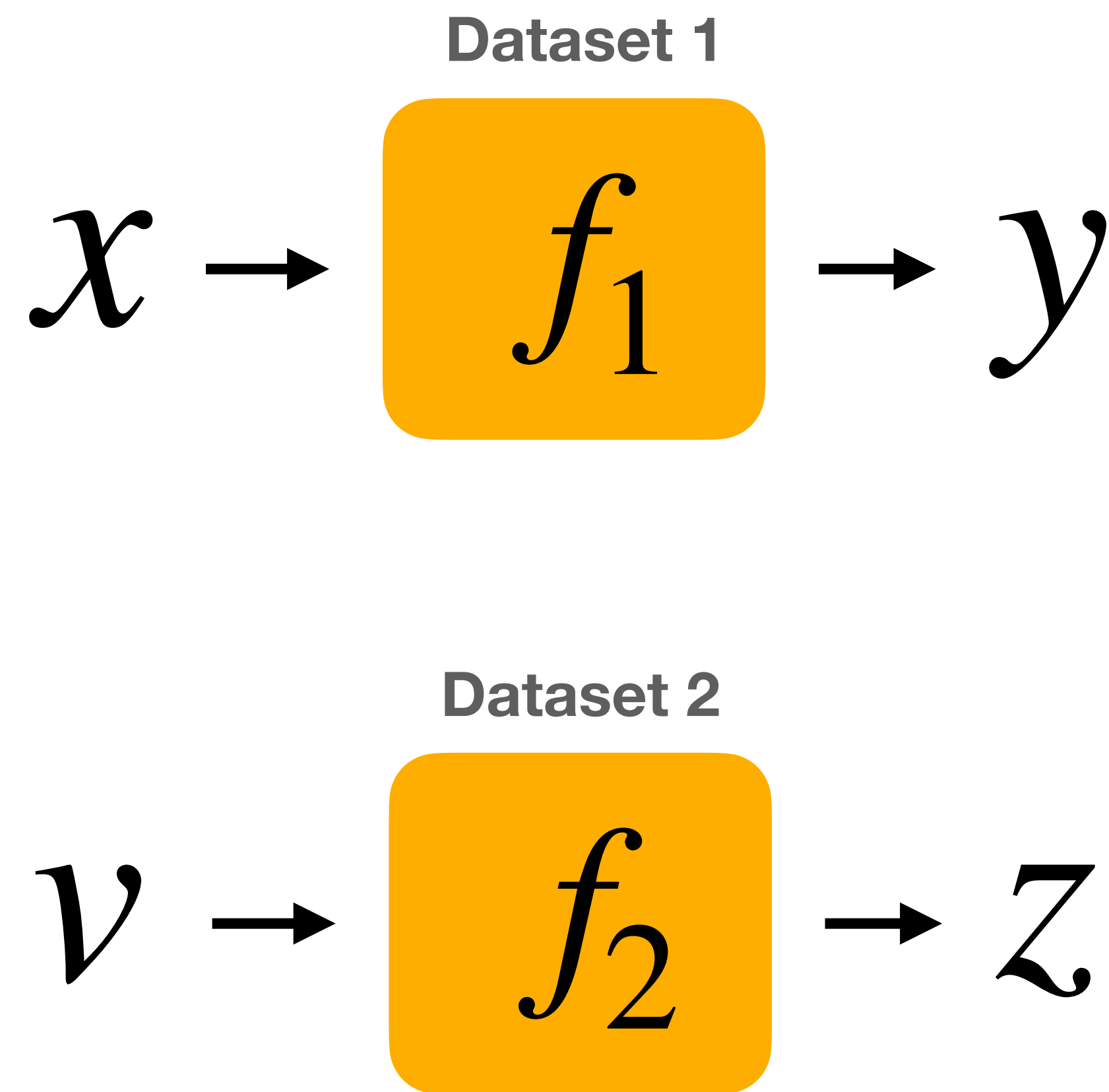


Transfer Learning

Retrain model on your data and task
By changing the 'head' of the network



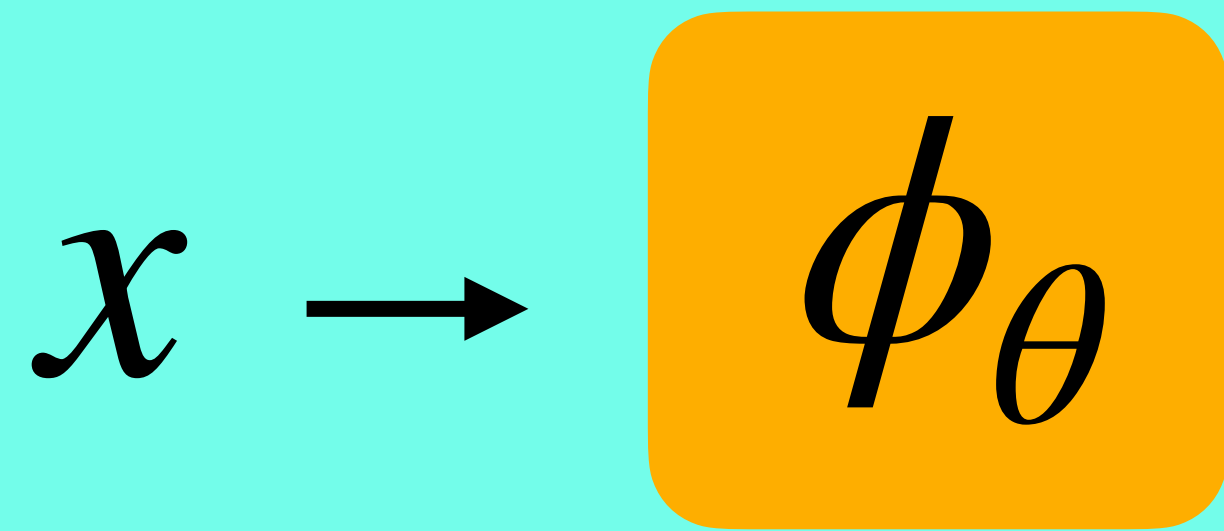
Transfer Learning



A Paradigm Shift

Self-Supervised Learning

Pre-Training



Pre-train a large models on a large scale unlabeled dataset

Adaptation



Adapt the retrained model to a wide range of tasks

Self-Supervised Learning

Pre-Training

Pre-Training

$x \rightarrow \phi_{\theta}$

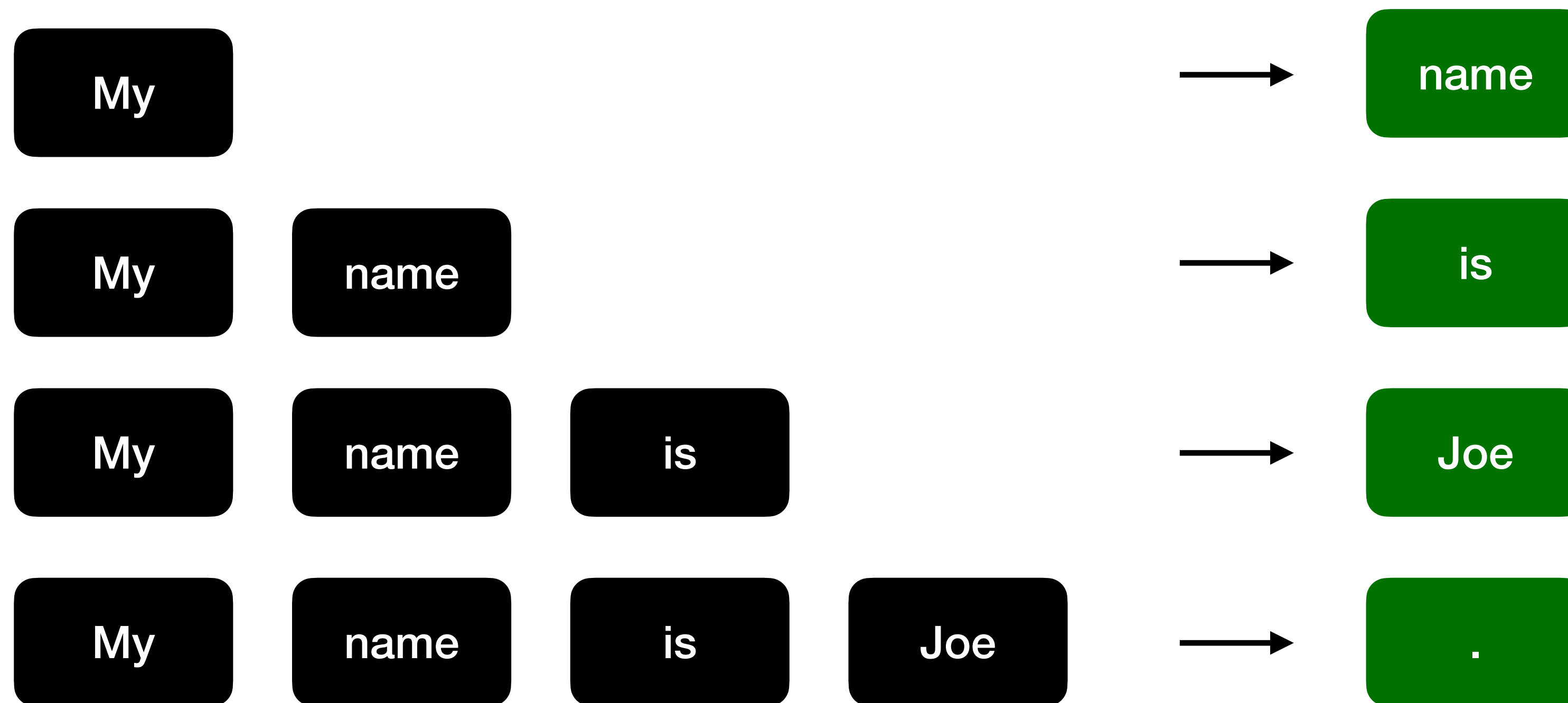
Pre-train a large models on a large scale unlabeled dataset

- Data: $\{x^{(1)}, \dots, x^{(n)}\}$
- Feature Map: $\phi_{\theta}(x) \in \mathbb{R}^m$ (learned in some cases)
- Pre-training Loss

$$L_{pre}(\theta) = \frac{1}{n} \sum_{i=1}^n l_{pre}(\theta, x^{(i)})$$

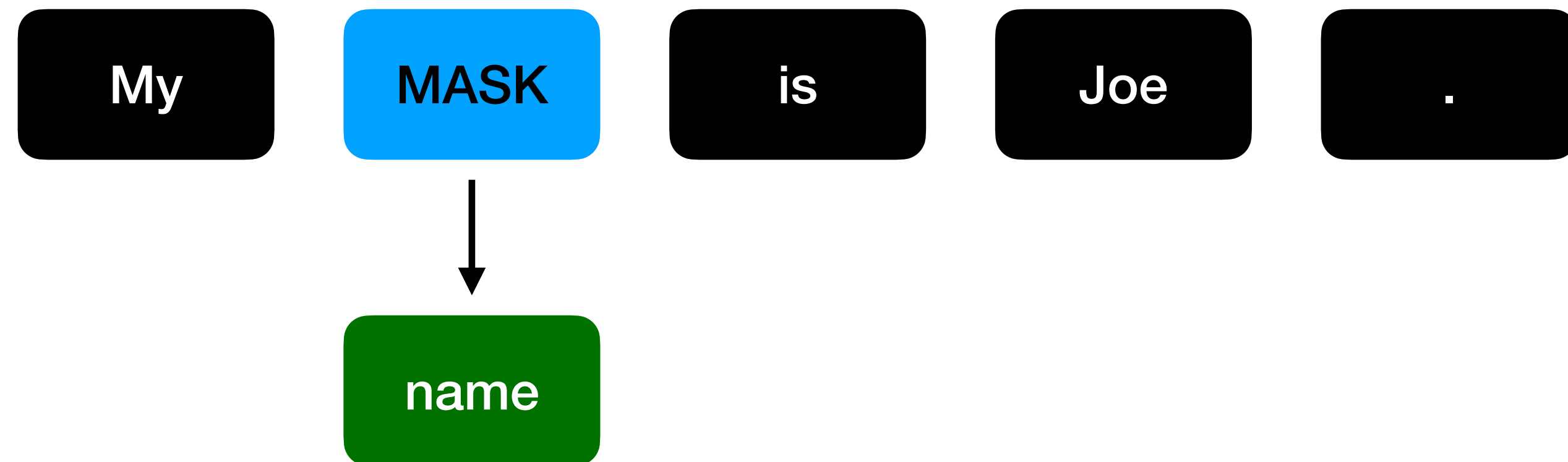
- Optimize $L_{pre}(\theta) \rightarrow \hat{\theta}$

Pre-Training and Transfer Learning



GPT-2 was retrained on 40GB of internet text posted by users on Reddit

Pre-Training and Transfer Learning



BERT was retrained on the content of the English wikipedia and 11'000 unpublished books

Pre-Training and Transfer Learning

Learning the bias is also transferred

He was very...

Personable

Large

Lazy

Fantastic

She was very...

Beautiful

Gorgeous

Naughty

Bubbly

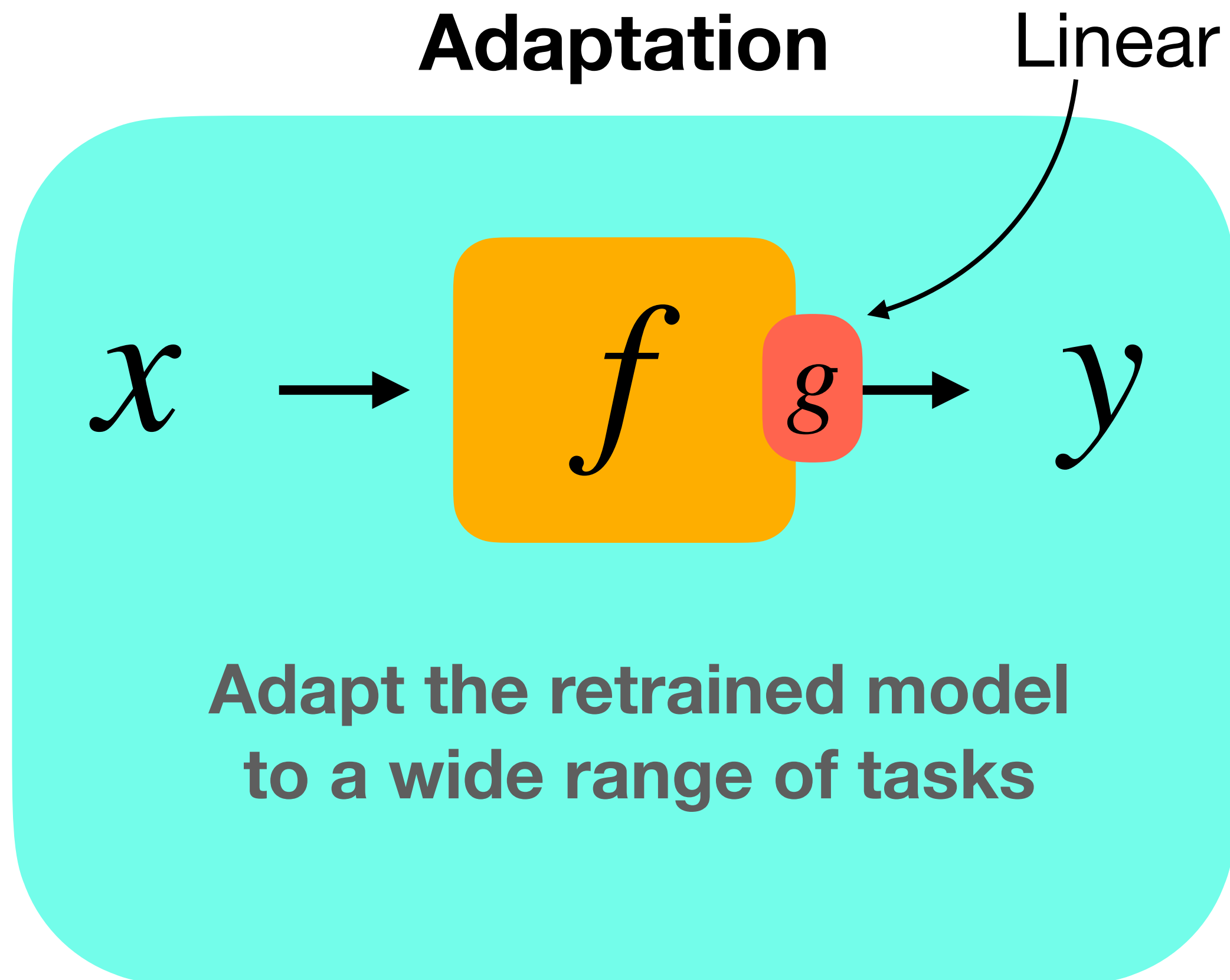
Adaptation



**Adapt the retrained model
to a wide range of tasks**

- Data:
 $\{(x^{(1)}, y^{(1)}), \dots, (x^{(n)}, y^{(n)})\}$
- $n = 0$: zero-shot learning
- n is small: few-shot learning

Adaptation: Linear Probe/Head



Prediction Model

$$f = w^{\top} \phi_{\hat{\theta}}(x)$$

Train w with loss function

$$\min_w \frac{1}{n_t} \sum_{i=1}^{n_t} l_{task}(y_t^{(i)}, w^{\top} \phi_{\hat{\theta}}(x_t^{(i)}))$$

Adaptation: Fine-tuning

Adaptation



Adapt the retrained model
to a wide range of tasks

Prediction Model

$$f(w, \theta) = w^{\top} \phi_{\hat{\theta}}(x)$$

Optimize both w, θ on downstream task

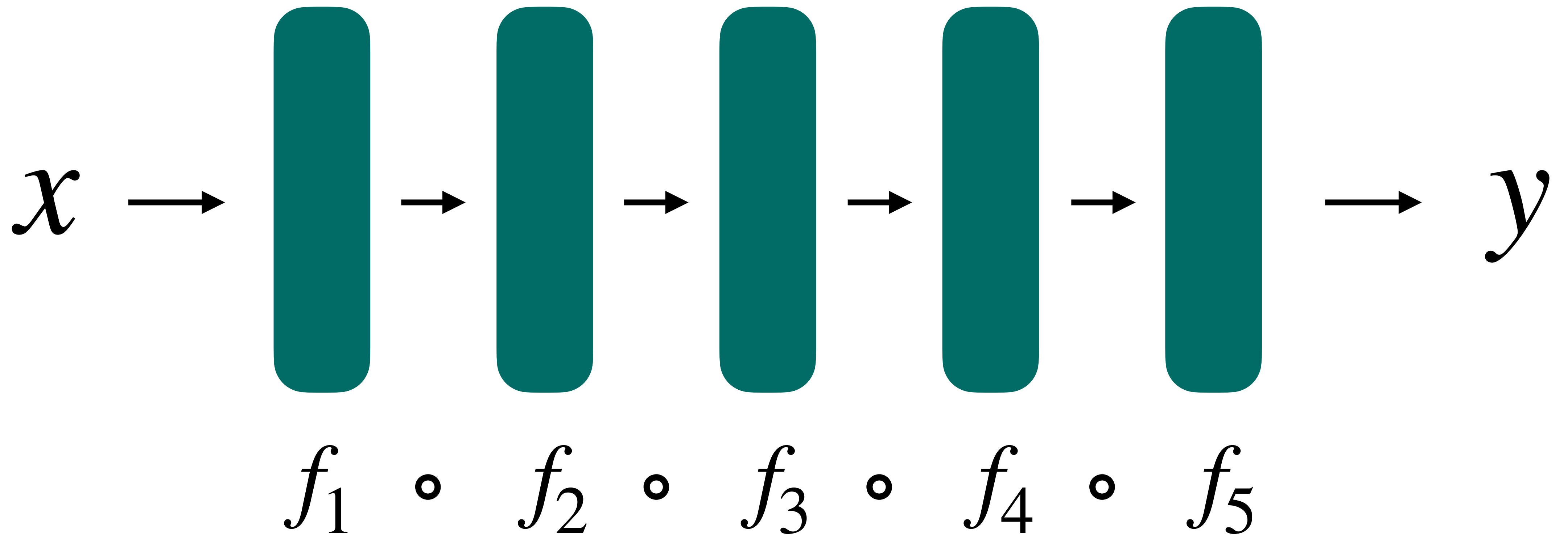
$$\min_{w, \theta} \frac{1}{n} \sum_{i=1}^{n_t} l_{task}(y^{(i)}, w^{\top} \phi_{\hat{\theta}}(x^{(i)}))$$

Initialize

$$\theta \leftarrow \hat{\theta} \text{ and } w \leftarrow \text{random}$$

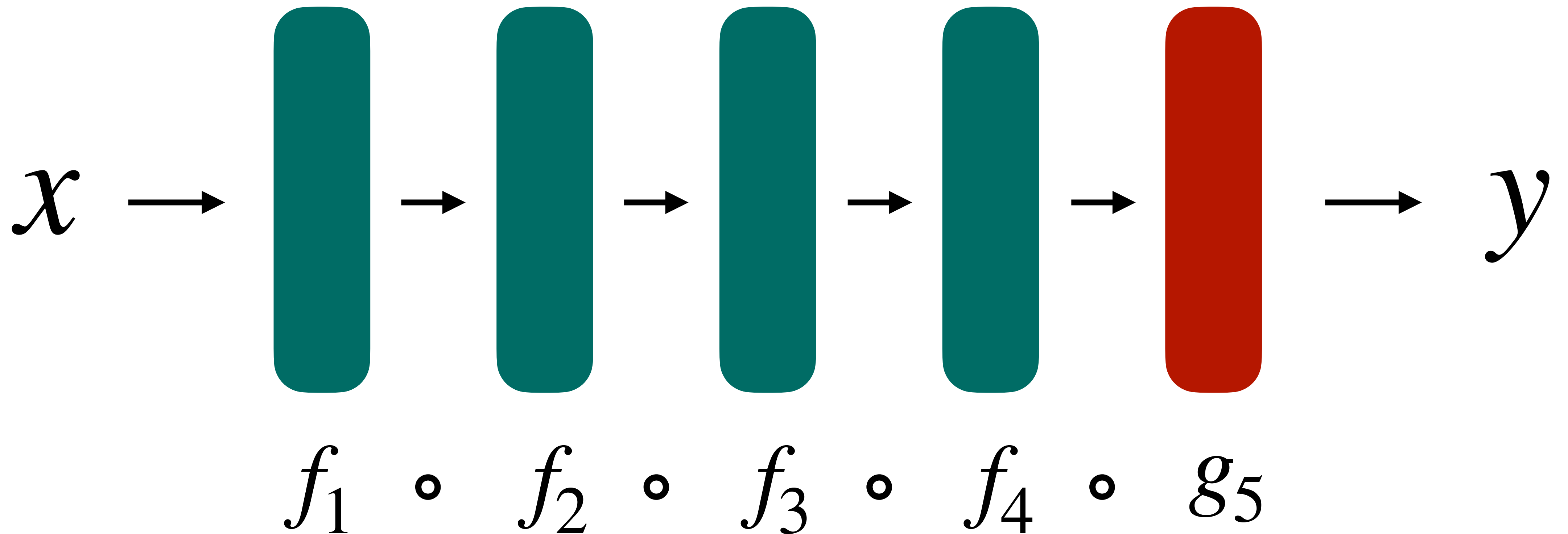
Computer Vision: supervised pretraining

Train Model on Big Dataset of Images



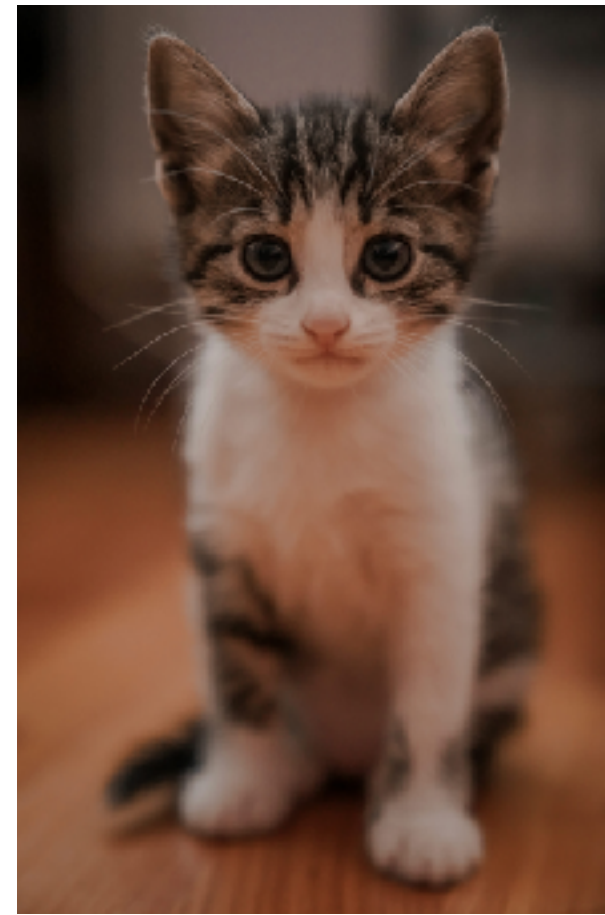
Computer Vision: supervised pretraining

Train linear head on last layer

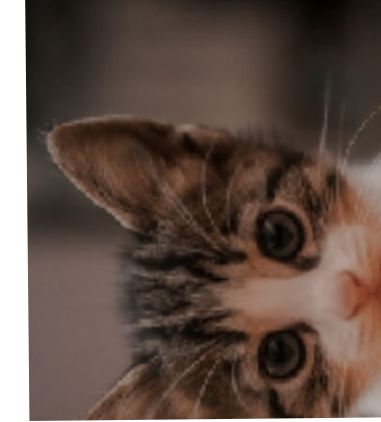


Contrastive Learning (no labels)

- **Data Augmentation**
 - Image random crop
 - Flip
 - Color Transformation



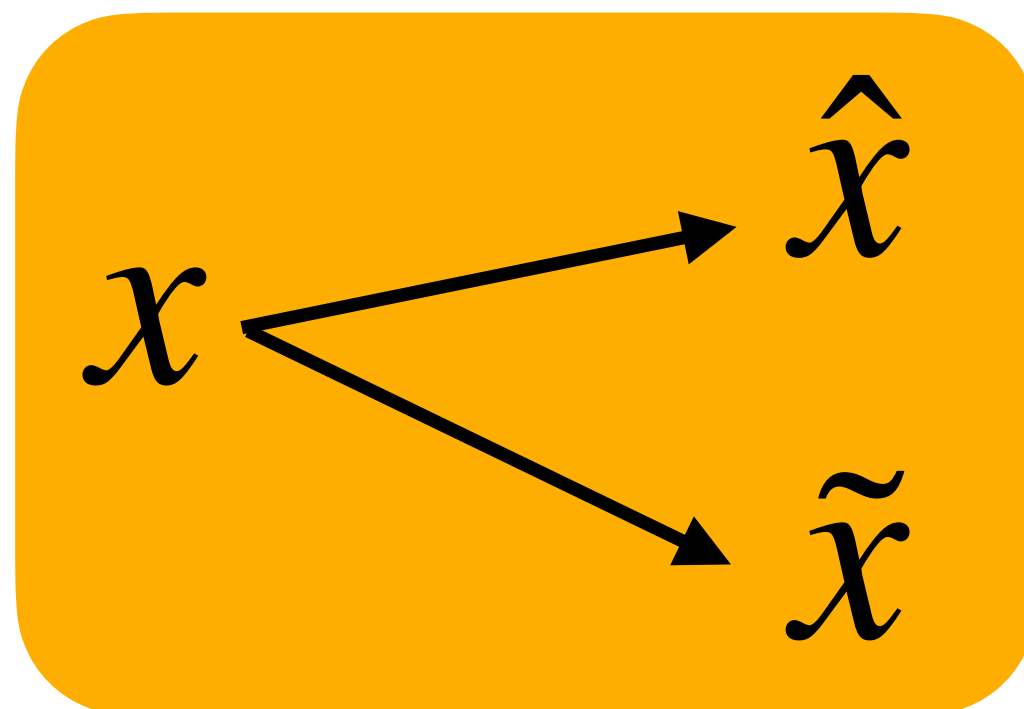
$\hat{x}$



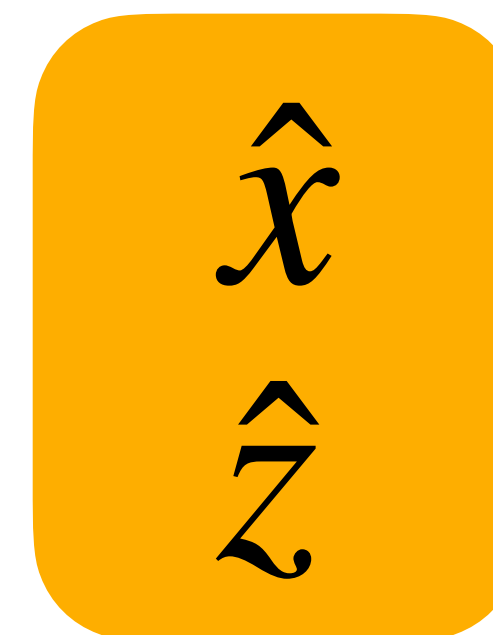
$\tilde{x}$



$\hat{z}$



Make $\phi_{\theta}(\hat{x})$ and $\phi_{\theta}(\tilde{x})$
have similar
representations

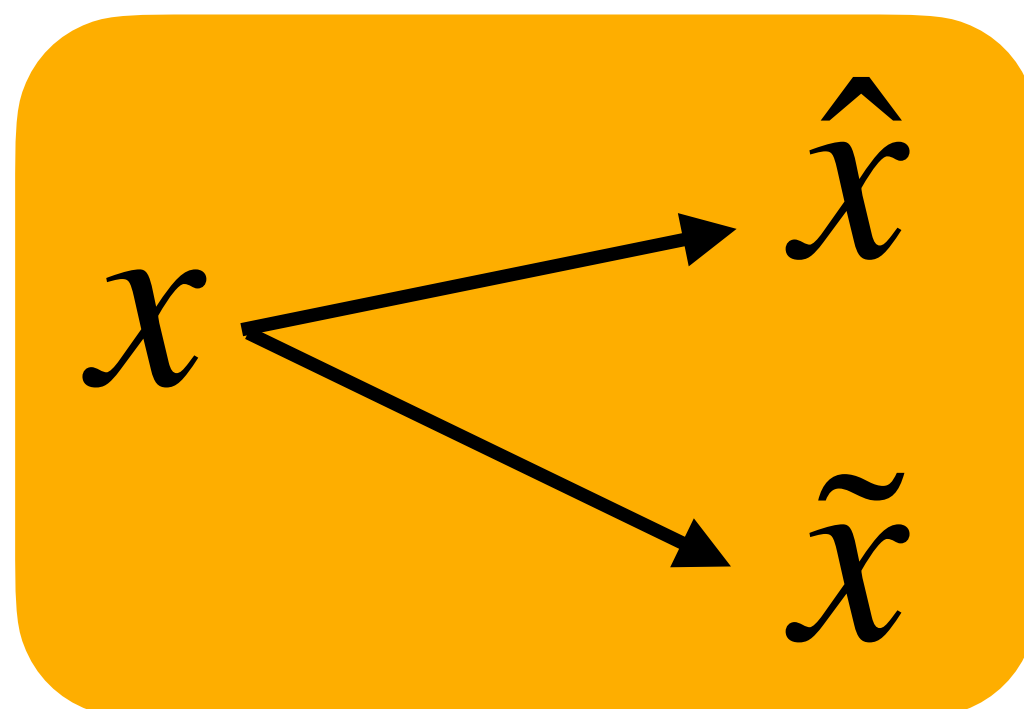


Make $\phi_{\theta}(\hat{x})$ and $\phi_{\theta}(\hat{z})$
far from each other

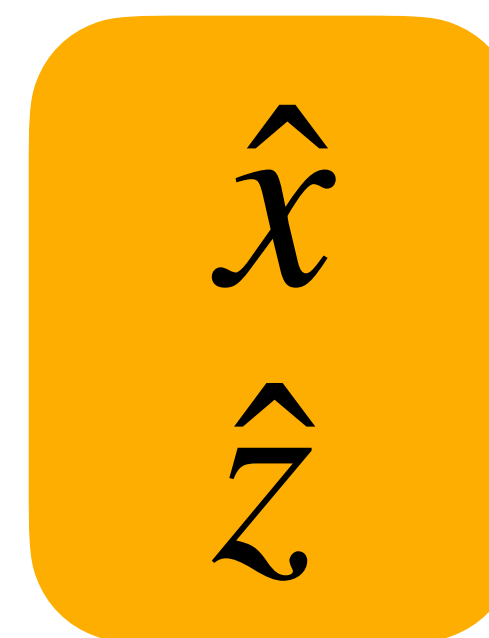
Random/negative Pair

Contrastive Learning (no labels)

Random/negative Pair



Make $\phi_{\theta}(\hat{x})$ and $\phi_{\theta}(\tilde{x})$ have similar representations



Make $\phi_{\theta}(\hat{x})$ and $\phi_{\theta}(\hat{z})$ far from each other

SIMCLR Loss Function

$x^{(1)} \quad x^{(2)} \quad \dots \quad x^{(B)}$

$\hat{x}^{(1)} \quad \hat{x}^{(2)} \quad \dots \quad \hat{x}^{(B)}$

$\tilde{x}^{(1)} \quad \tilde{x}^{(2)} \quad \dots \quad \tilde{x}^{(B)}$

$$-\sum_{i=1}^B \log \frac{\exp \left(\phi_{\theta}(\hat{x}^{(i)})^{\top} \phi_{\theta}(\tilde{x}^{(i)}) \right)}{\exp \left(\phi_{\theta}(\hat{x}^{(i)})^{\top} \phi_{\theta}(\tilde{x}^{(i)}) \right) + \sum_{j \neq i} \exp \left(\phi_{\theta}(\hat{x}^{(i)})^{\top} \phi_{\theta}(\tilde{x}^{(j)}) \right)}$$

$$-\sum_{i=1}^B \log \frac{A}{A + B}$$

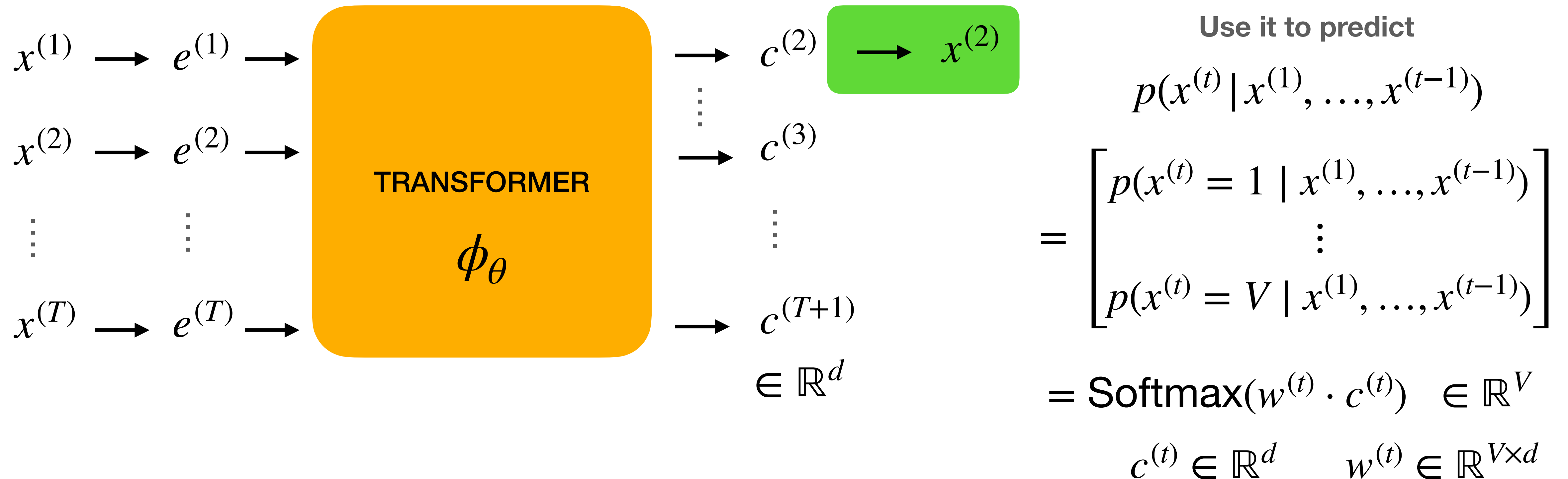
We want **B** to be small and **A** to be large

Large Language Models

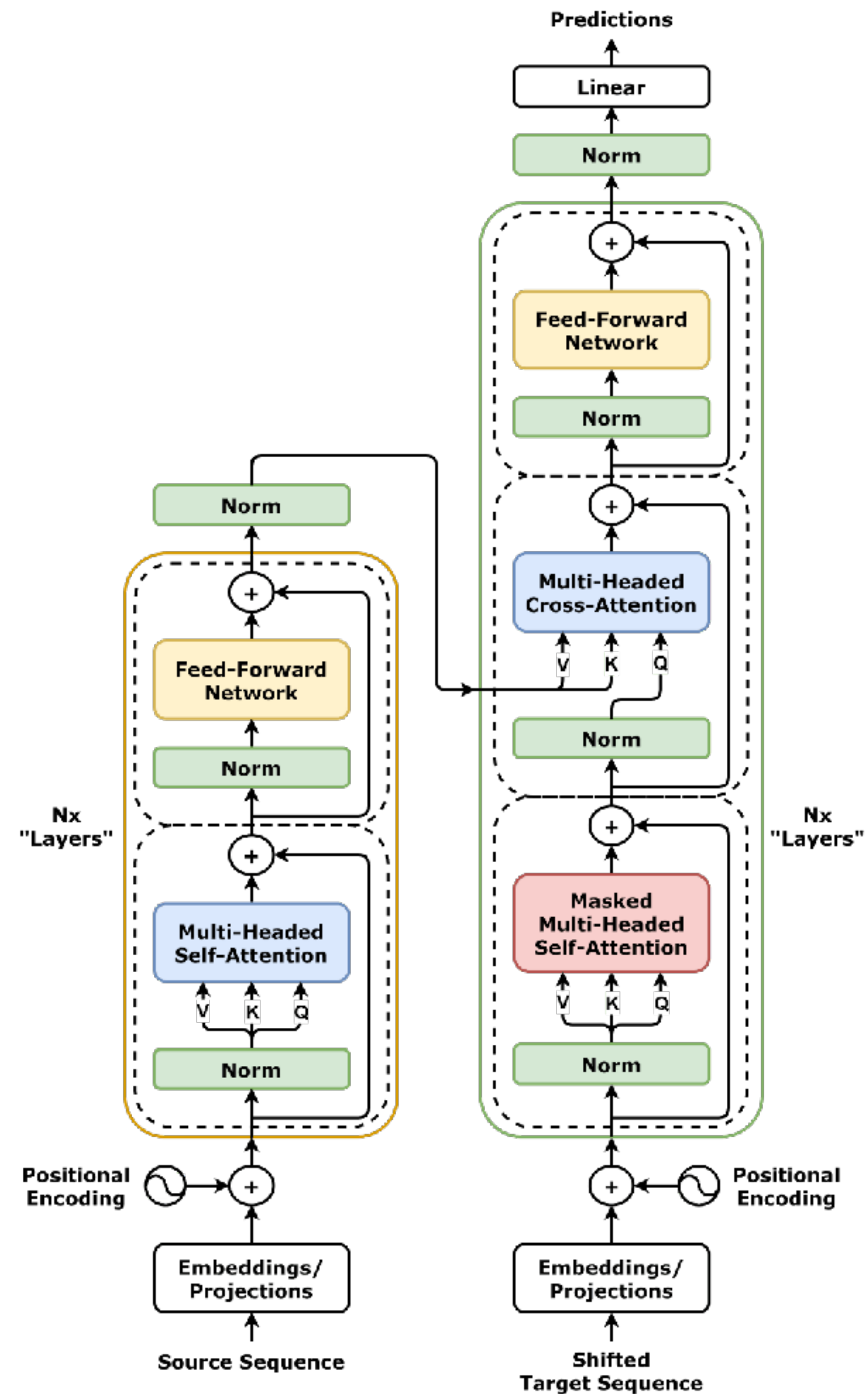
- Given a series of words, $(x^{(1)}, \dots, x^{(T)})$ in a vocabulary: $x^{(i)} \in \{1, \dots, V\}$
- **A Language Model** is a probabilistic model for $p(x^{(1)}, x^{(2)}, \dots, x^{(T)})$
- **Use Chain Rule:**
 - $p(x^{(1)}, x^{(2)}, \dots, x^{(T)}) = p(x^{(1)}) p(x^{(2)} | x^{(1)}) p(x^{(3)} | x^{(1)}, x^{(2)}) \dots$
- **Model:**
 - $p(x^{(t)} | x^{(1)}, \dots, x^{(t-1)})$ which is V dimensional instead of V^T dimensional

Large Language Models

- **Embedding** (learned) for each word $x^{(i)}$ into a vector $e_i \in \mathbb{R}^d$



Transformers

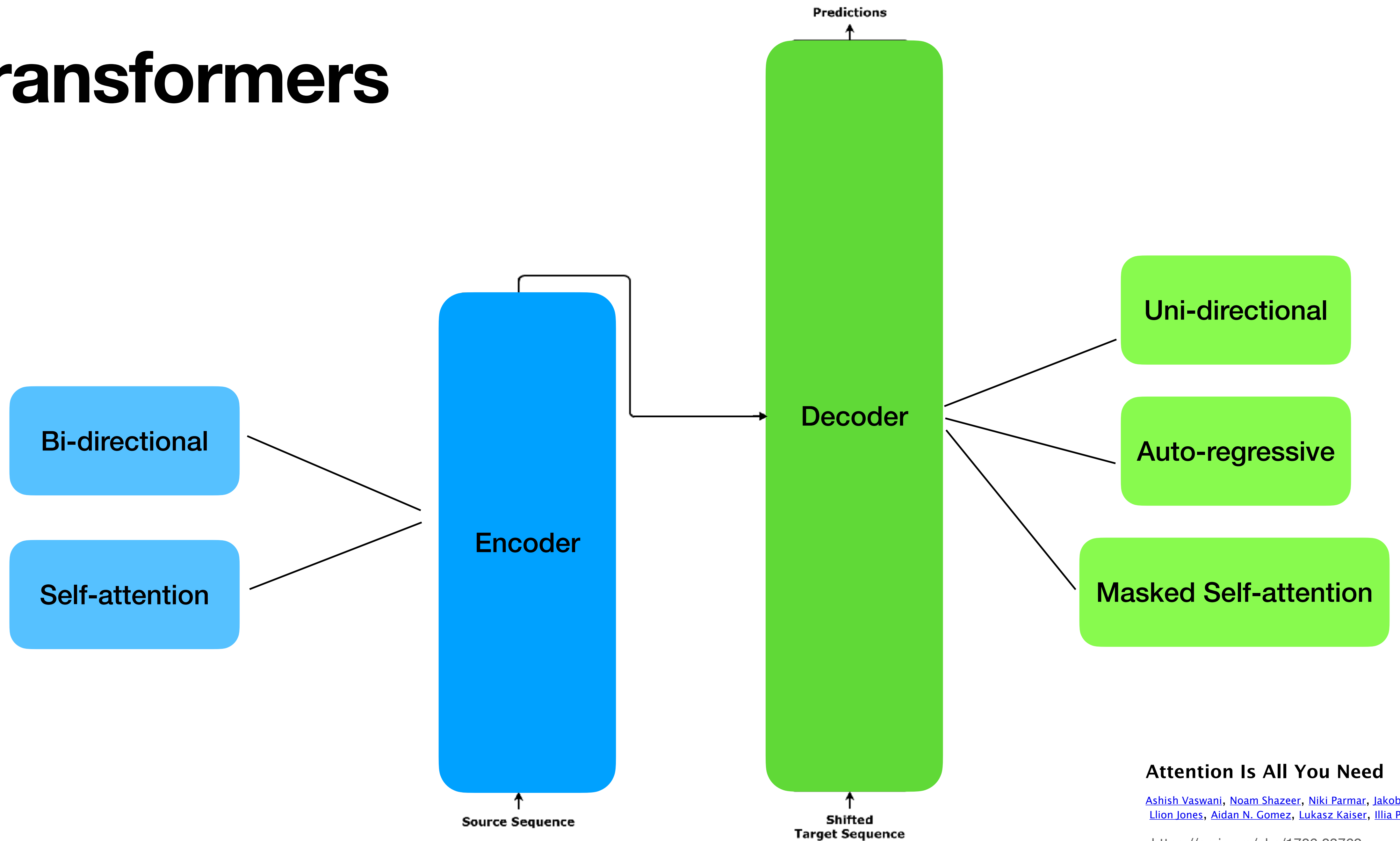


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<https://arxiv.org/abs/1706.03762>

Transformers



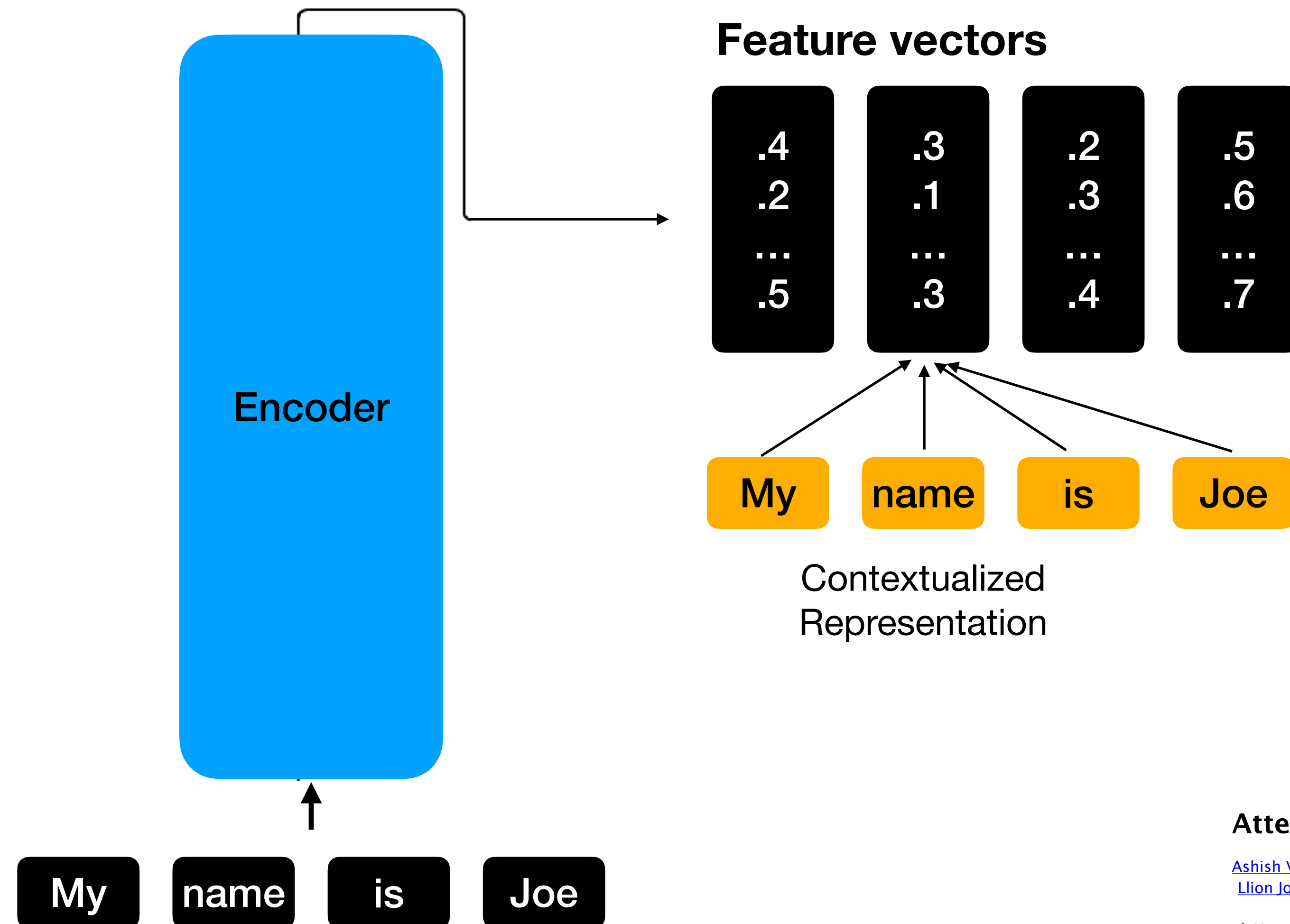
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Transformers

Encoder



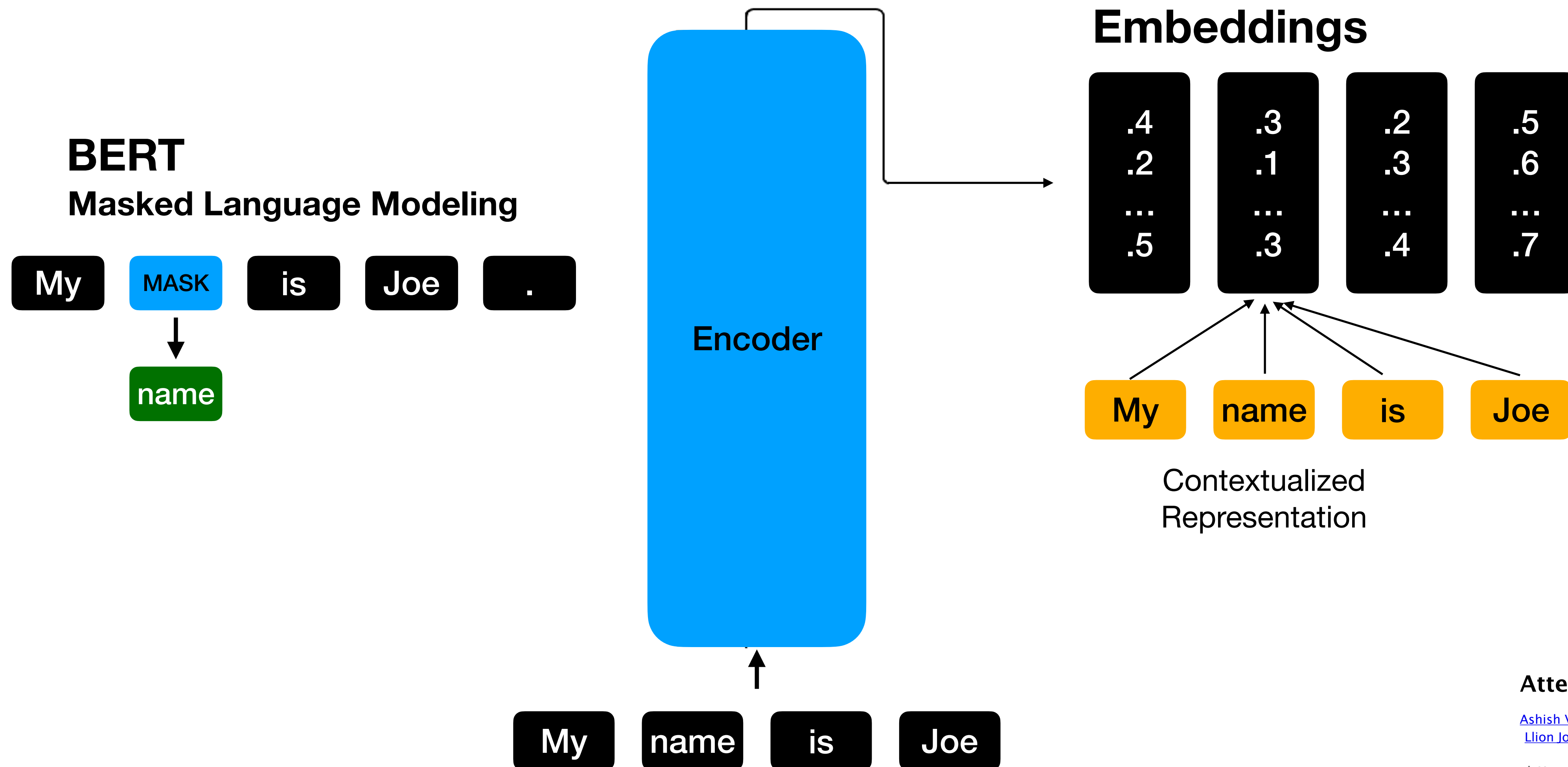
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Transformers

Encoder



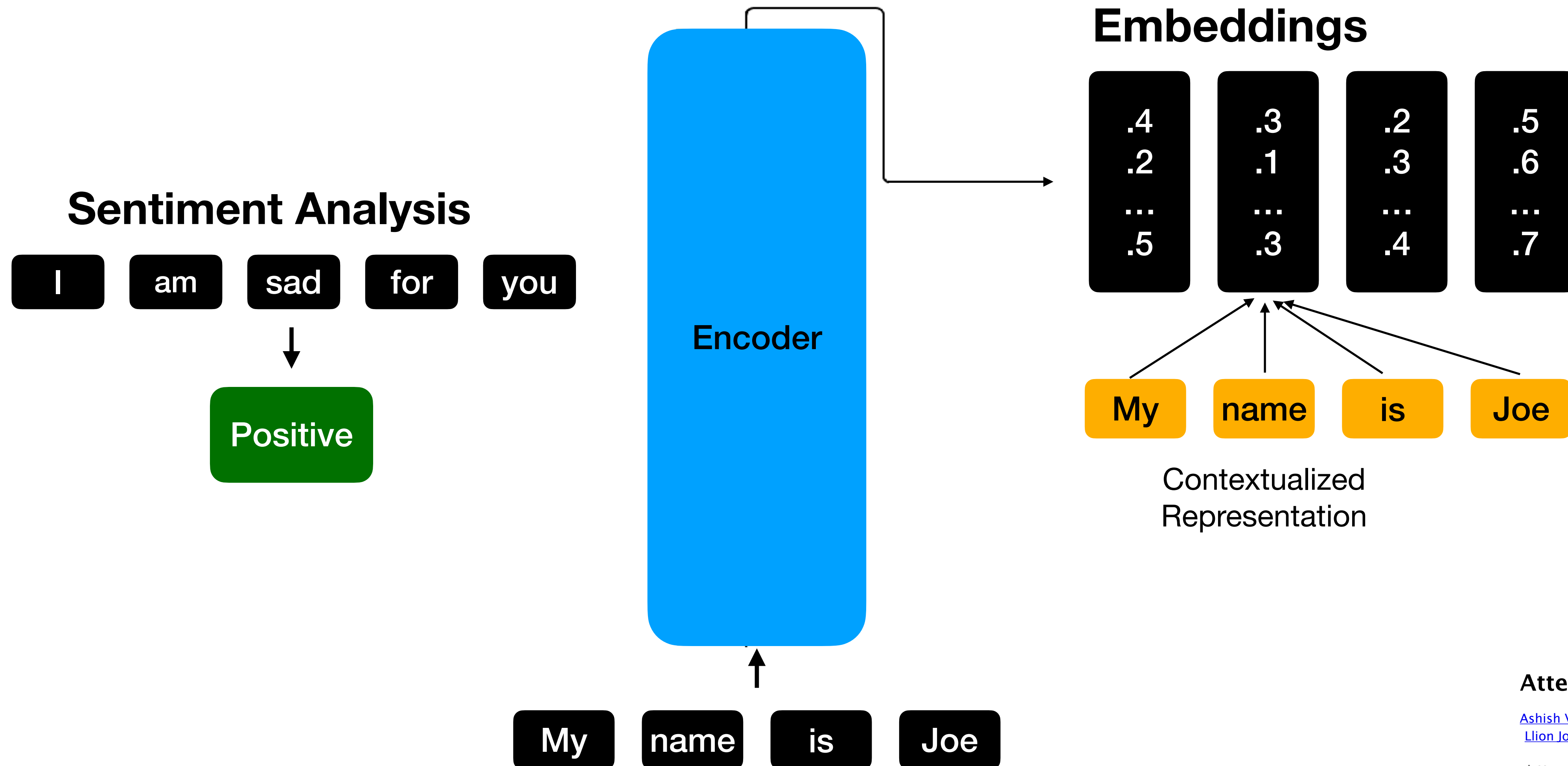
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Transformers

Encoder

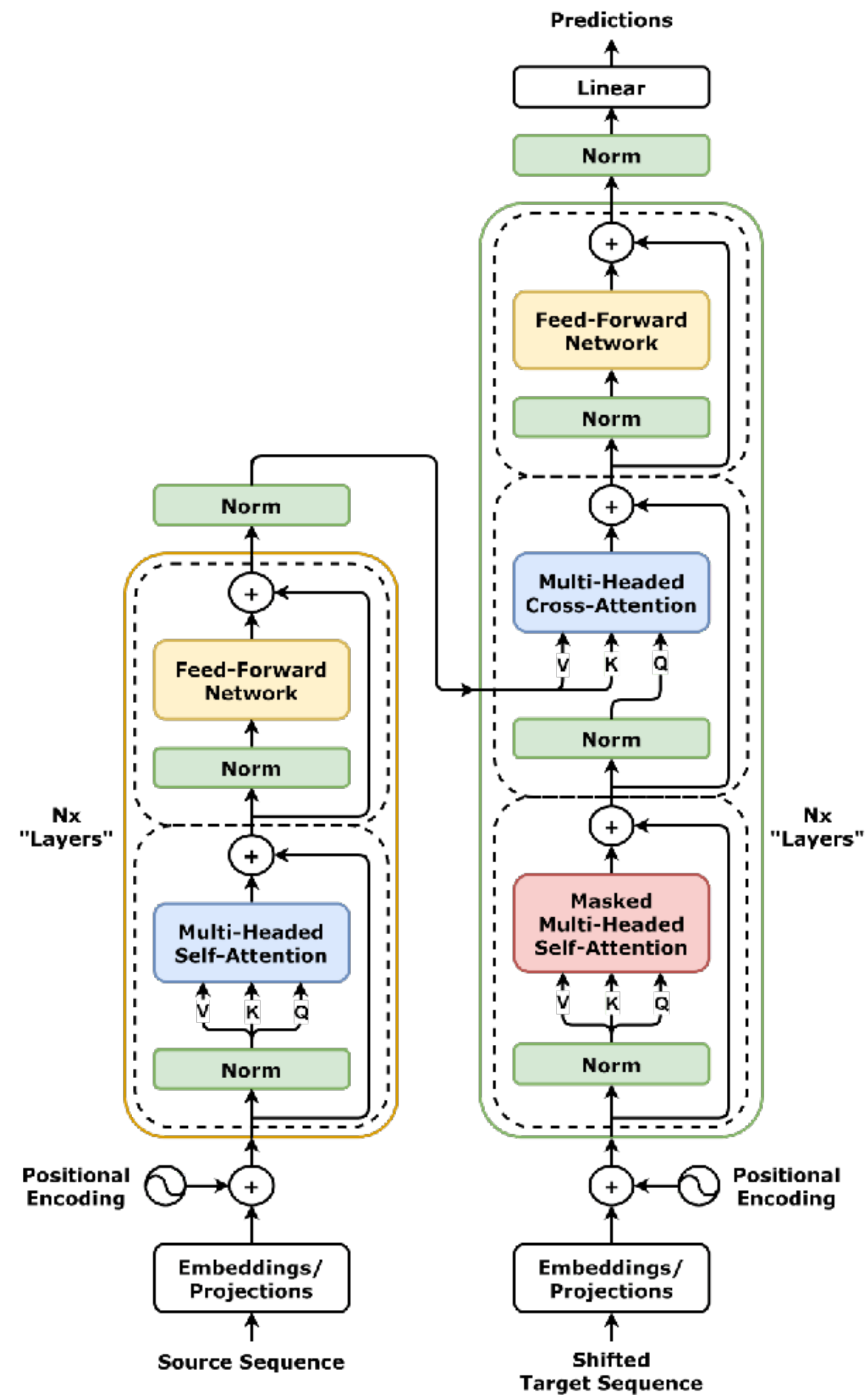


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Transformers



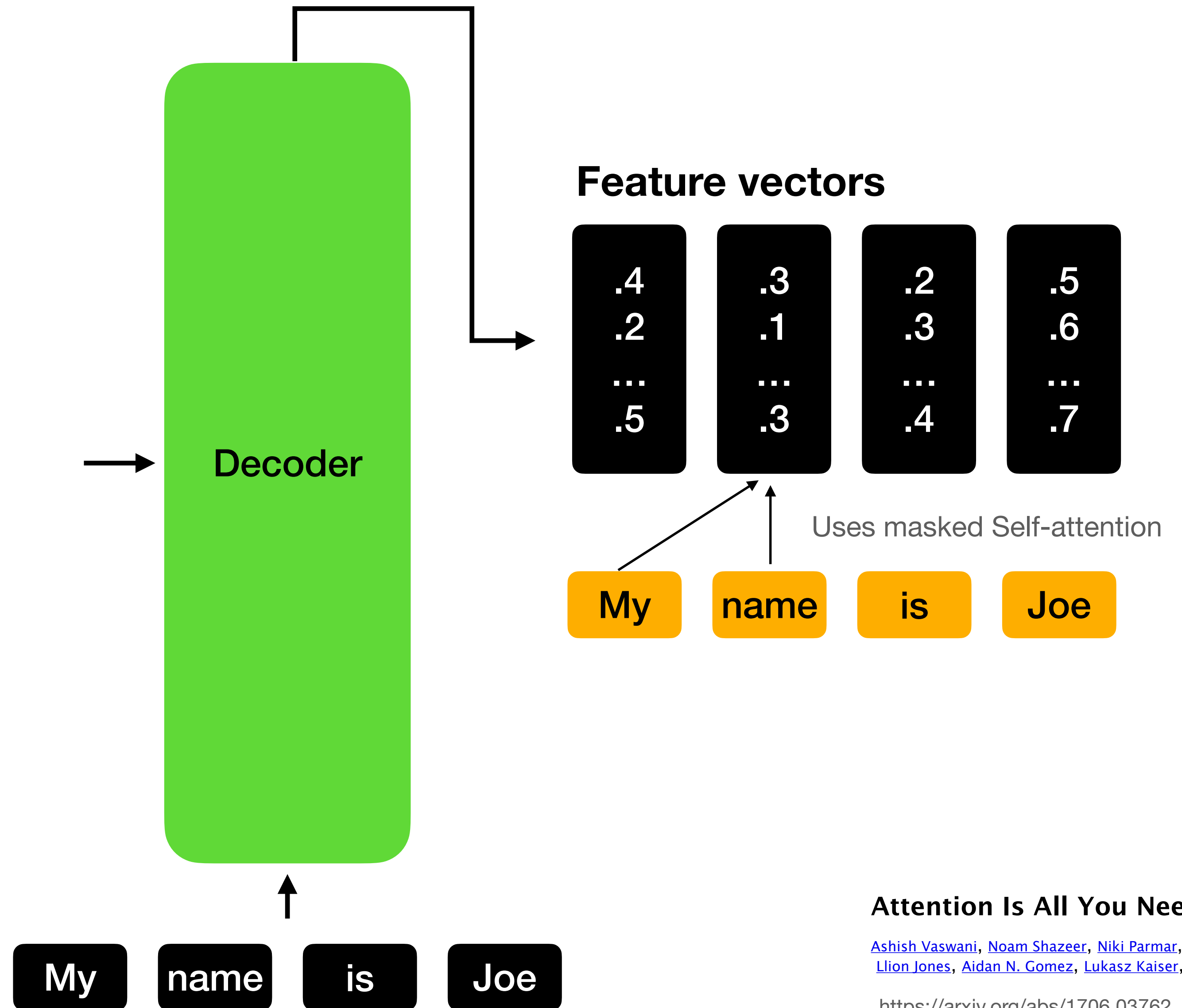
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Decoder



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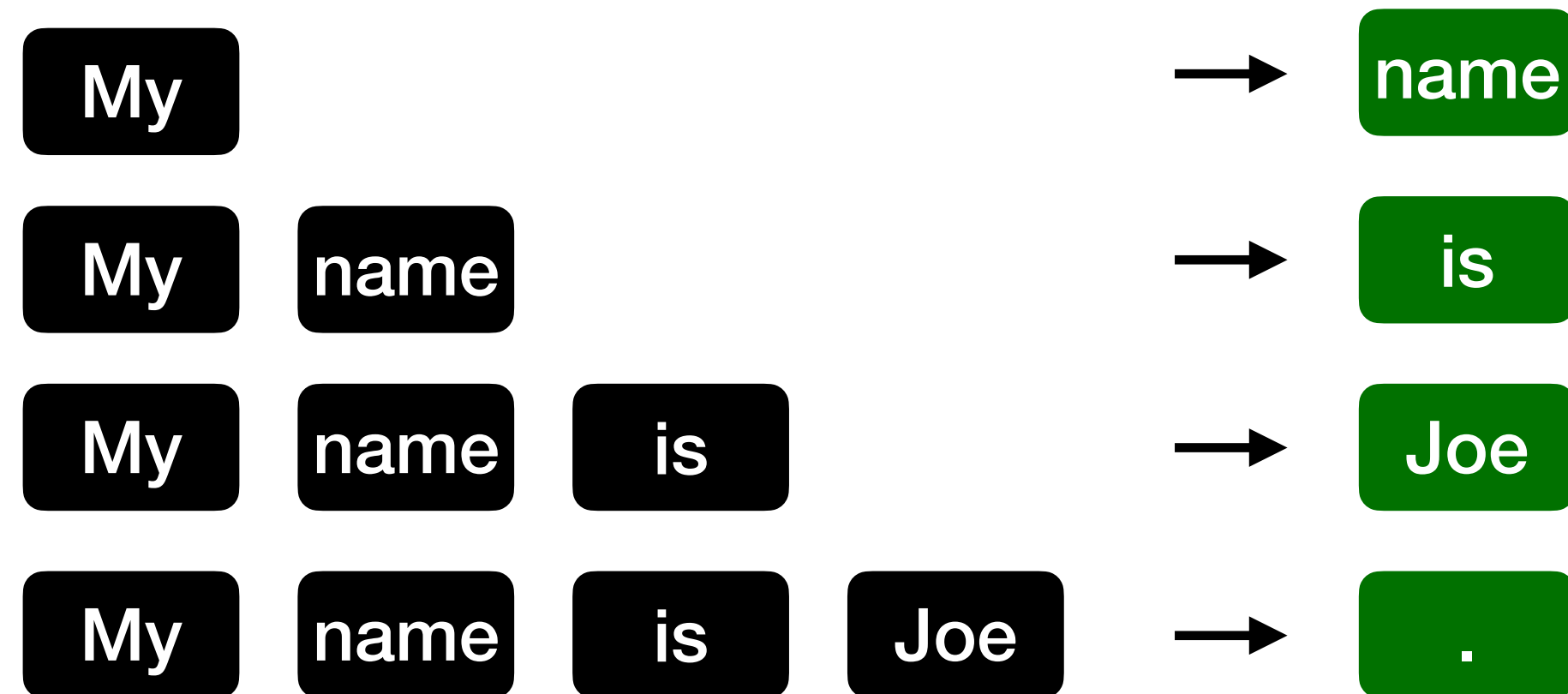
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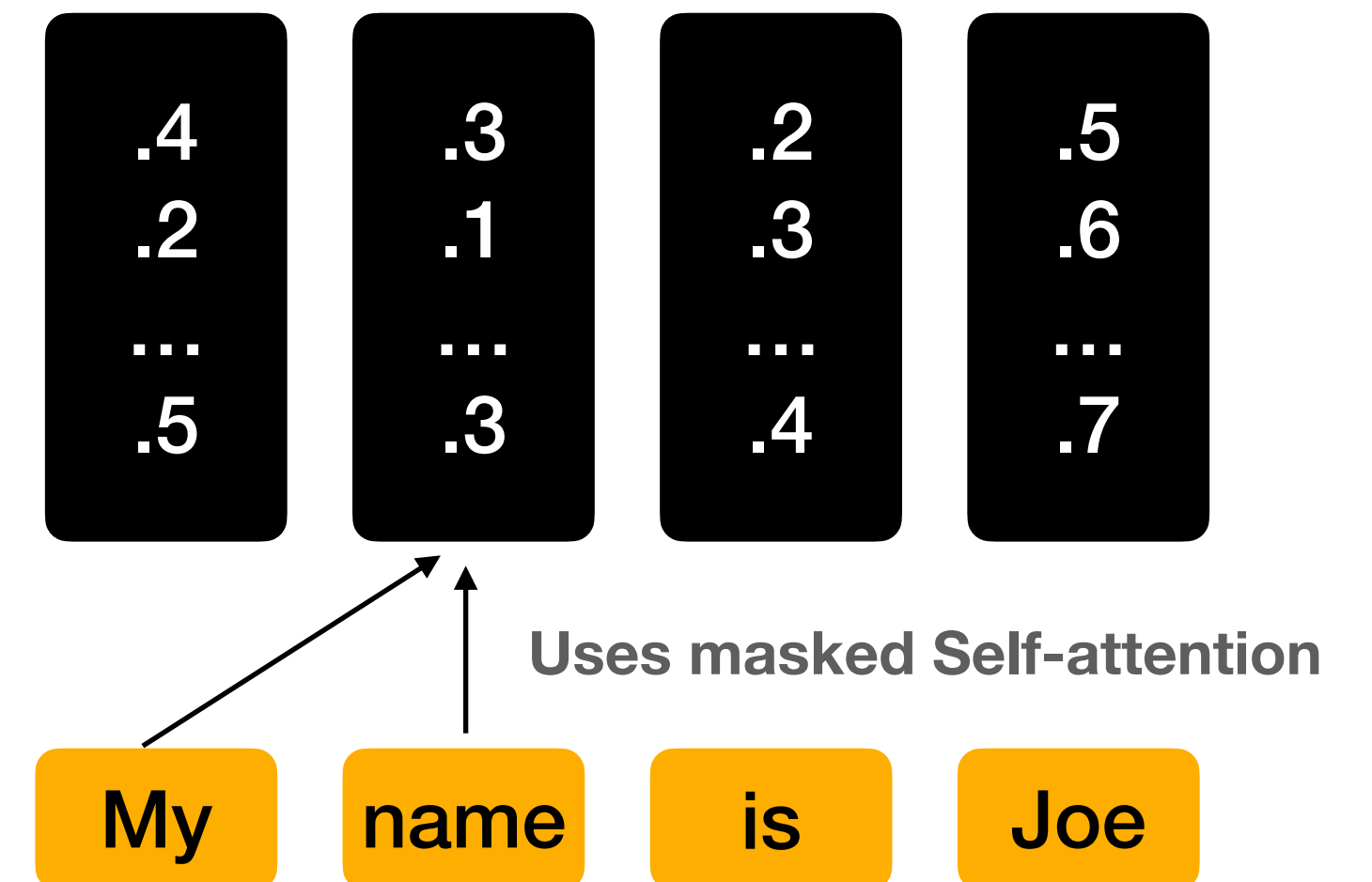
Transformers

Decoder

Used Auto-regressively



Feature vectors

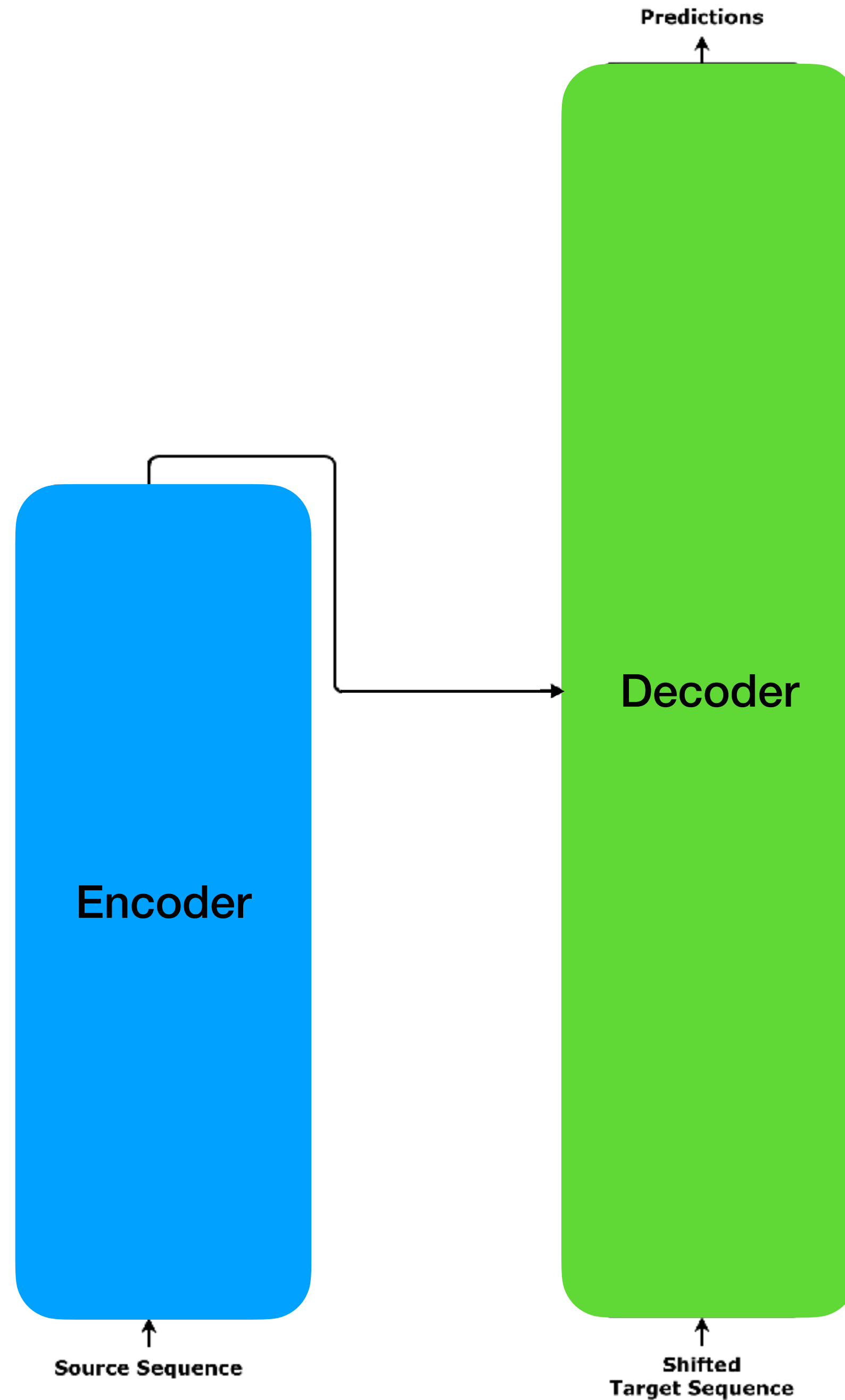


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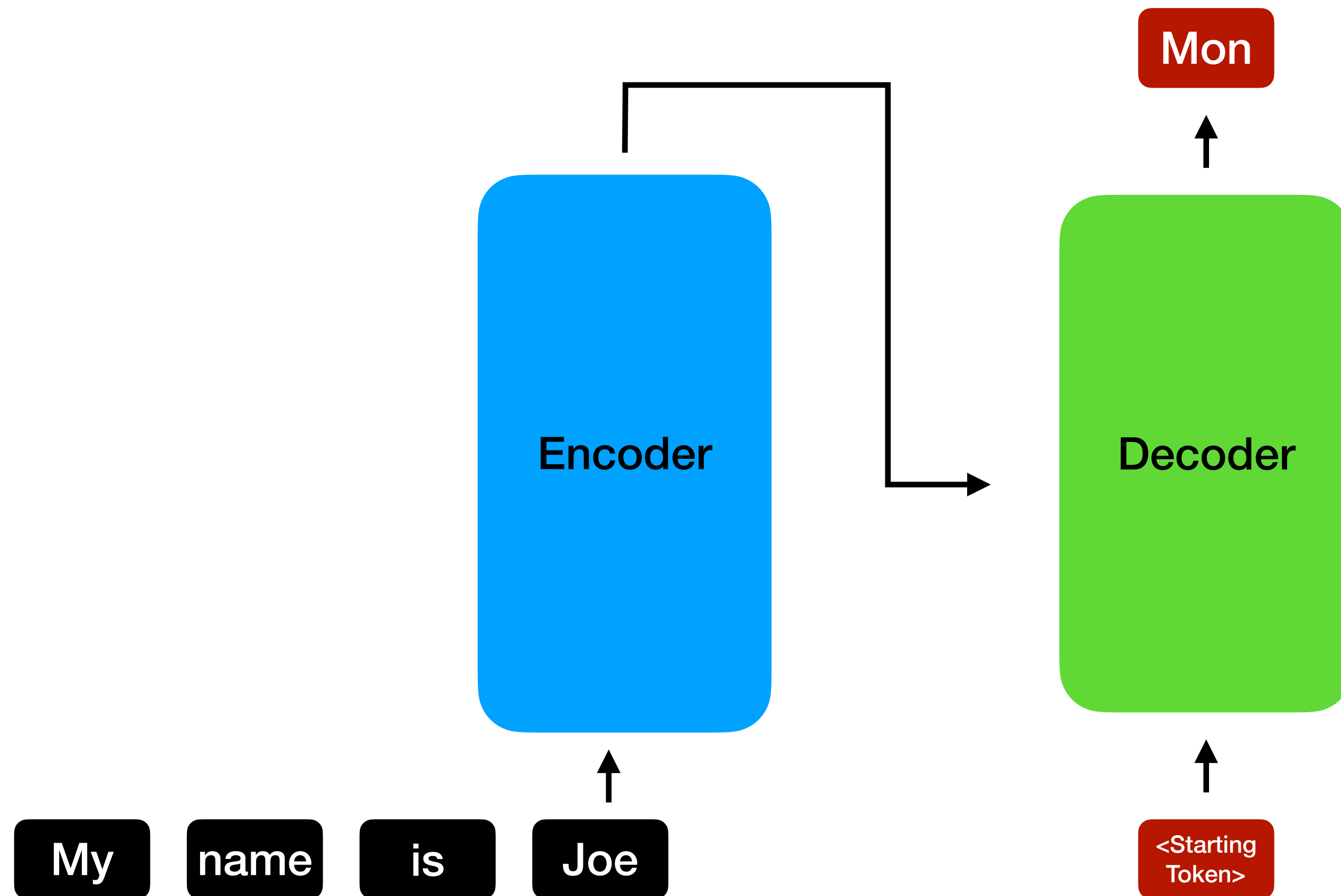
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Transformers

Translation



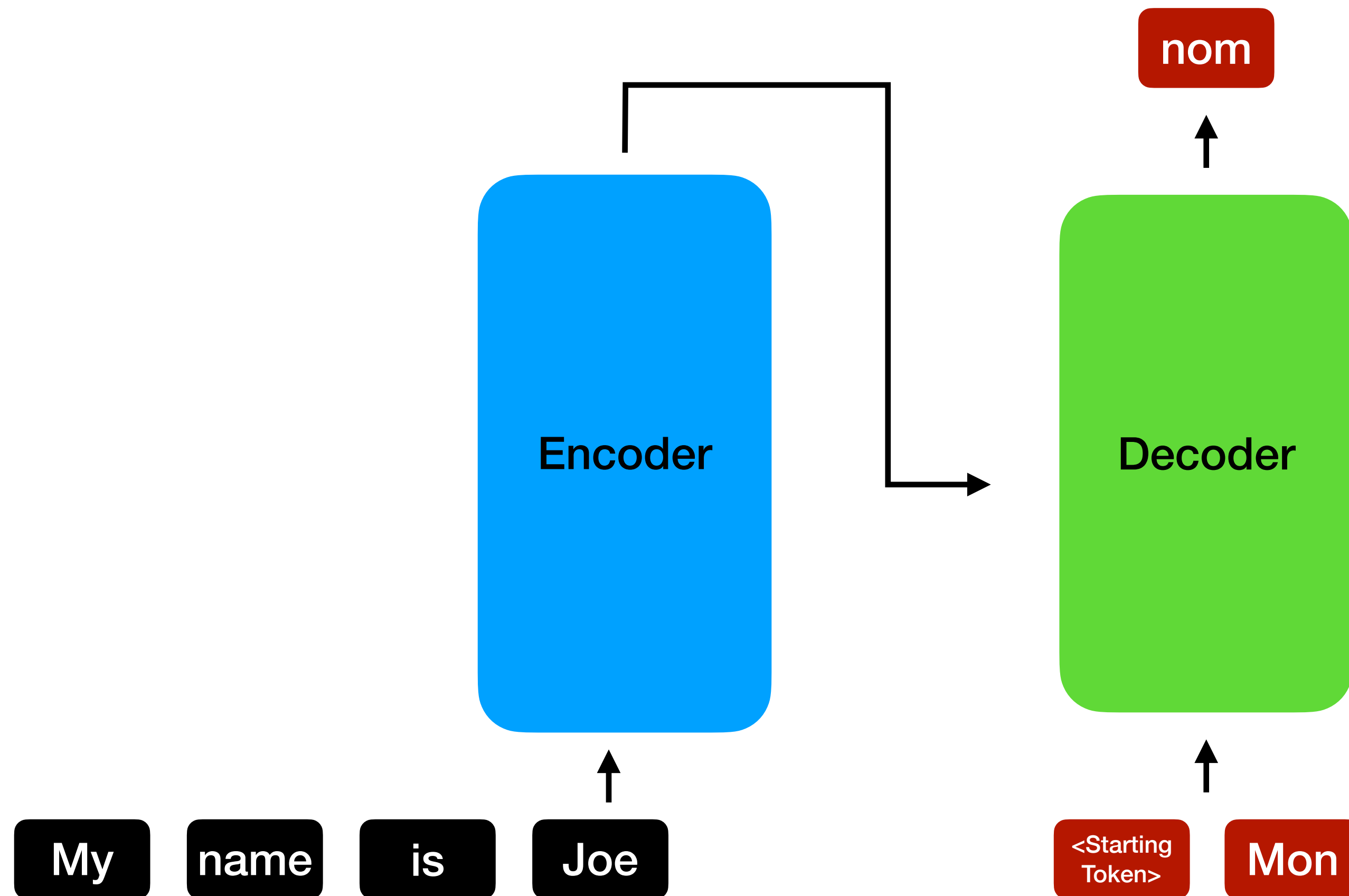
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Translation



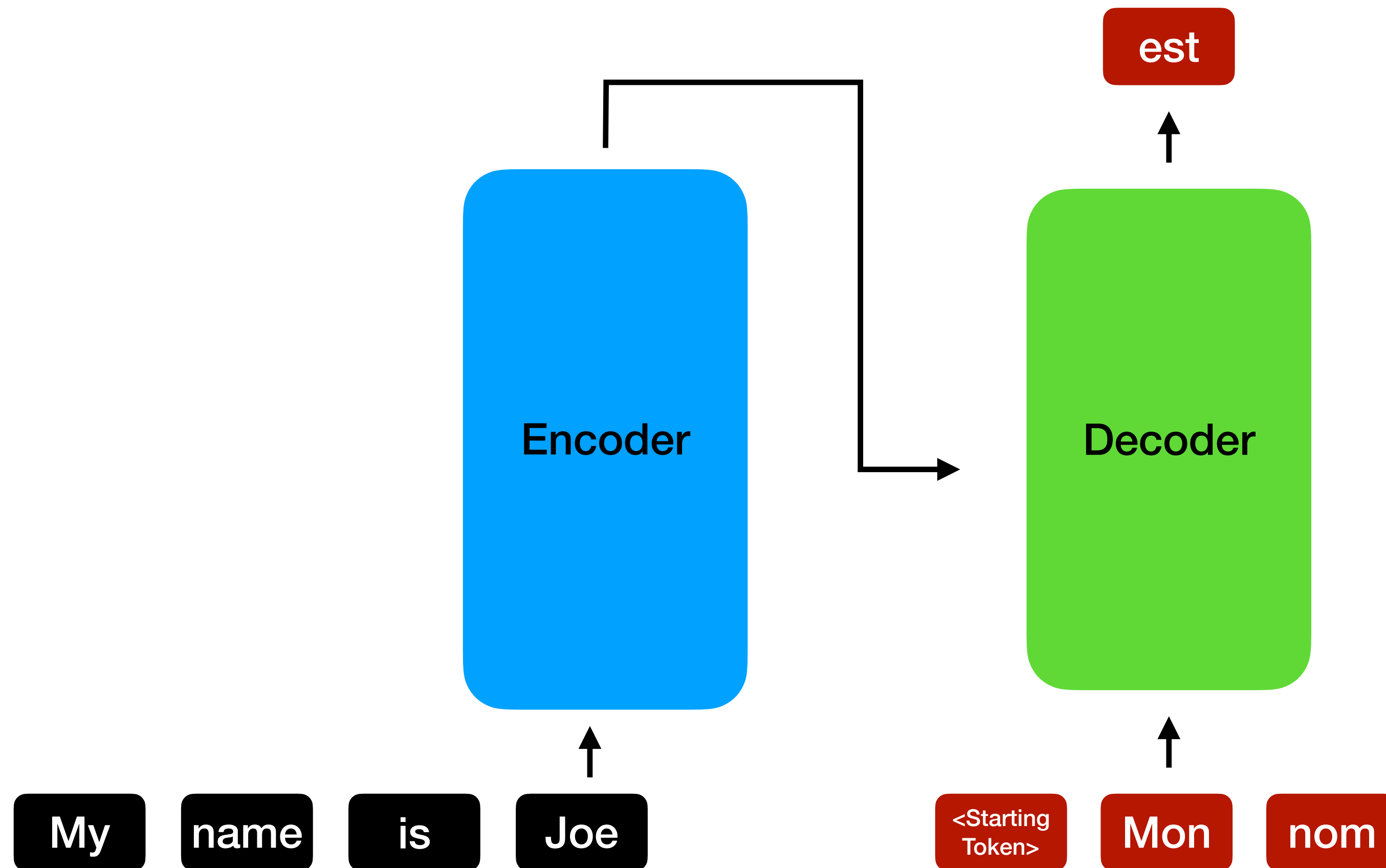
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Translation



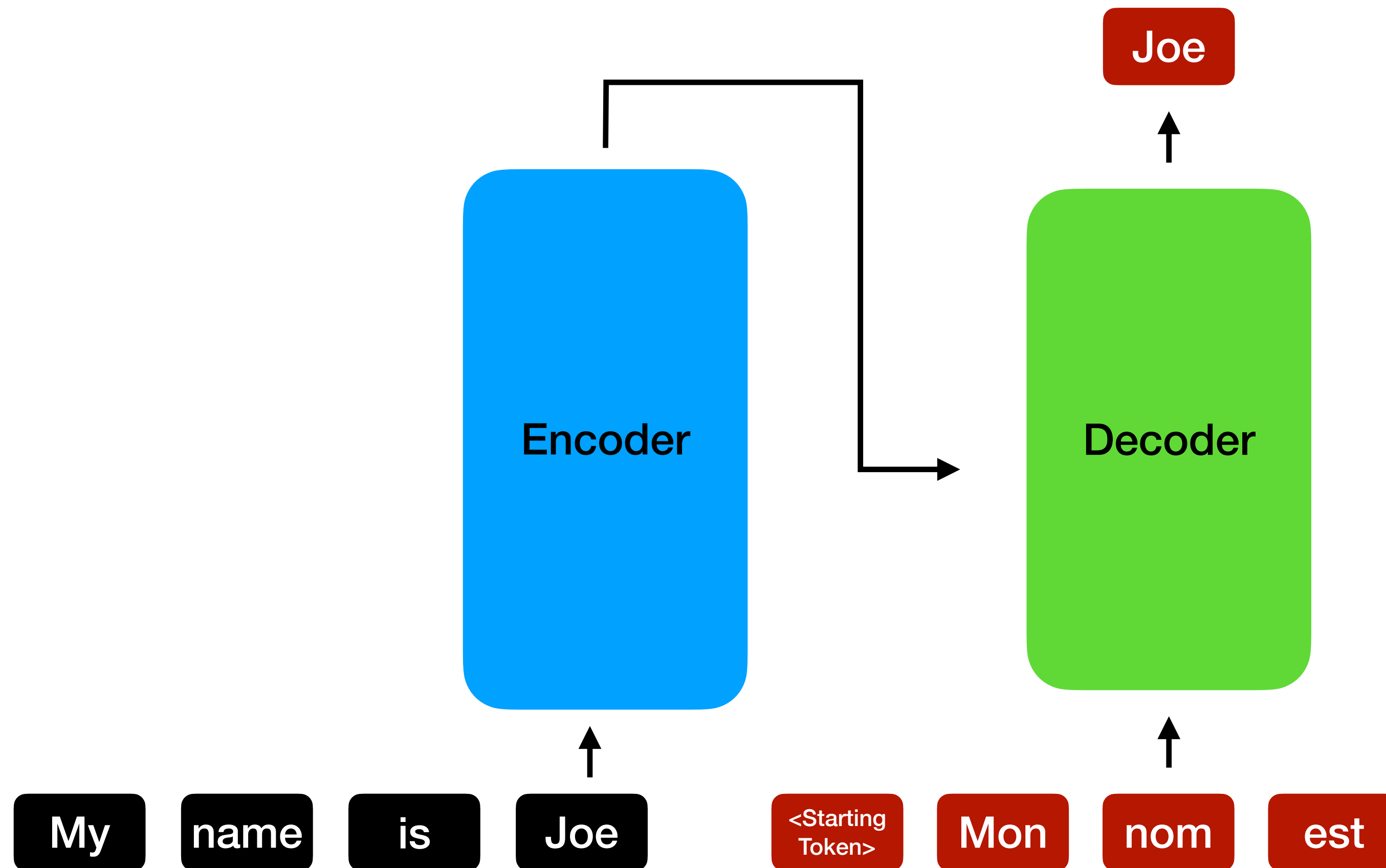
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Transformers

Translation



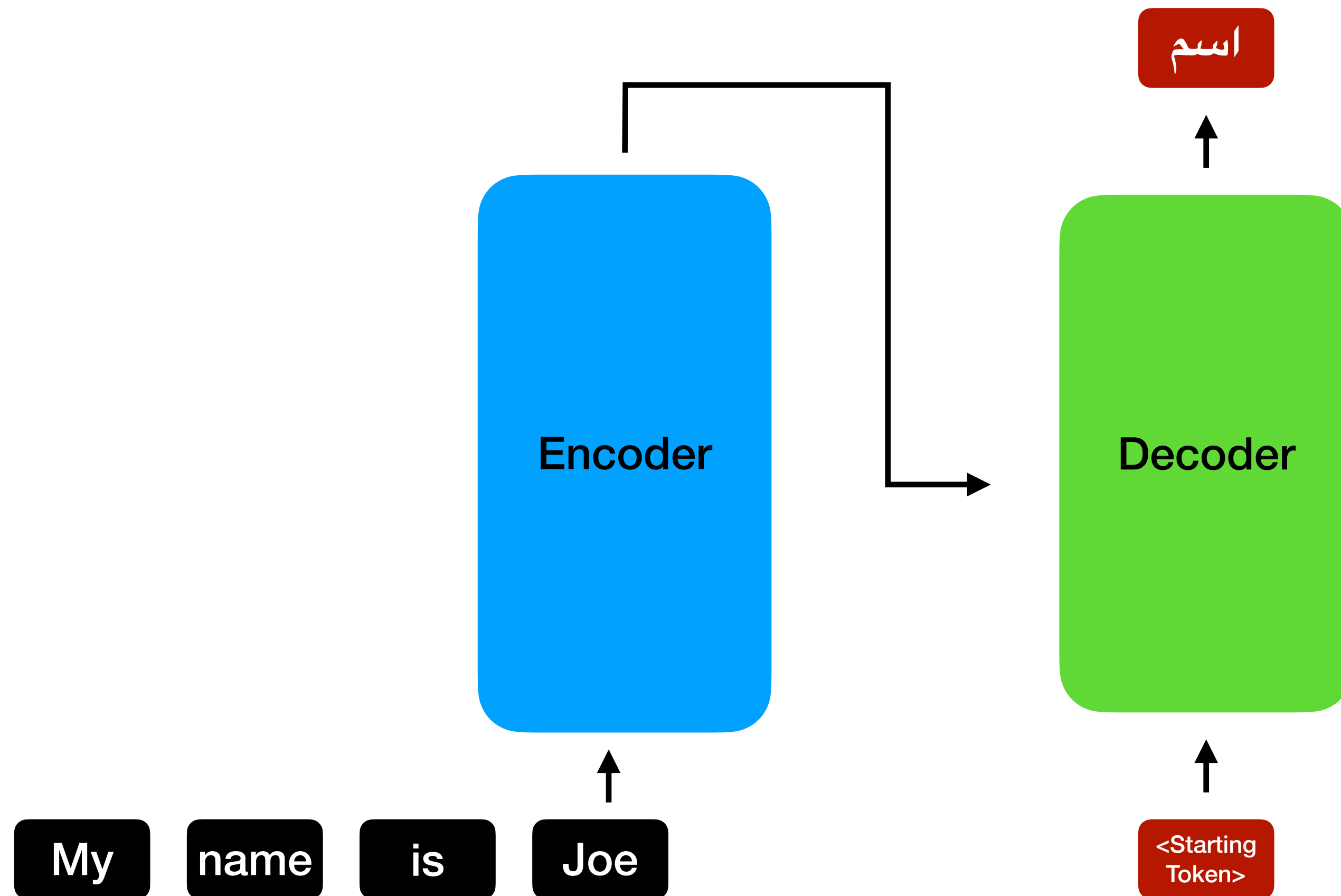
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Transformers

Translation



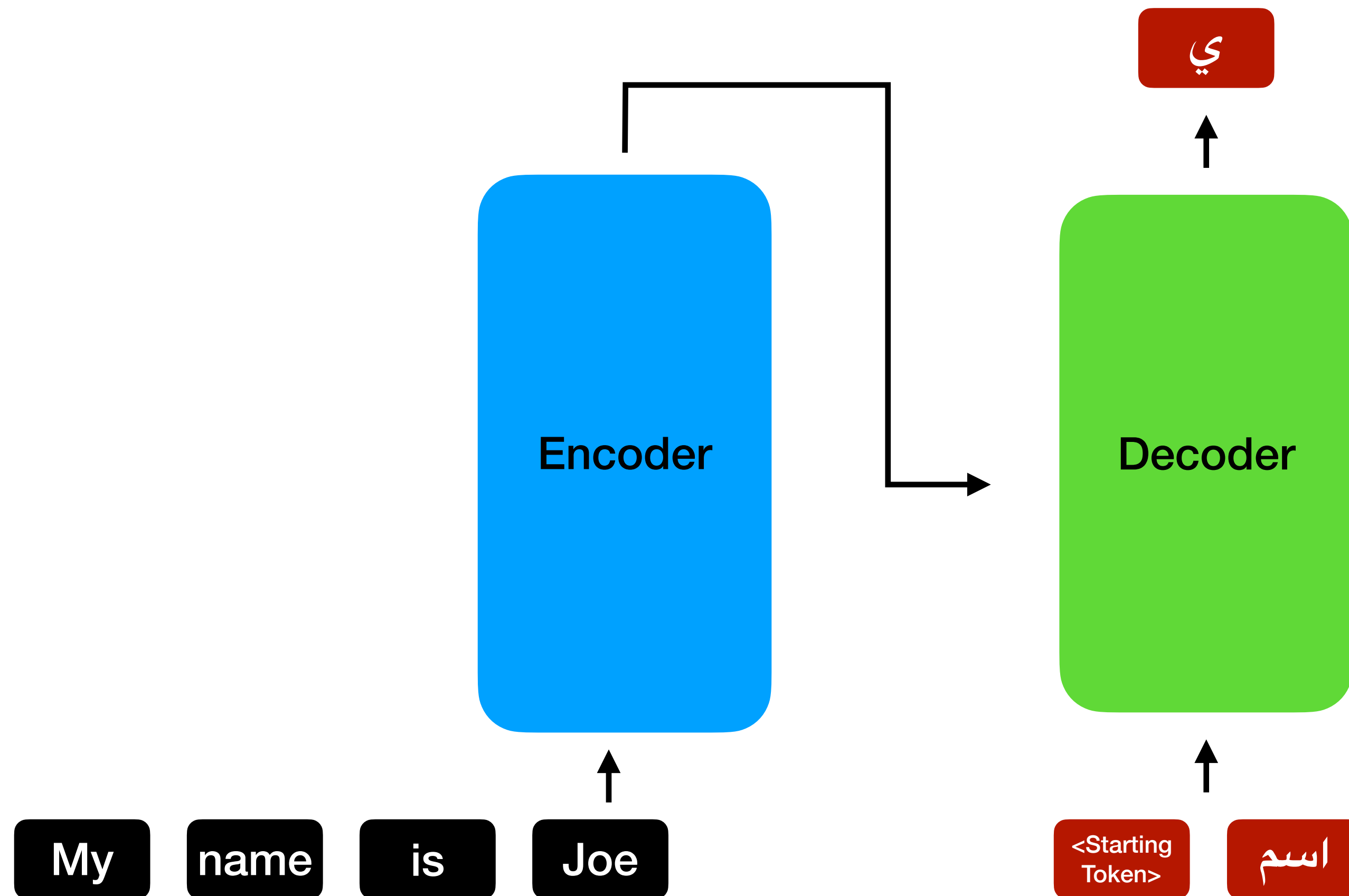
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Transformers

Translation



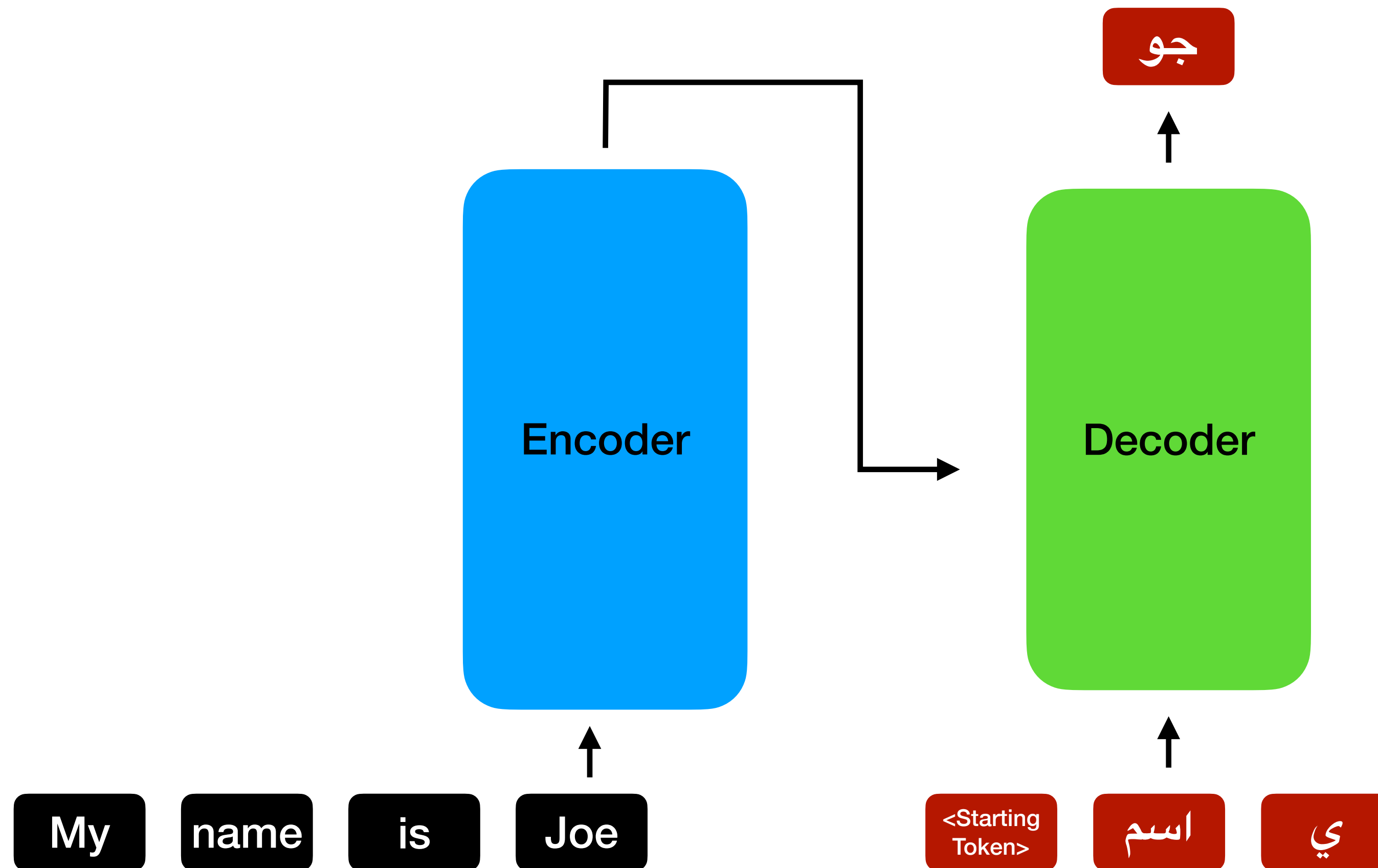
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Transformers

Translation

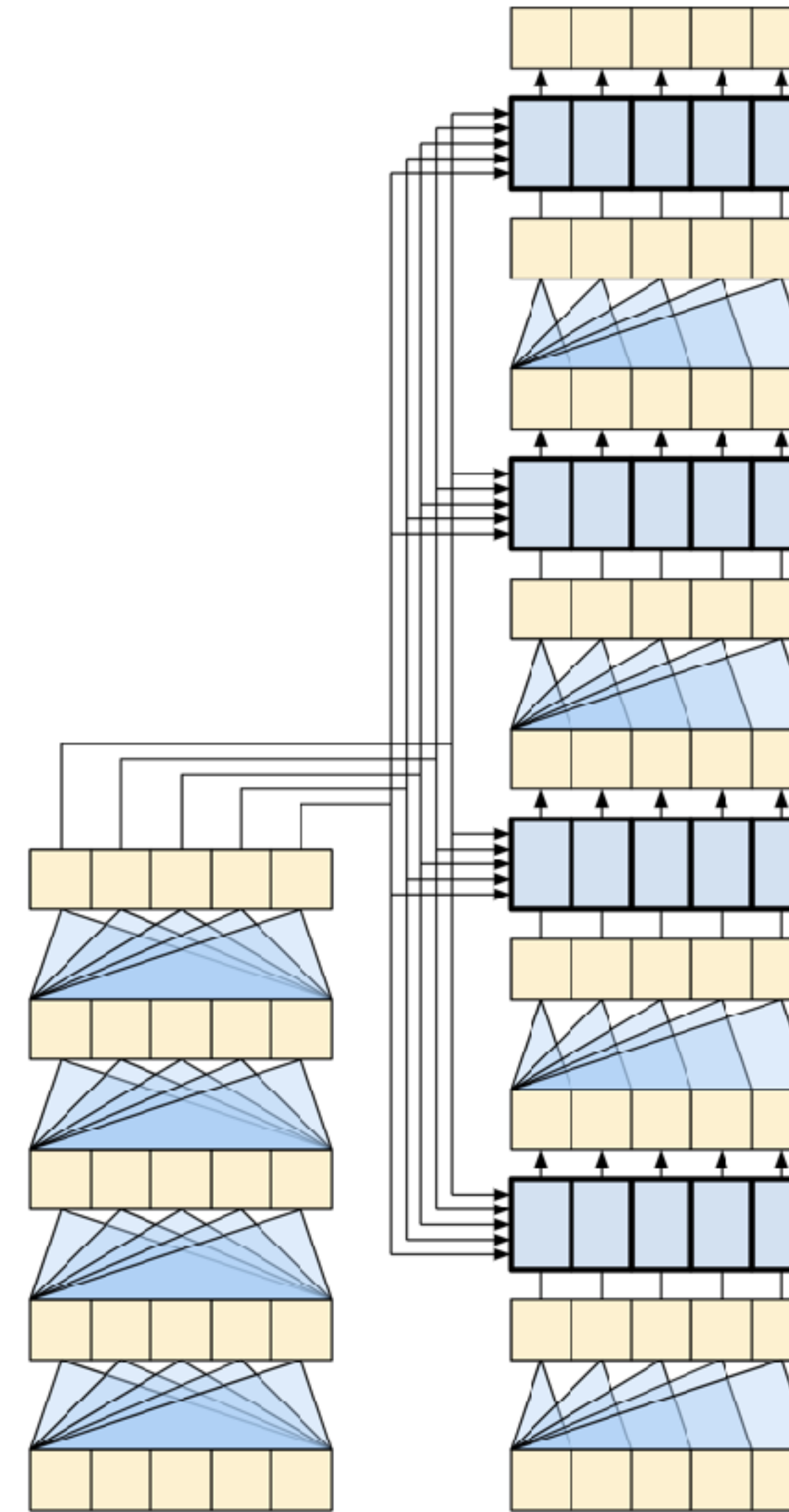
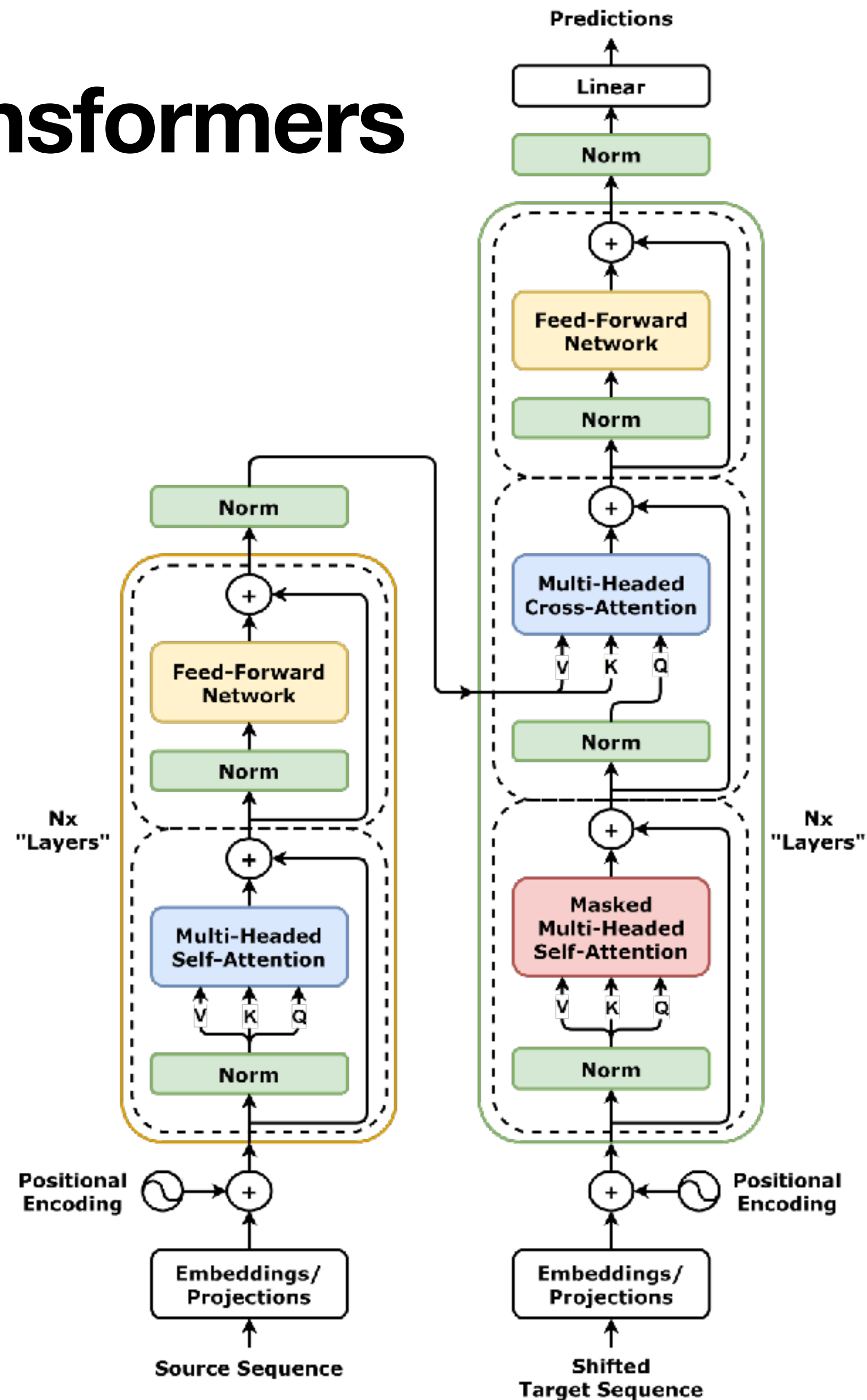


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Transformers

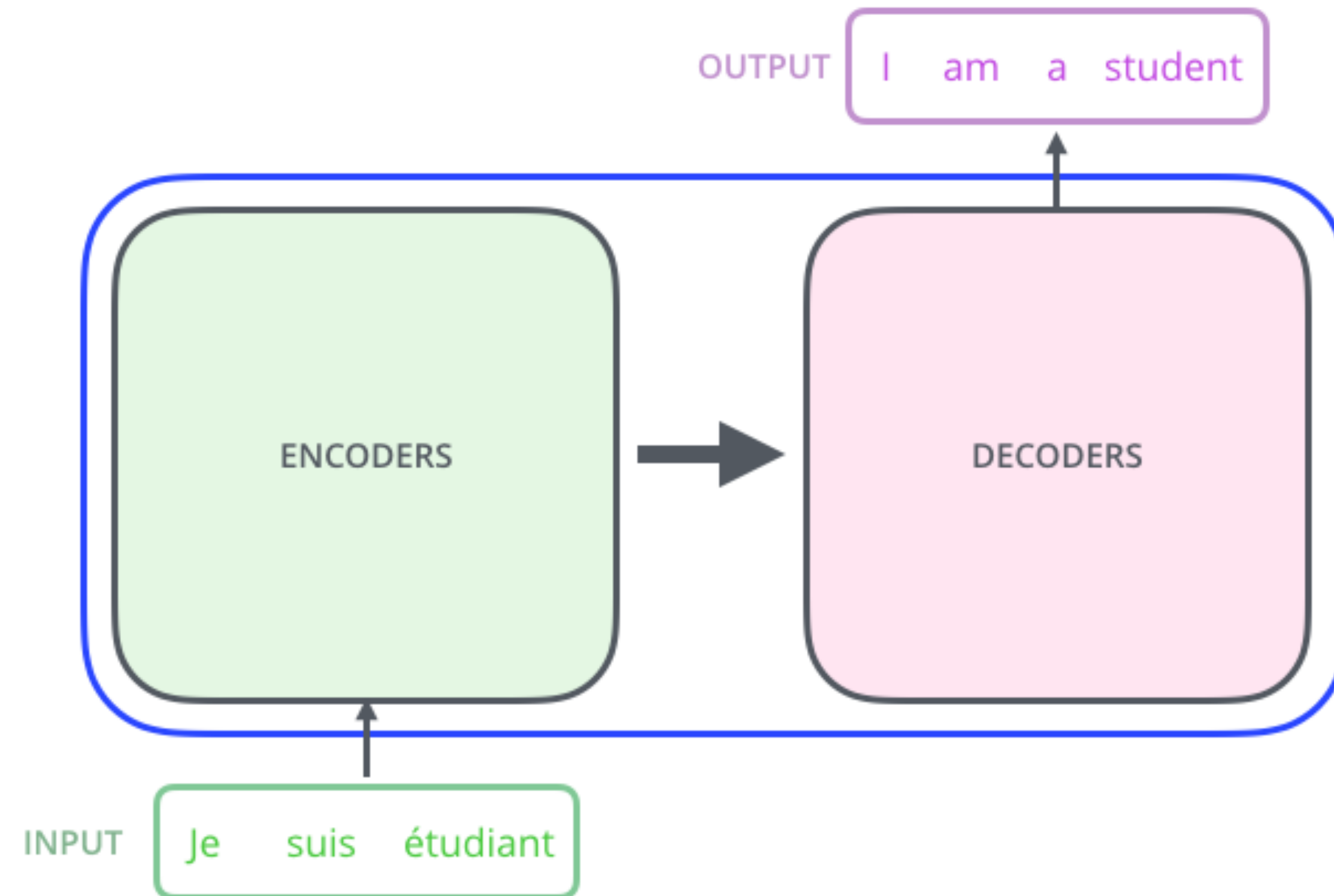


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Transformers



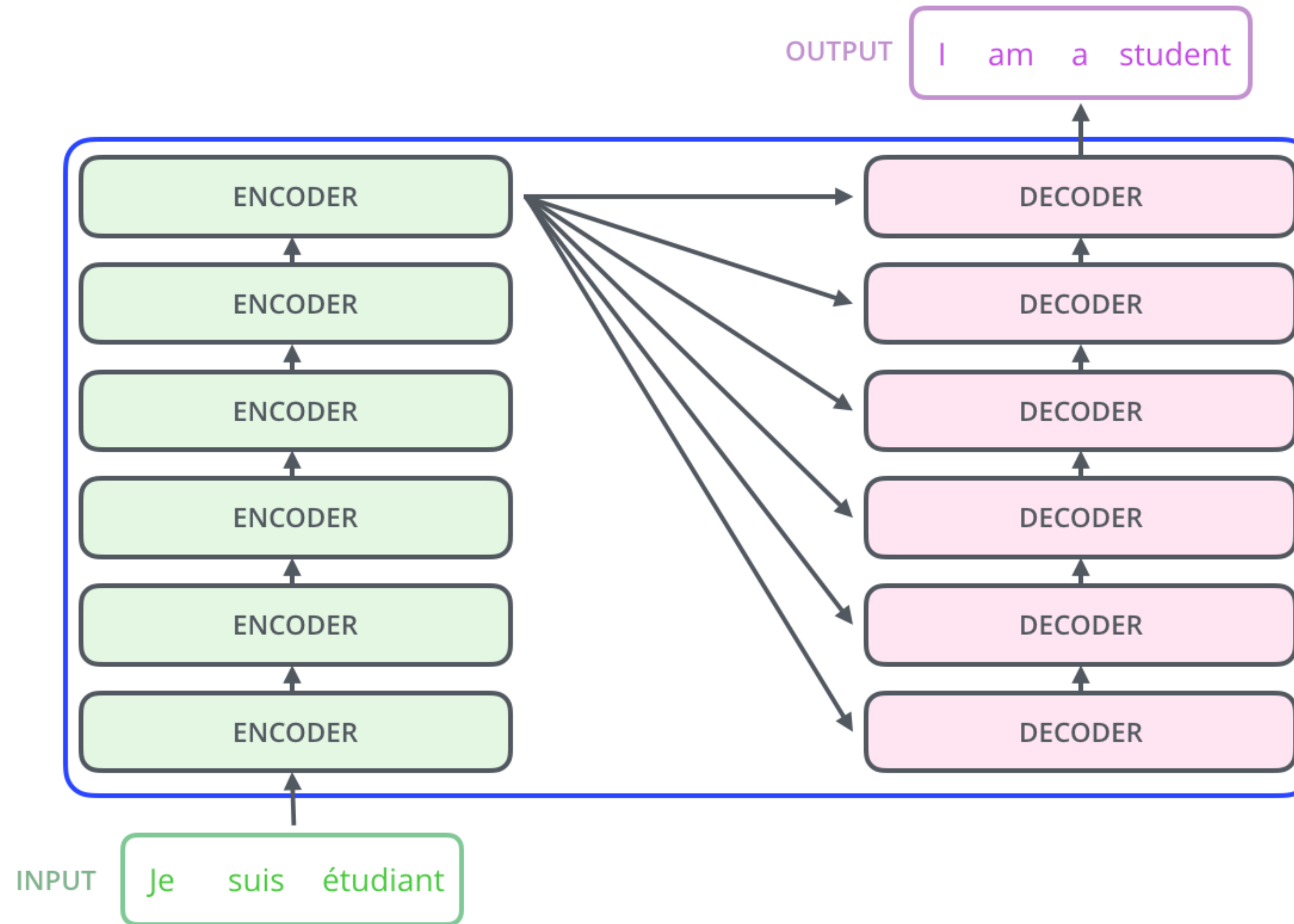
<https://arxiv.org/abs/1706.03762>

<https://jalammar.github.io/visualizing-neural-machine-translation-mechanics-of-seq2seq-models-with-attention/>

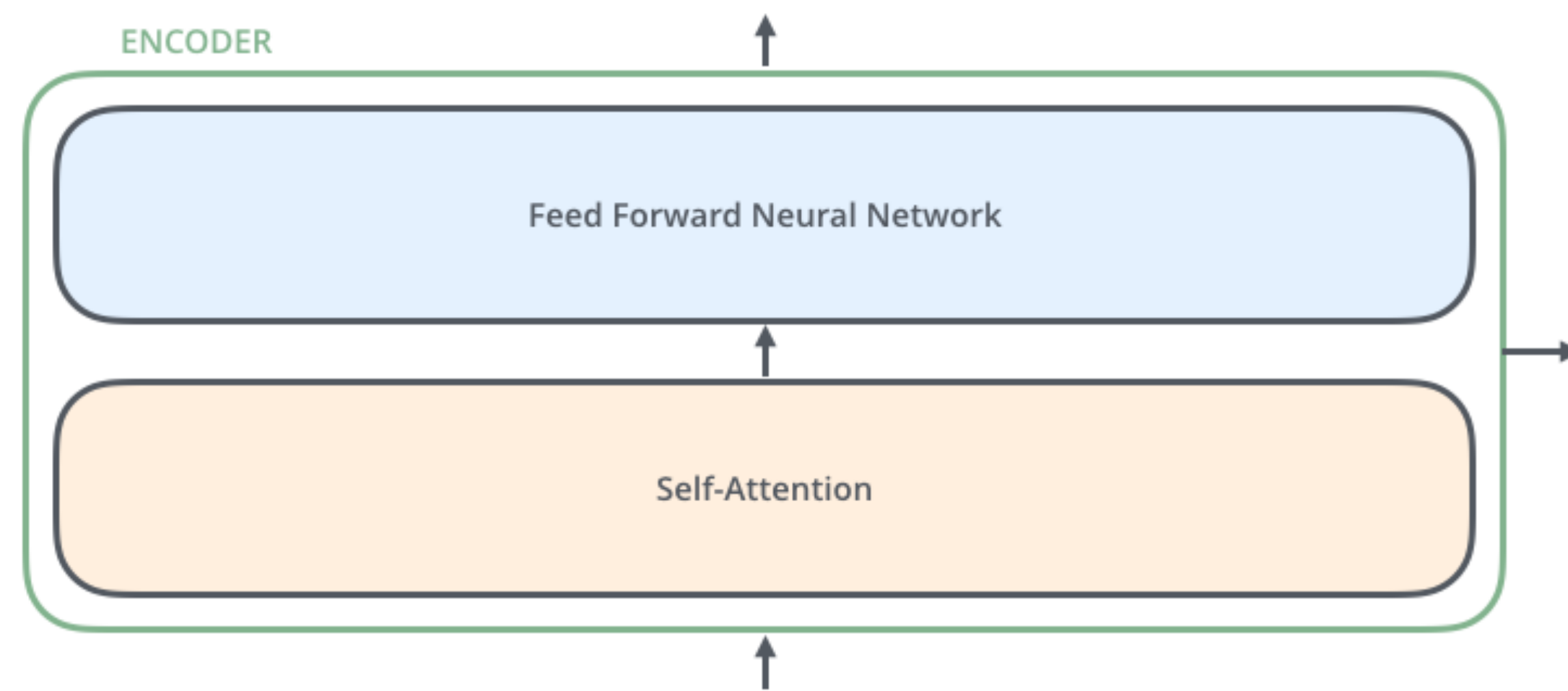
<https://www.tensorflow.org/text/tutorials/transformer>

<https://jalammar.github.io/illustrated-transformer/>

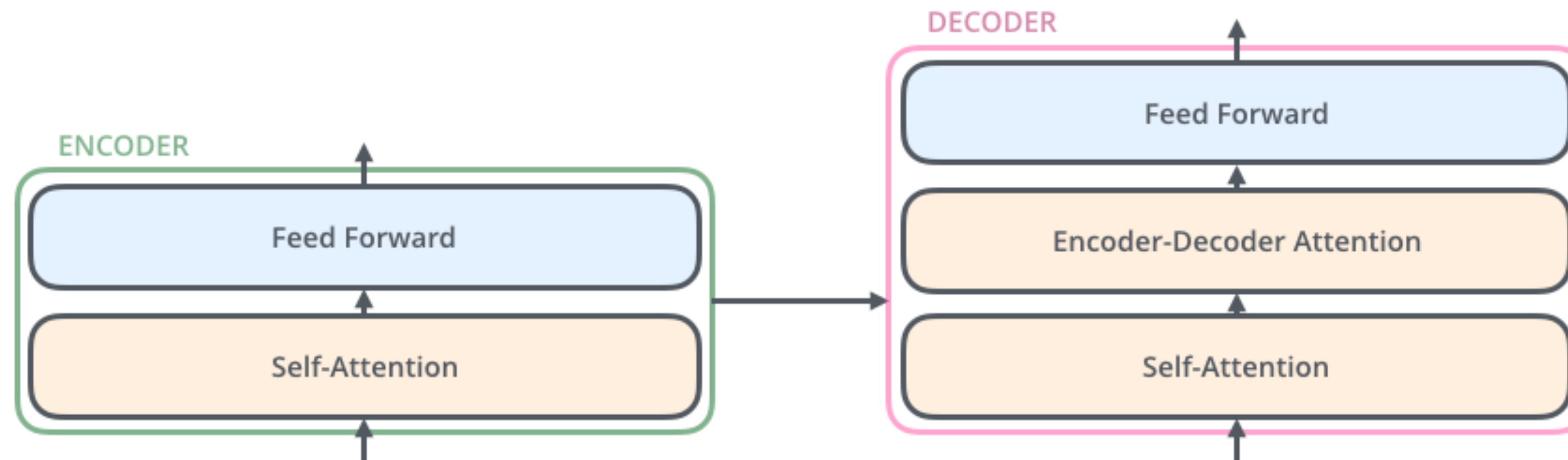
Transformers



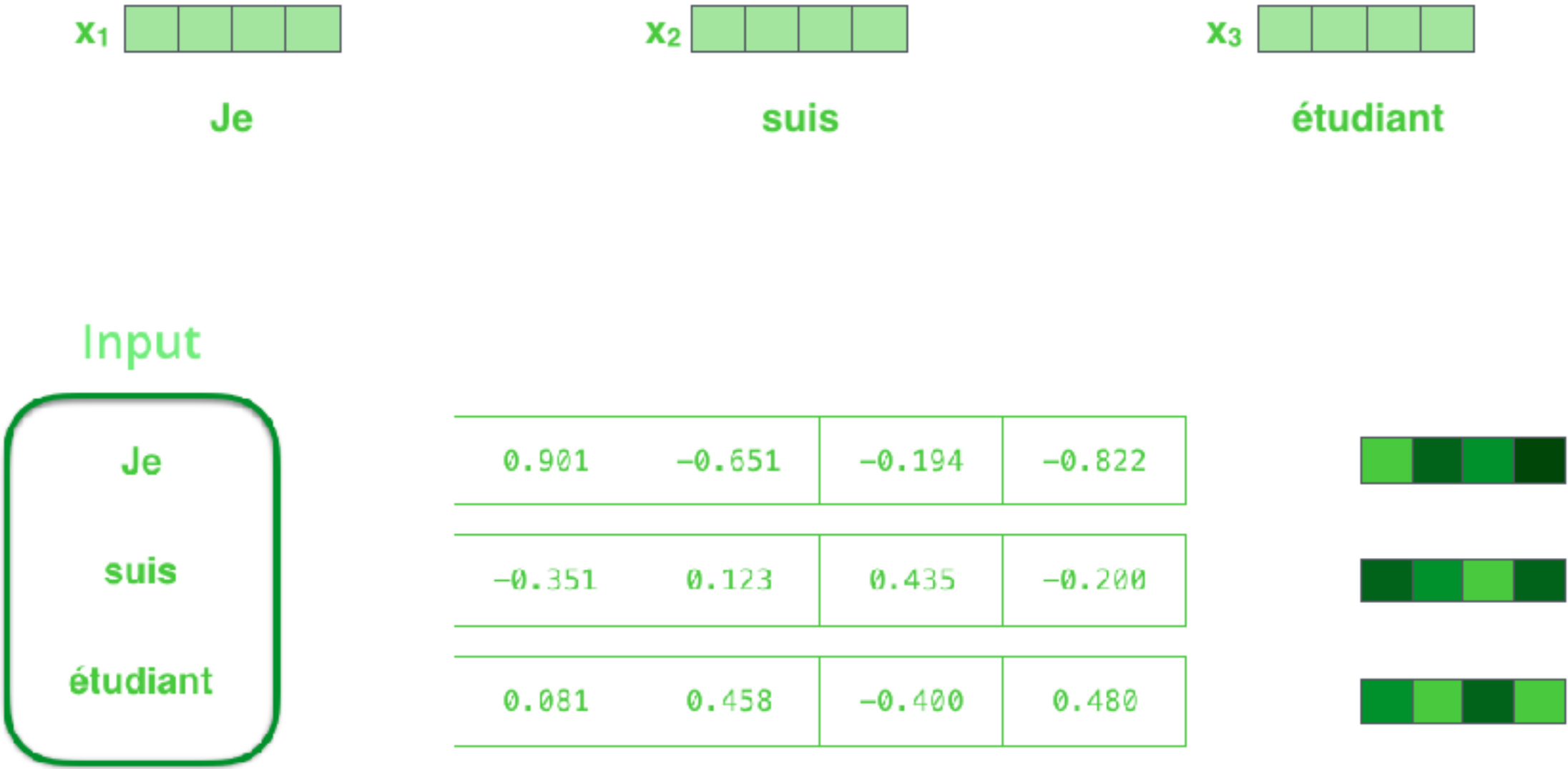
Transformers



Transformers

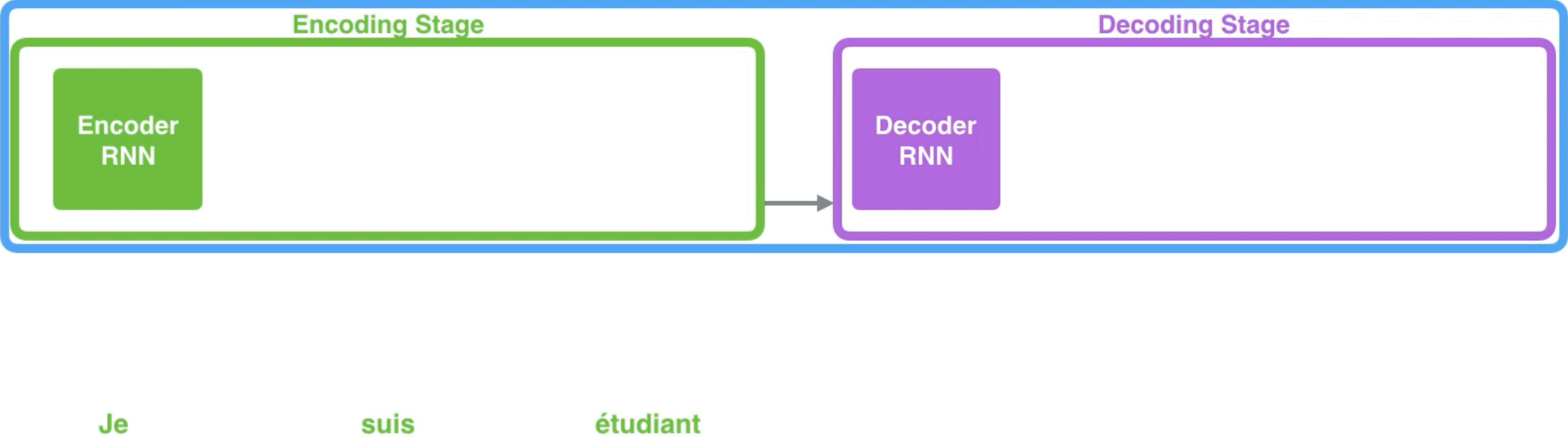


Transformers



Neural Machine Translation Model

Neural Machine Translation SEQUENCE TO SEQUENCE MODEL



Neural Machine Translation Model

Neural Machine Translation SEQUENCE TO SEQUENCE MODEL WITH ATTENTION



Je suis étudiant

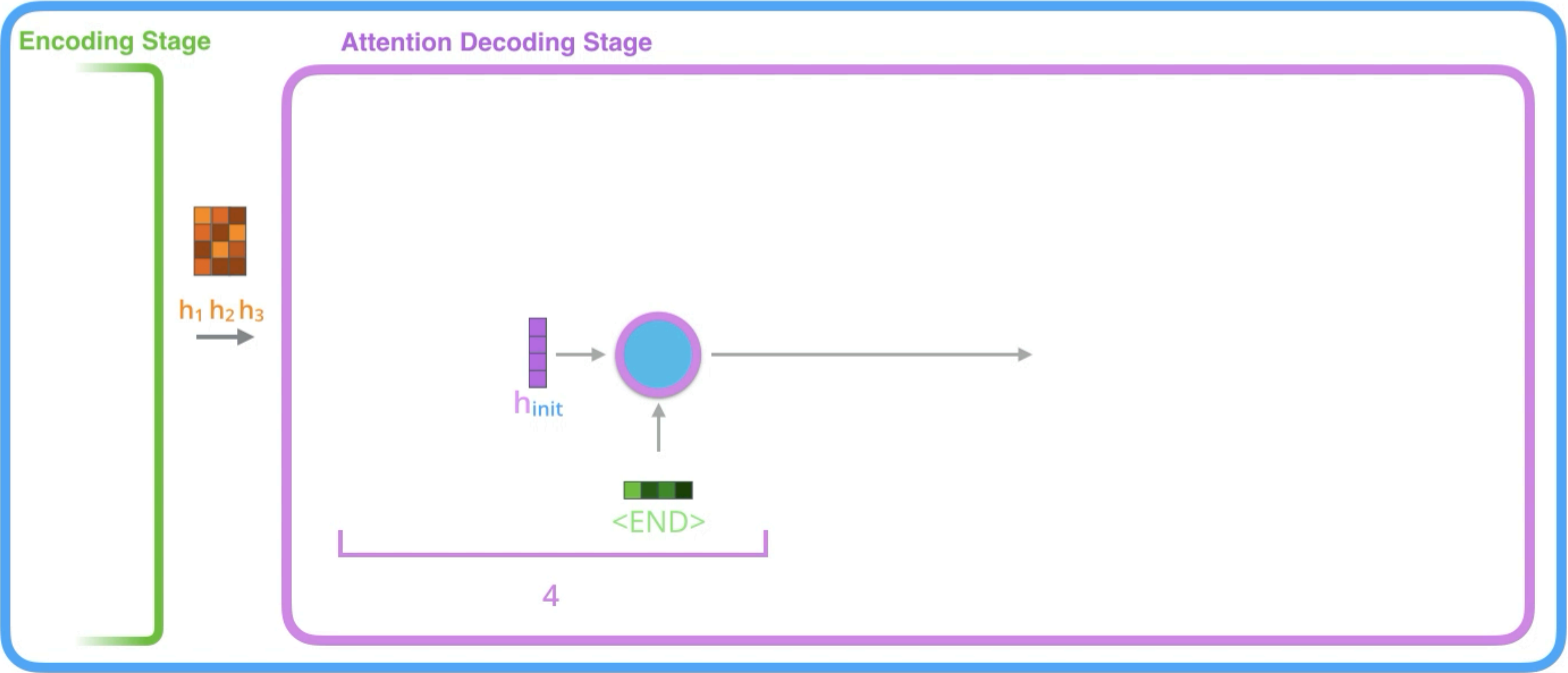
Neural Machine Translation Model

Attention at time step 4

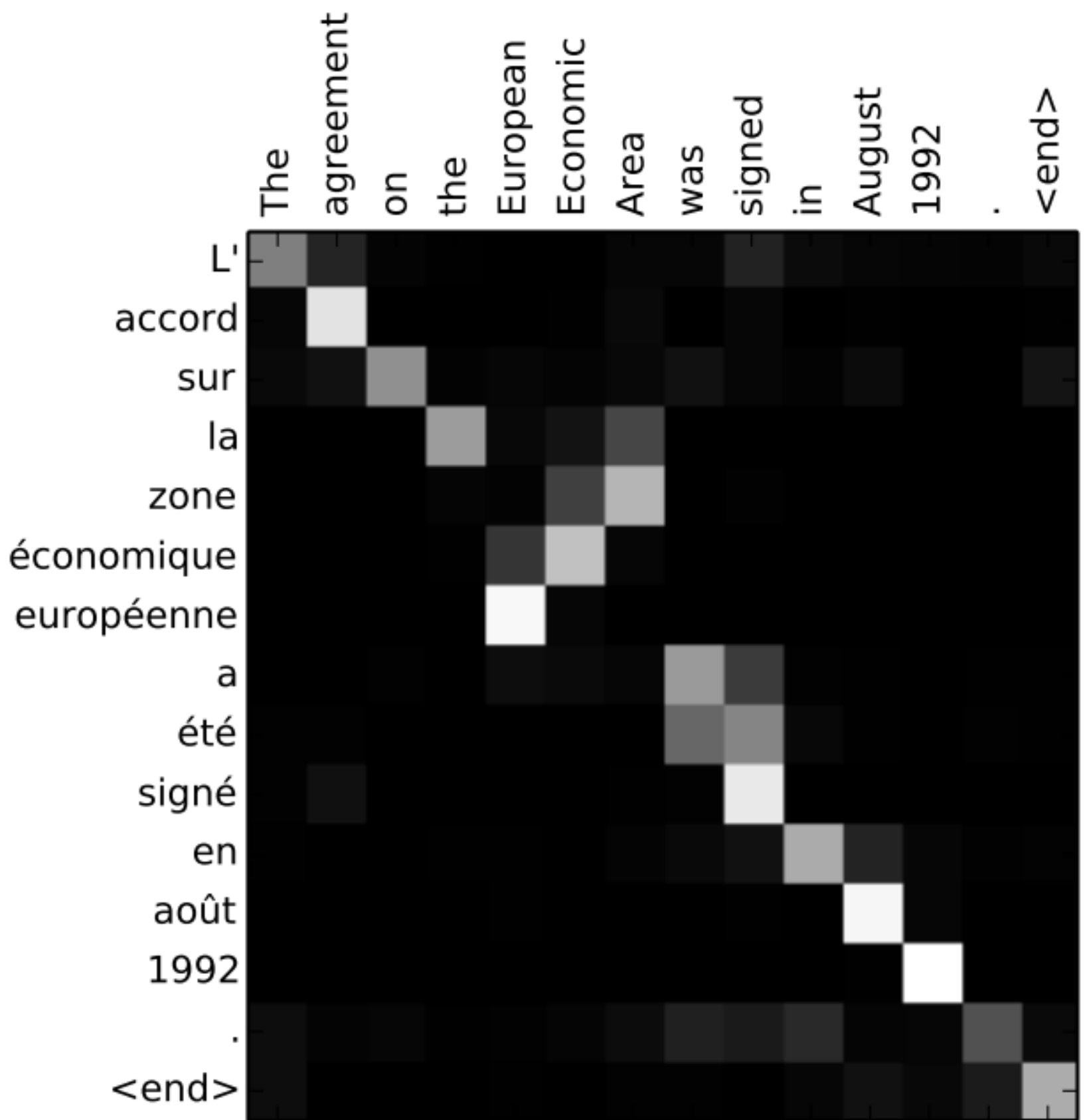
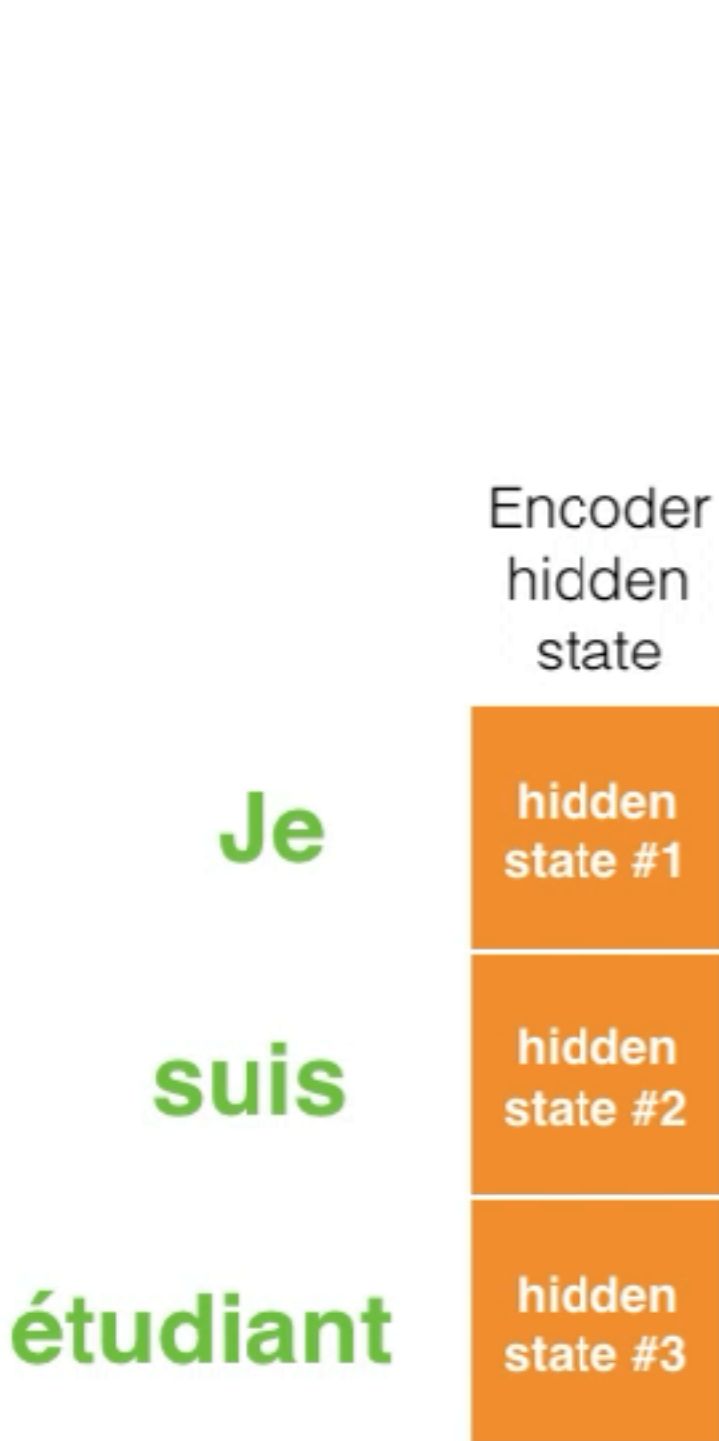


Neural Machine Translation Model

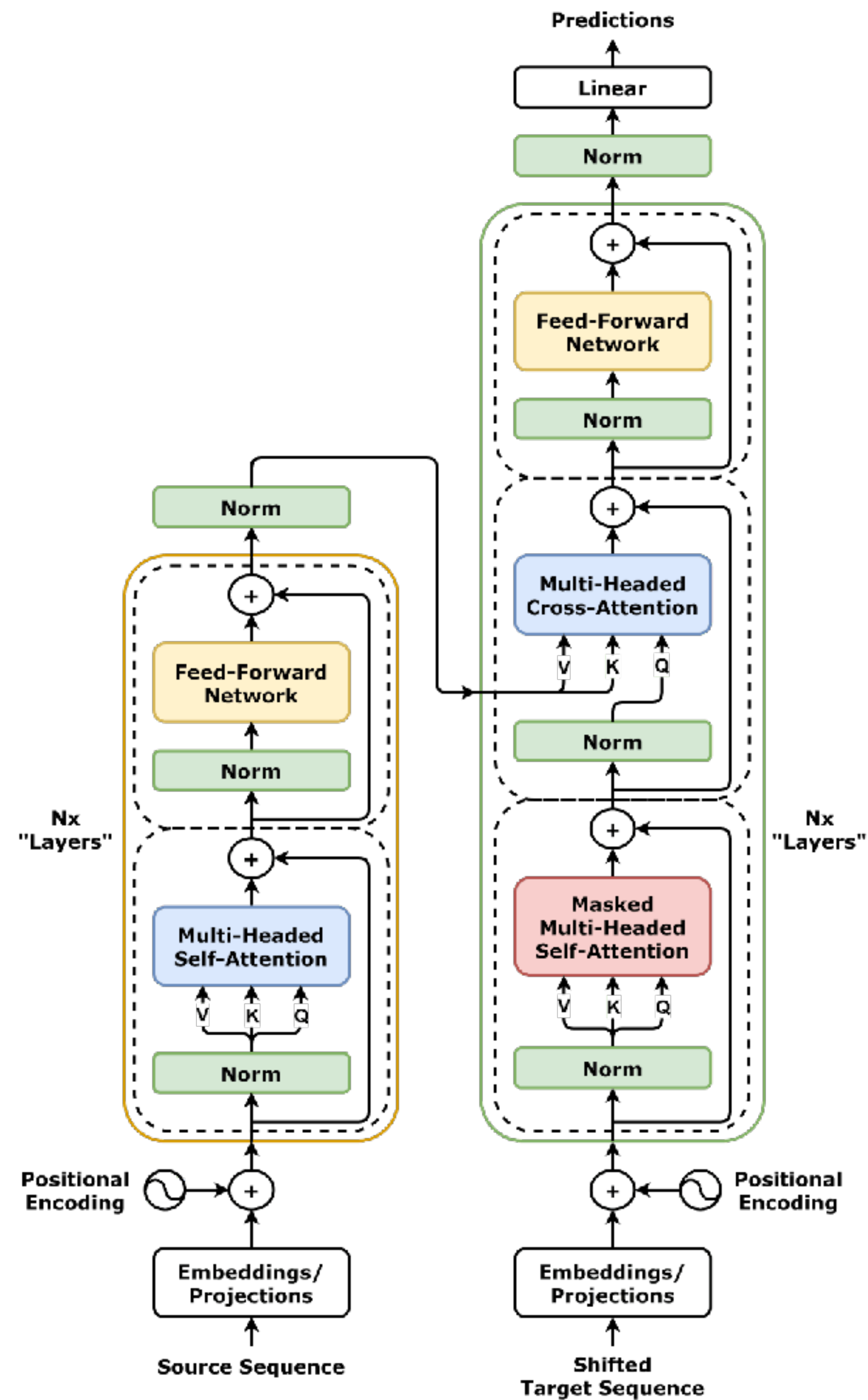
Neural Machine Translation SEQUENCE TO SEQUENCE MODEL WITH ATTENTION



Neural Machine Translation Model



Transformers

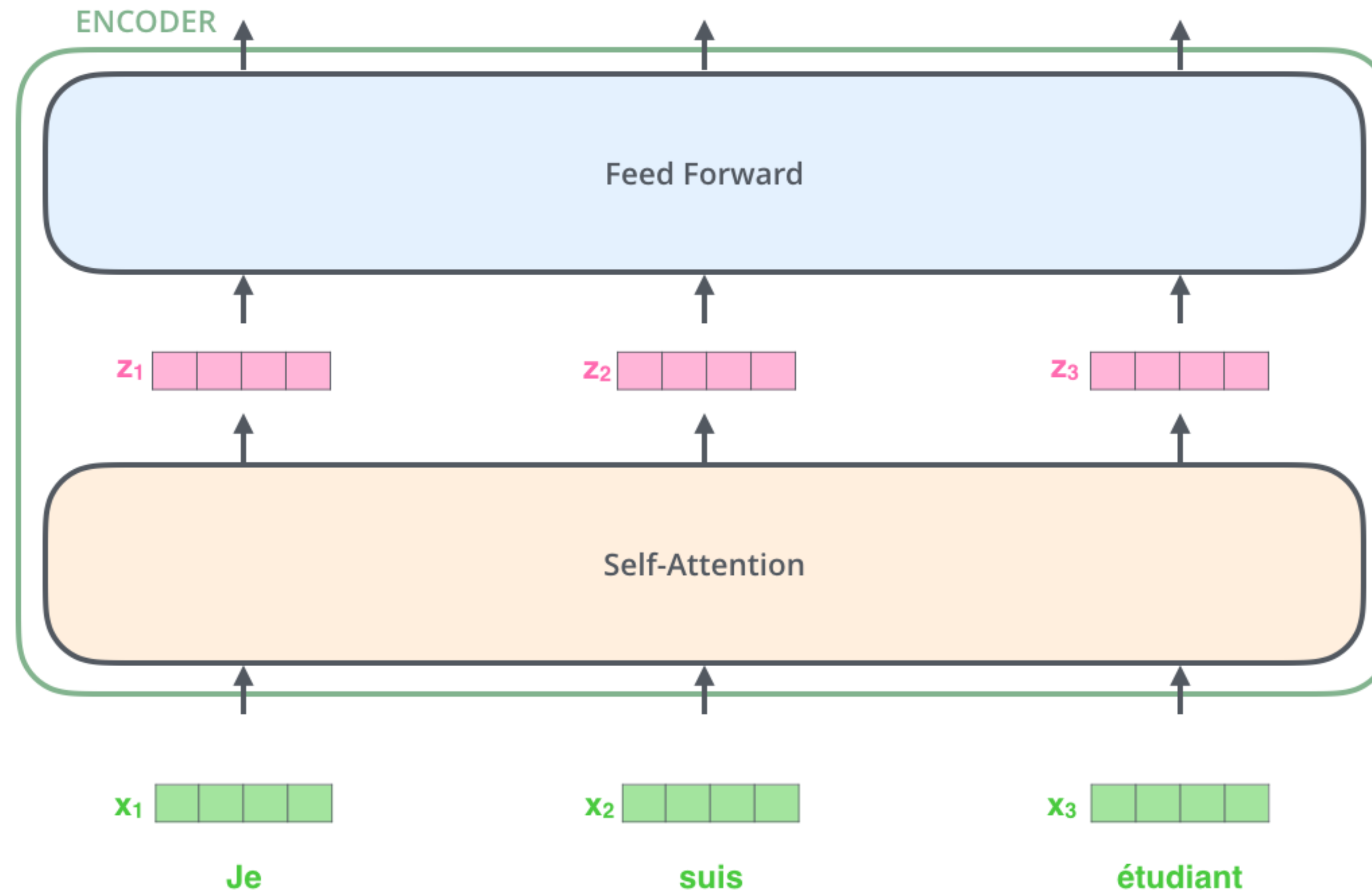


Attention Is All You Need

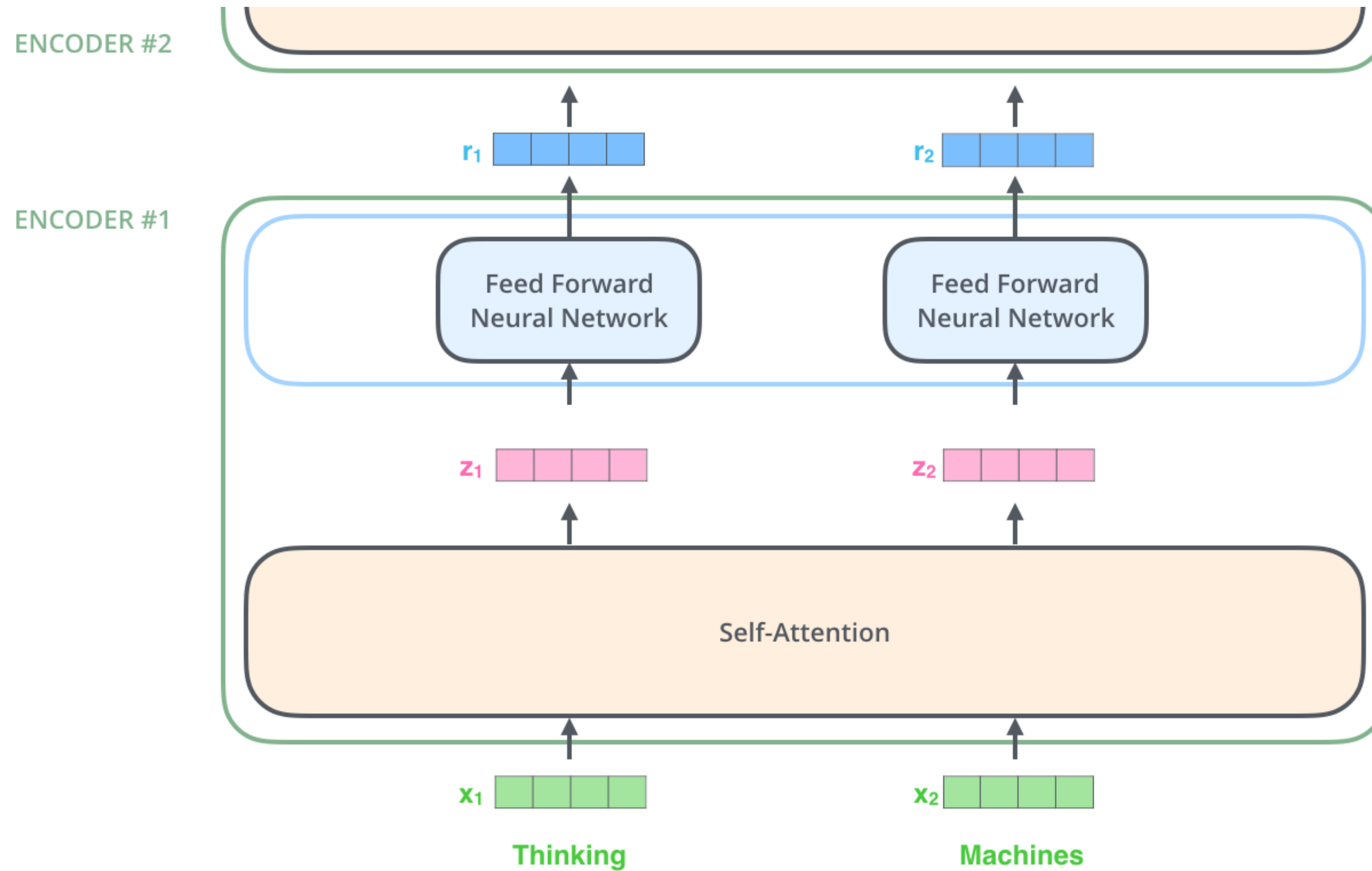
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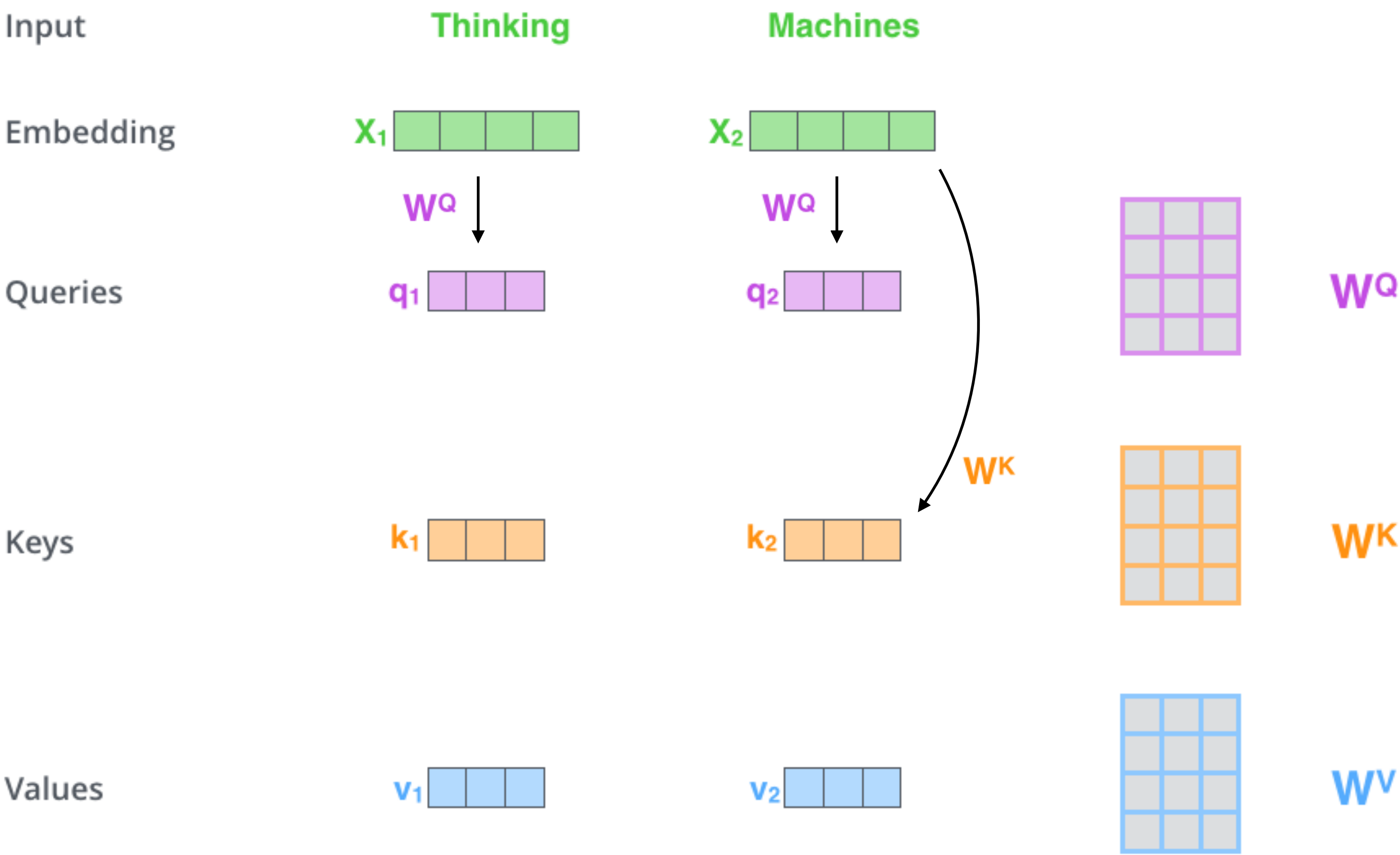
Transformers



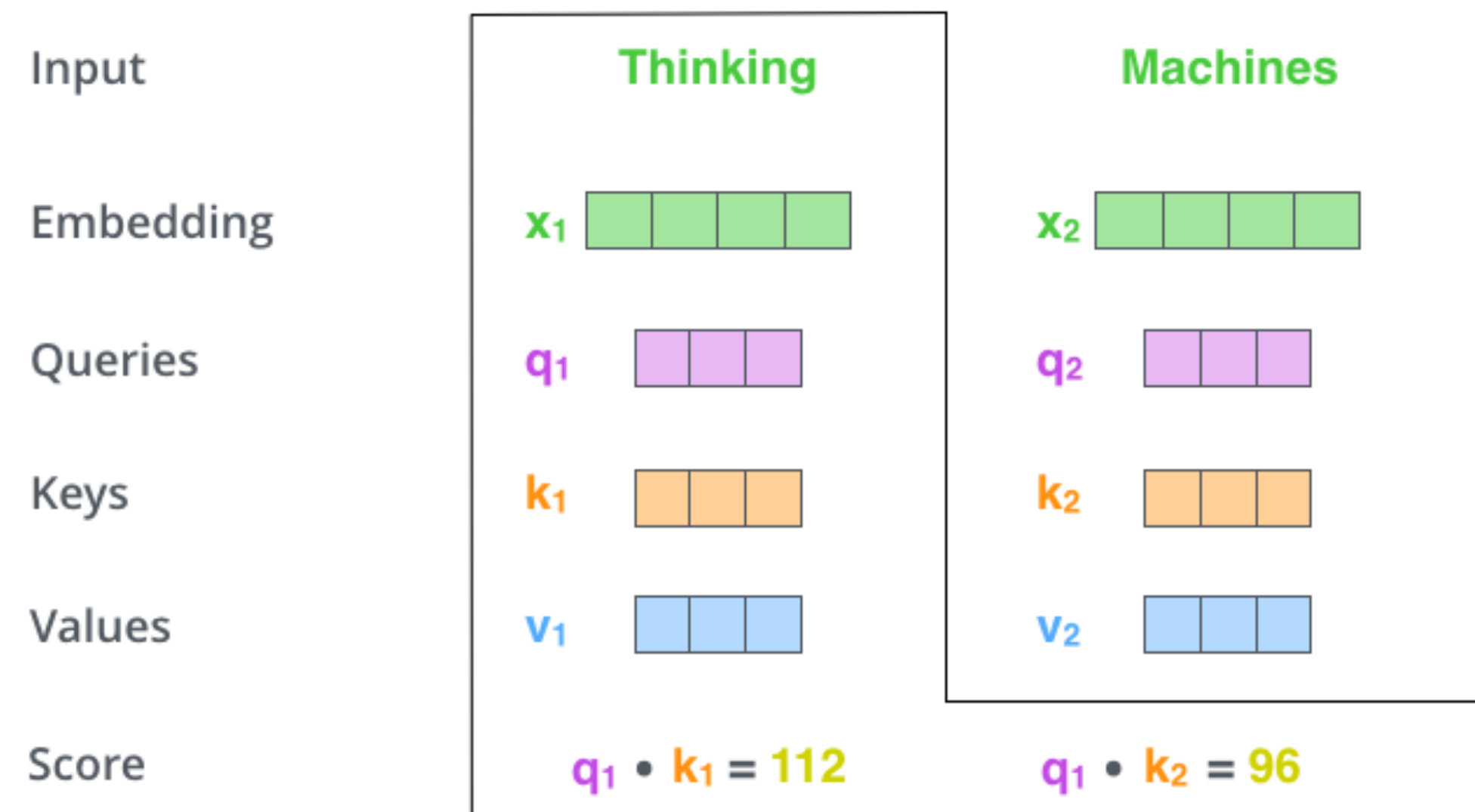
Transformers



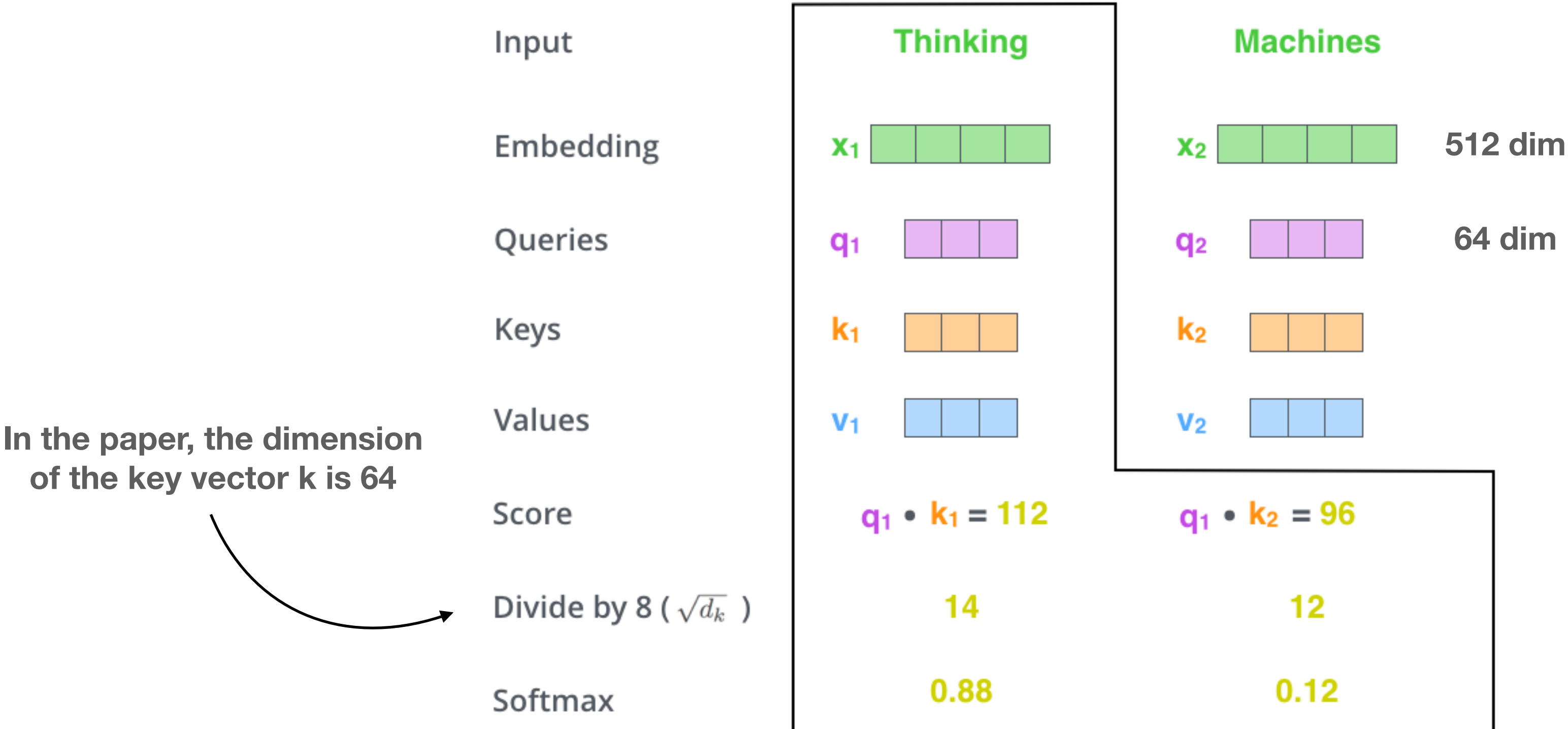
Transformers



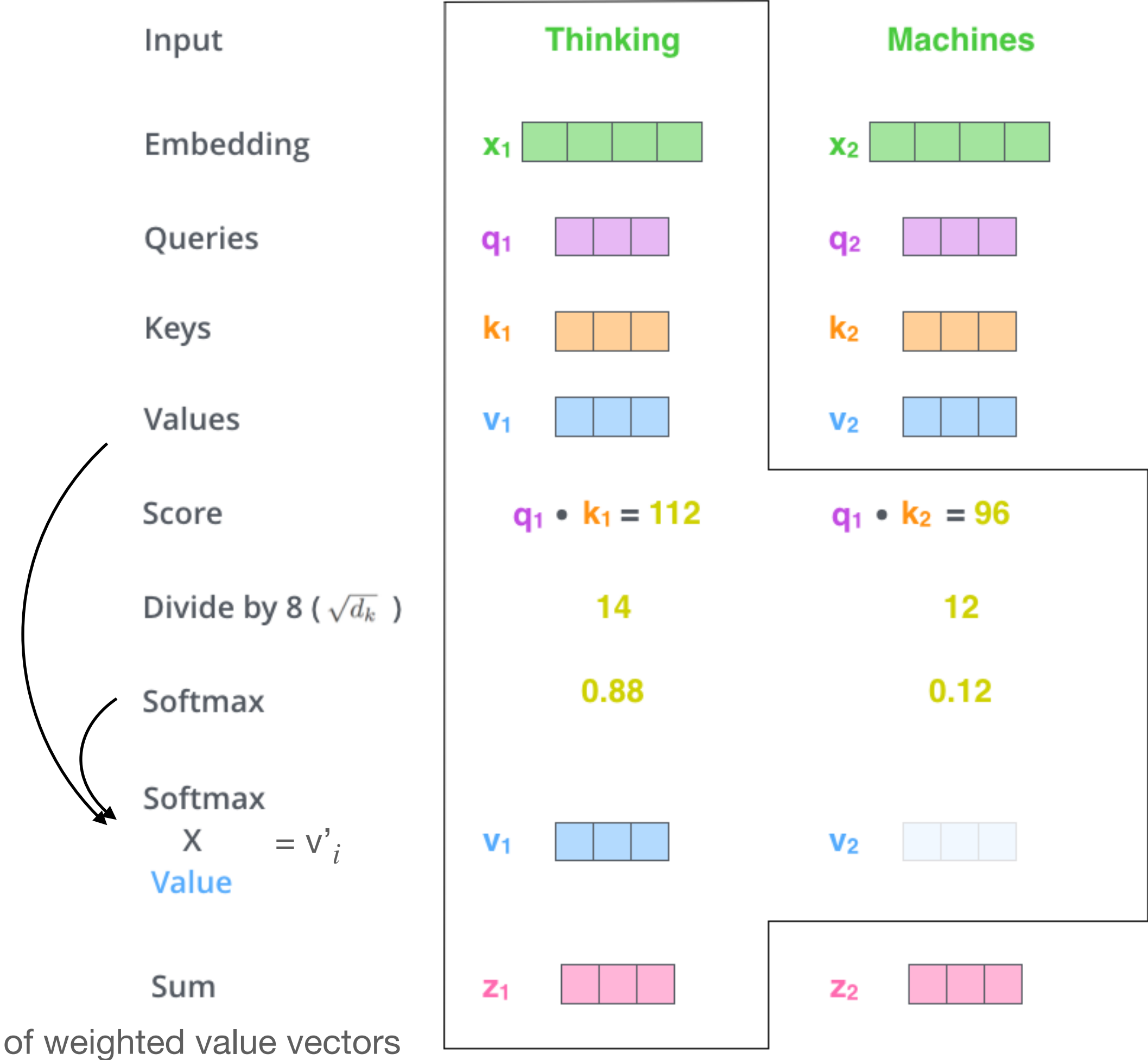
Transformers



Transformers

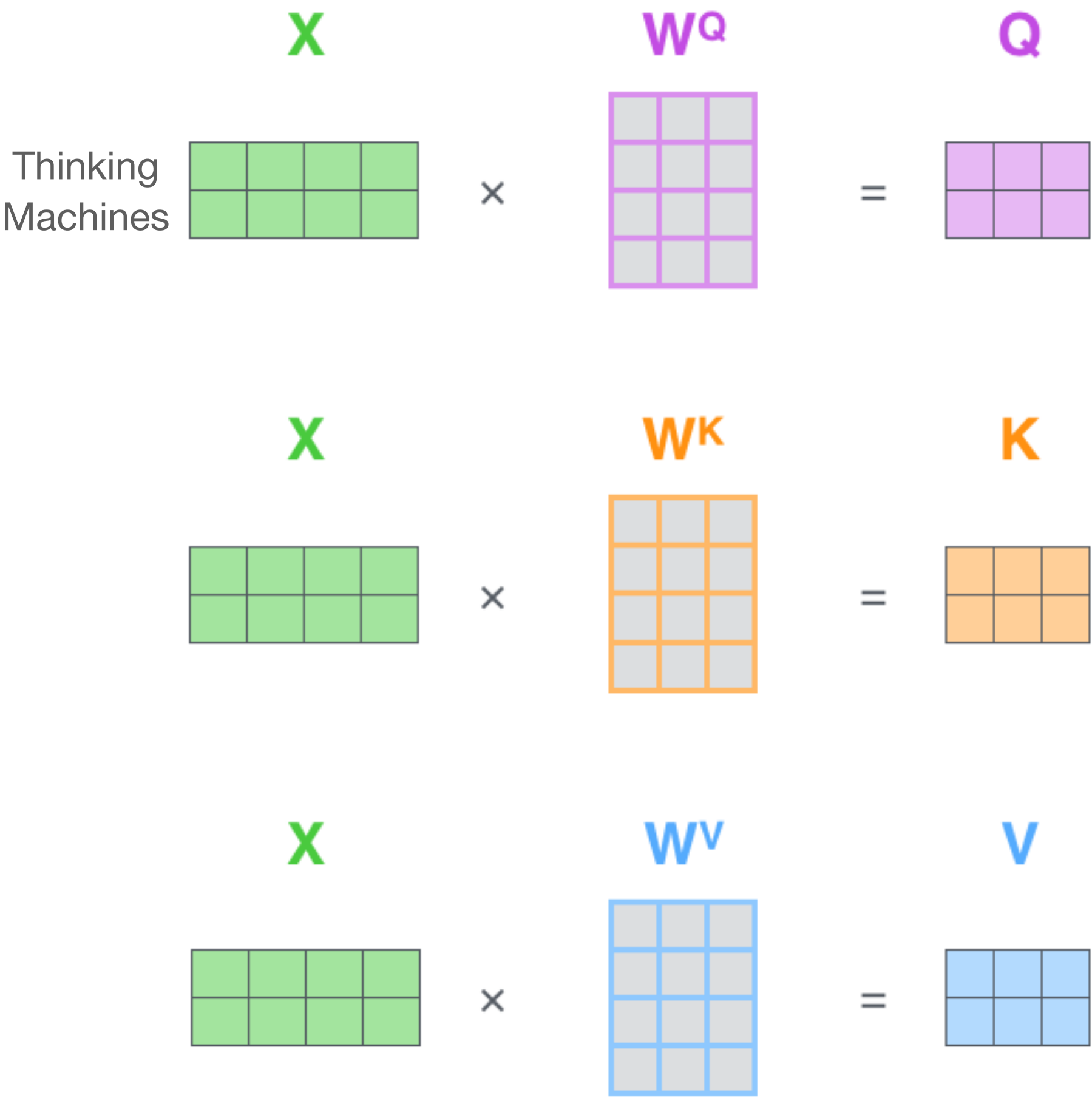


Transformers



$$z_j = \sum_i v'_i$$

Transformers



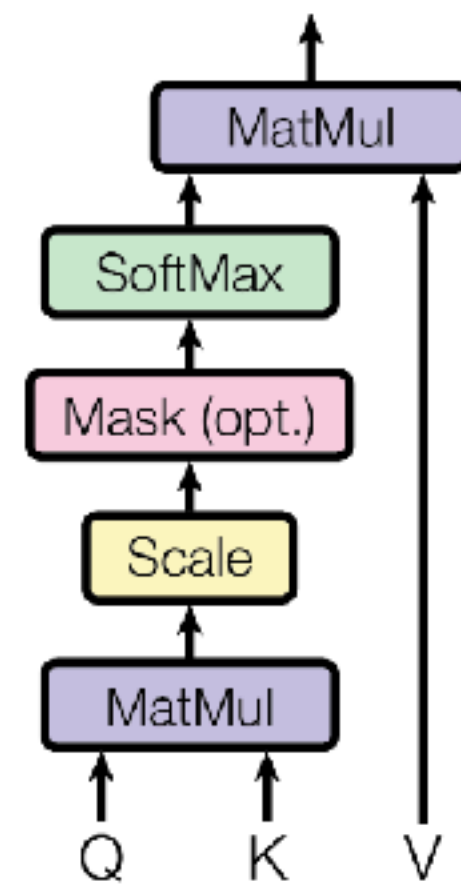
Attention Matrix

$$\text{Attention Matrix} = \text{softmax} \left(\frac{\overset{\text{Q}}{\begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \square & \square & \square \\ \hline \end{array}} \times \overset{\text{K}^T}{\begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \square & \square \\ \hline \end{array}} \right) \overset{\text{V}}{\begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \square & \square & \square \\ \hline \end{array}}$$

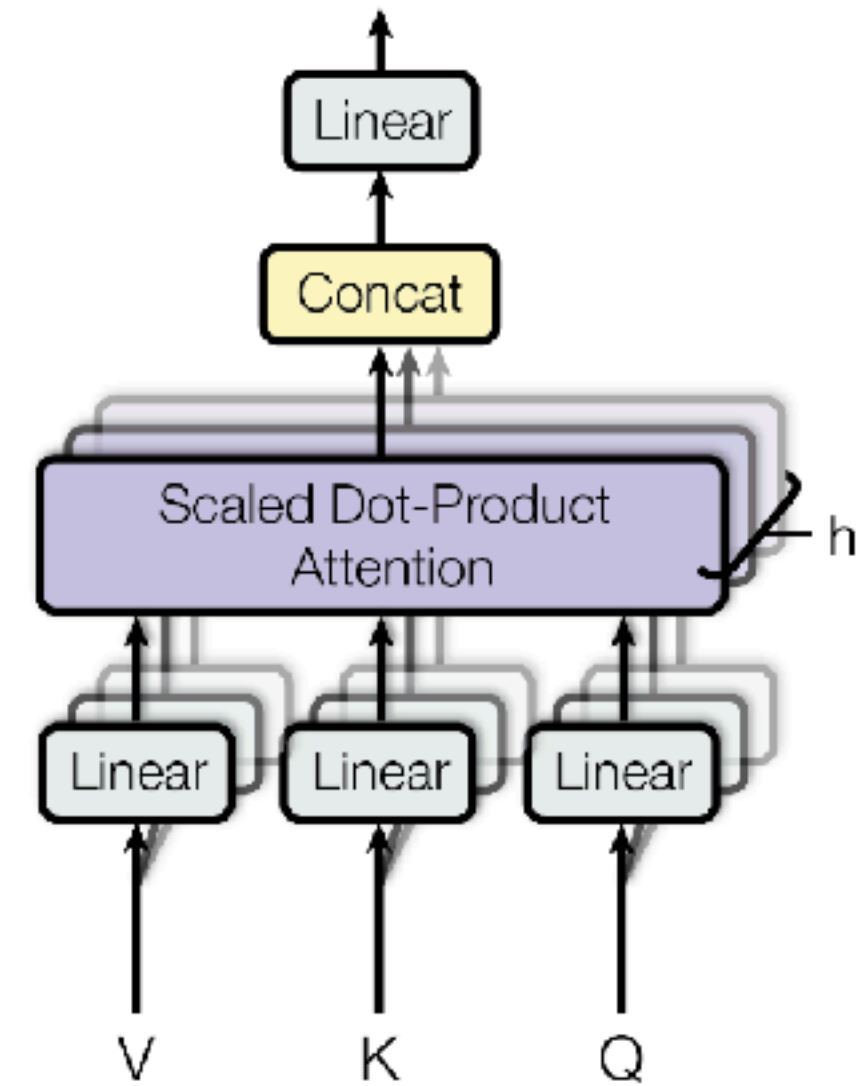
$= \overset{\text{Z}}{\begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \square & \square & \square \\ \hline \end{array}}$

Attention Matrix

Scaled Dot-Product Attention

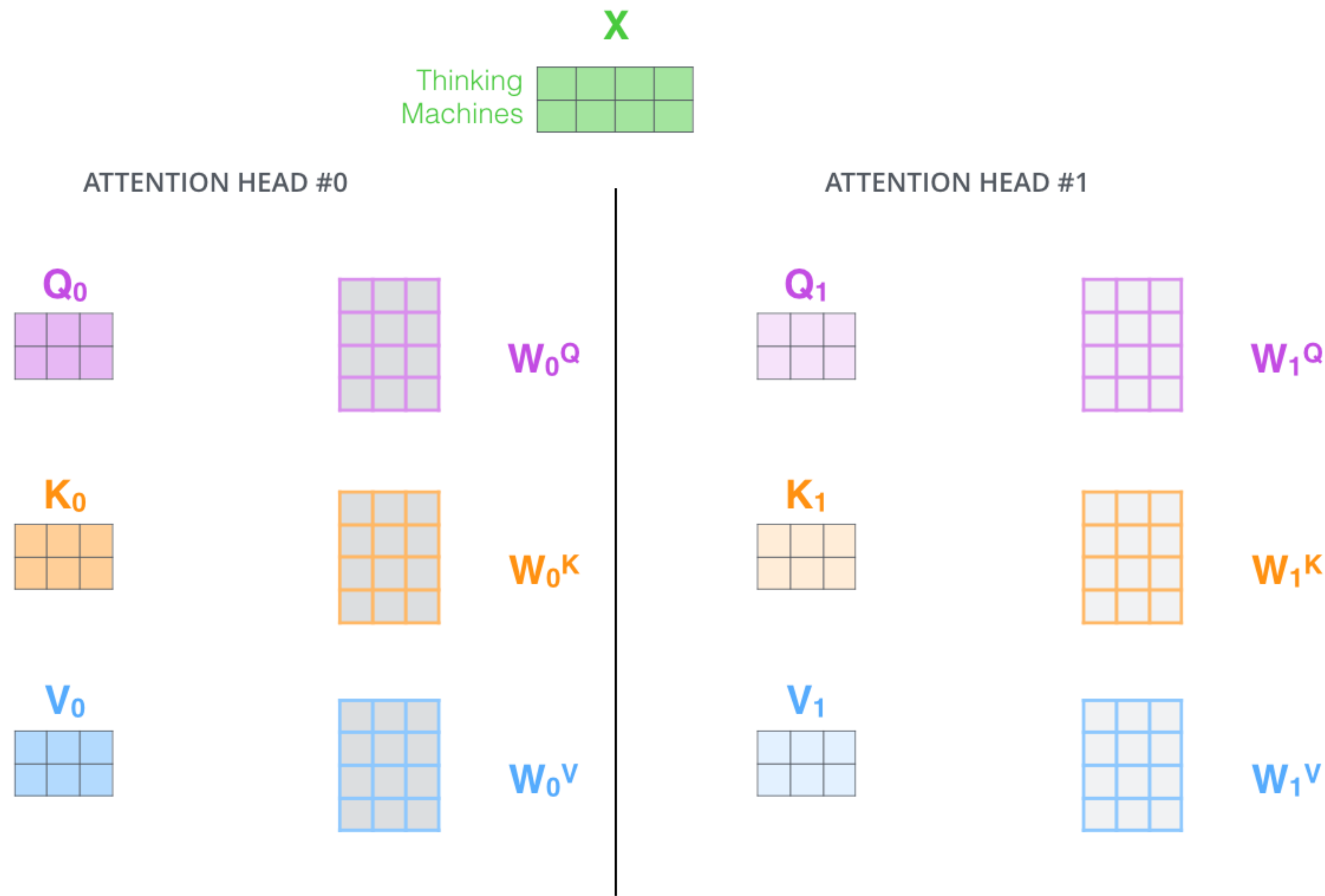


Multi-Head Attention

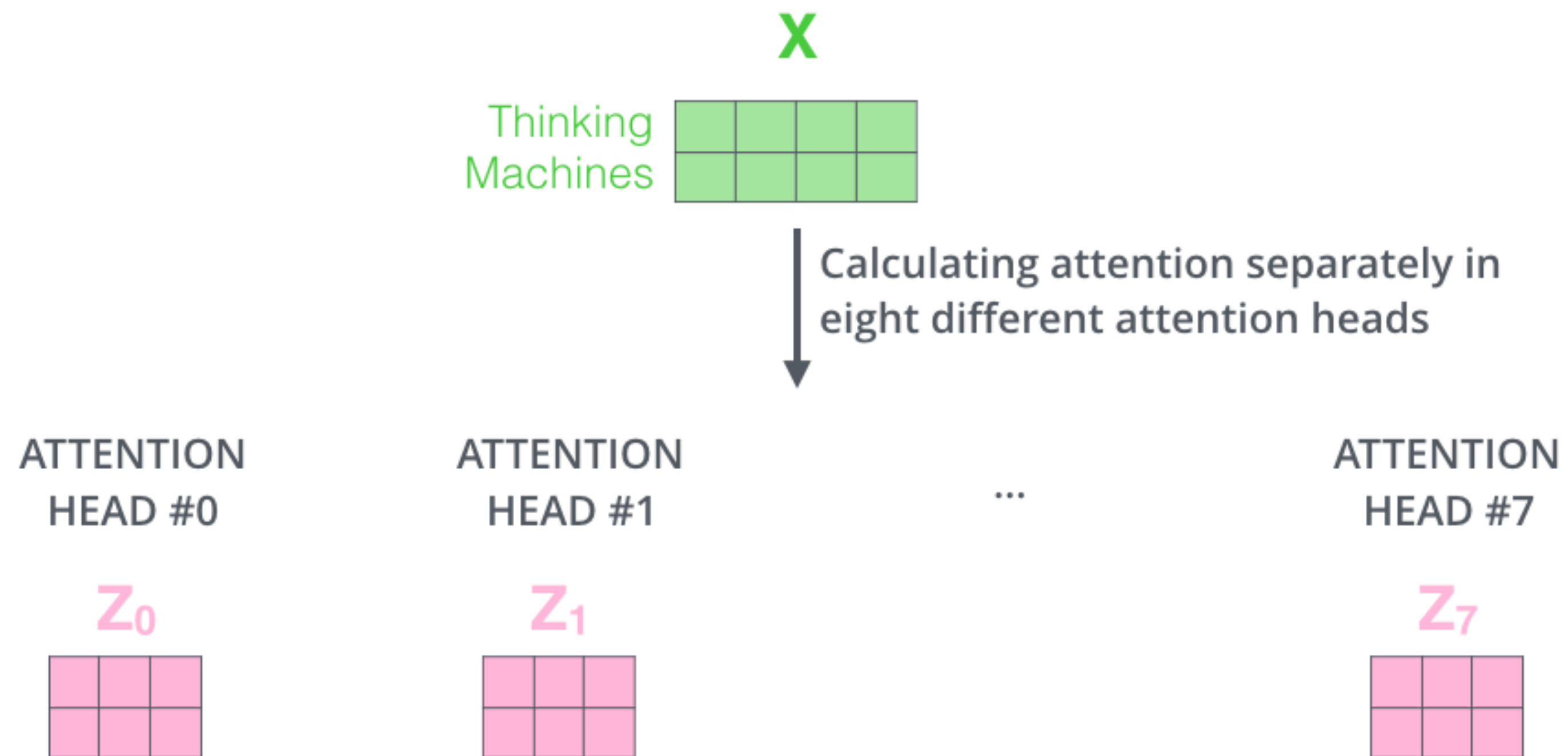


$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Transformers

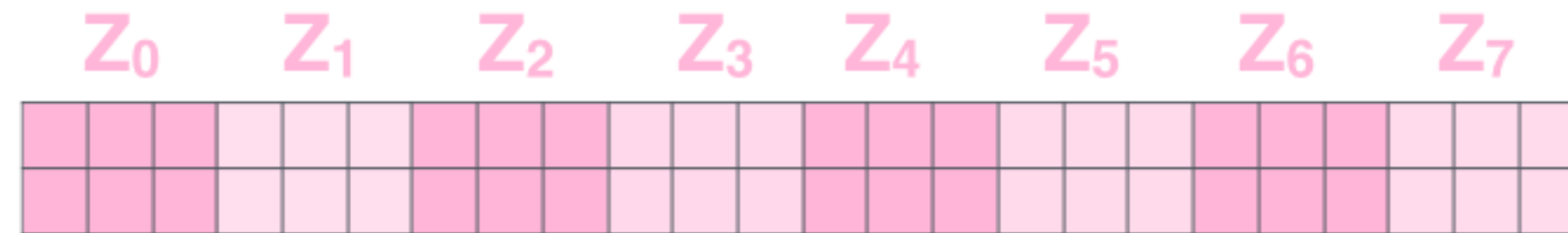


Transformers



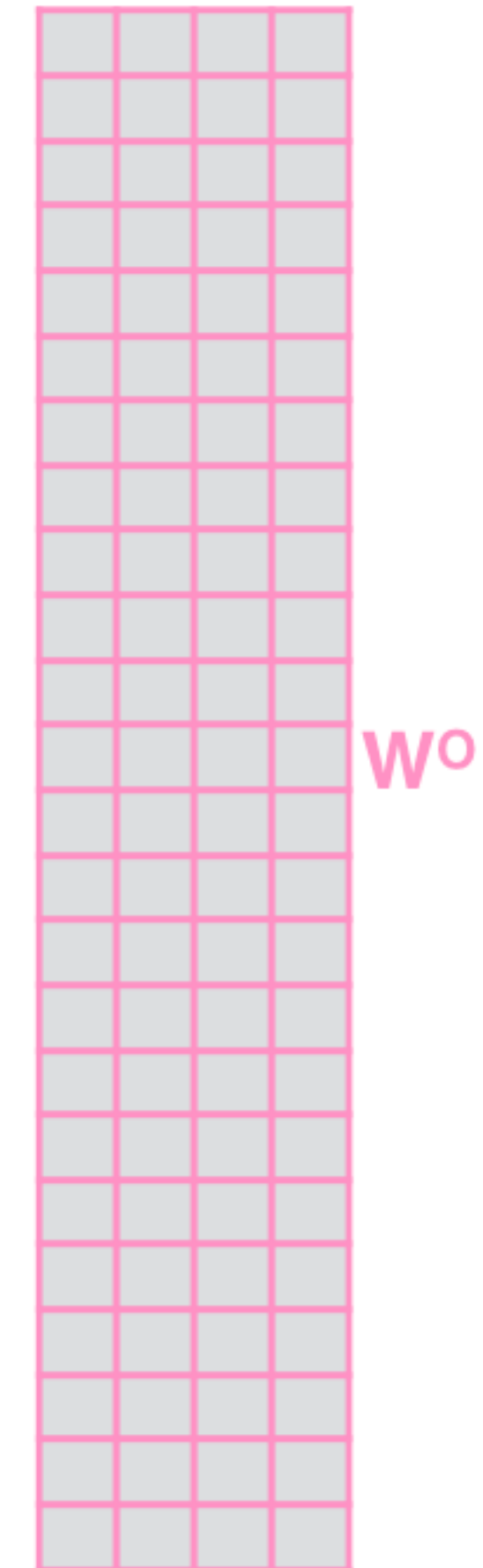
Transformers

1) Concatenate all the attention heads

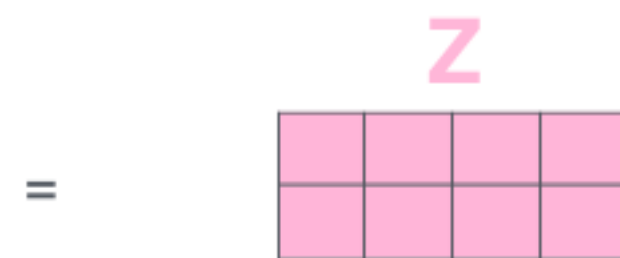


2) Multiply with a weight matrix W^O that was trained jointly with the model

$\times$



3) The result would be the Z matrix that captures information from all the attention heads. We can send this forward to the FFNN



Transformers

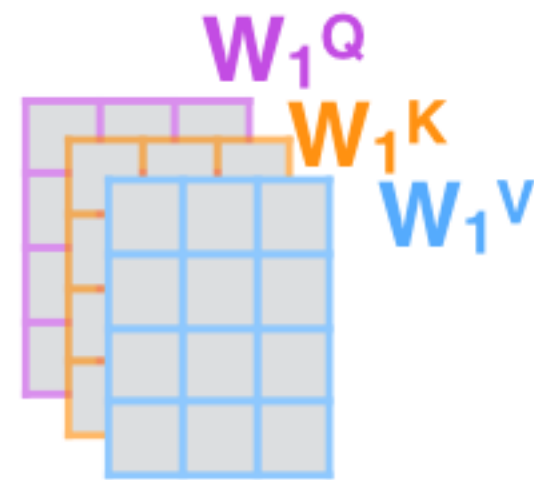
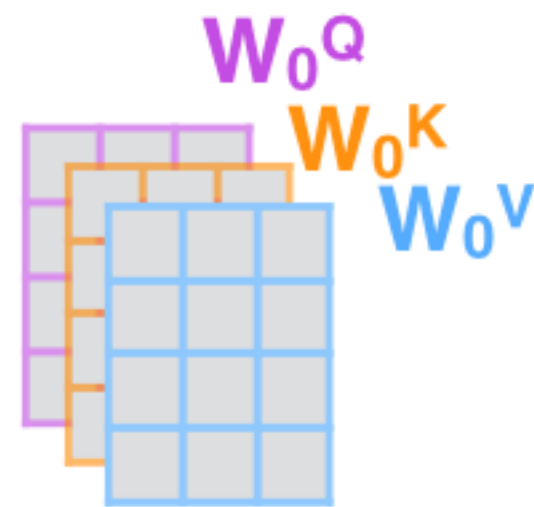
1) This is our input sentence*

Thinking
Machines

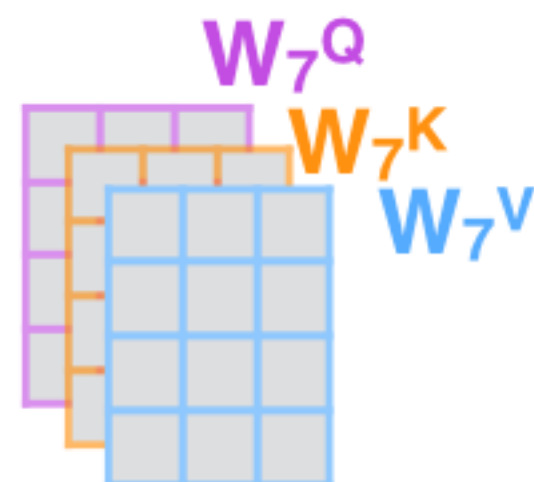
2) We embed each word*



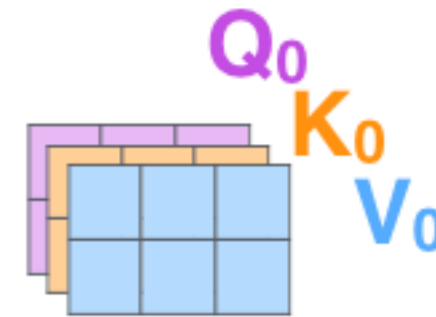
3) Split into 8 heads.
We multiply X or R with weight matrices



...



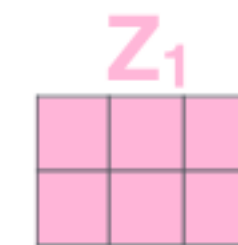
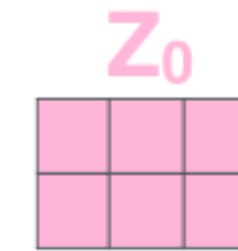
4) Calculate attention using the resulting $Q/K/V$ matrices



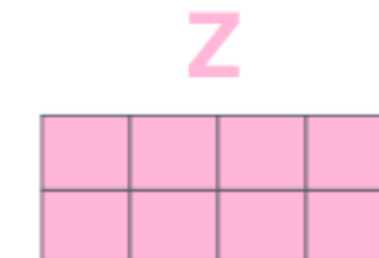
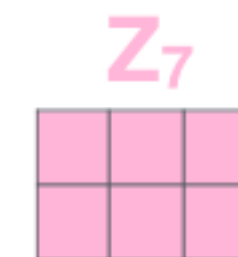
...



5) Concatenate the resulting Z matrices, then multiply with weight matrix W^O to produce the output of the layer



...

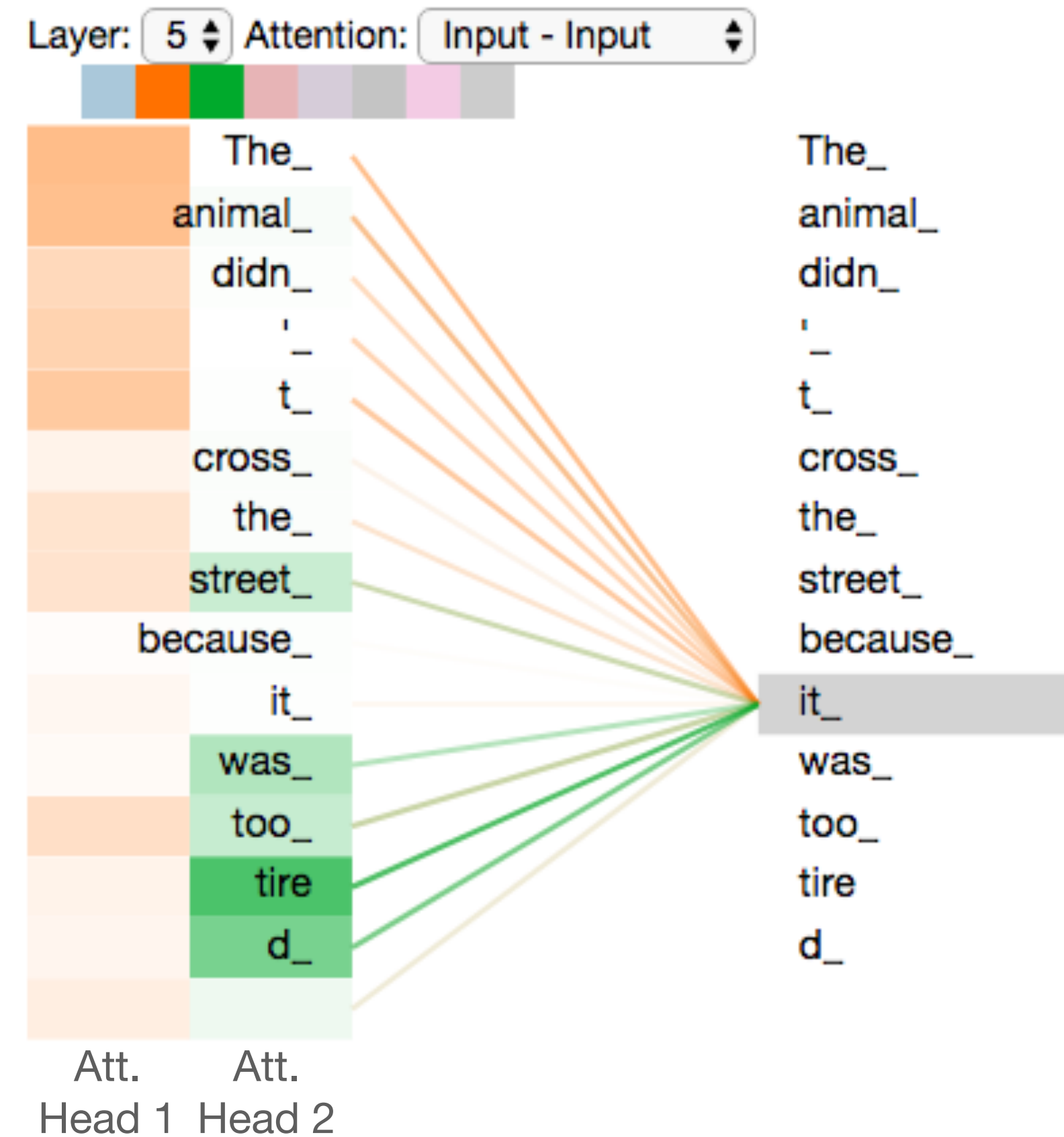


* In all encoders other than #0, we don't need embedding. We start directly with the output of the encoder right below this one

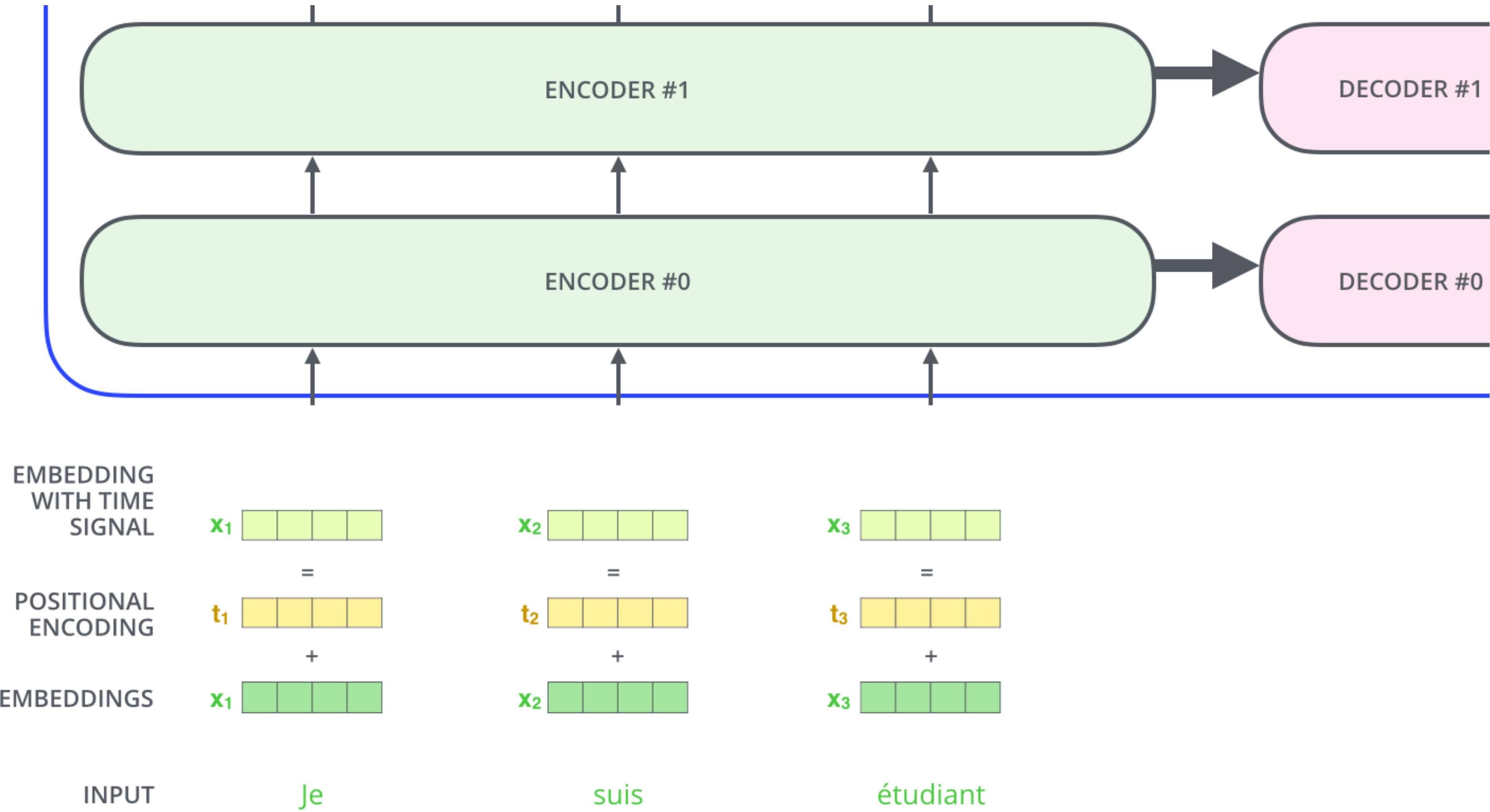


Transformers

"The animal didn't cross the street because it was too tired"

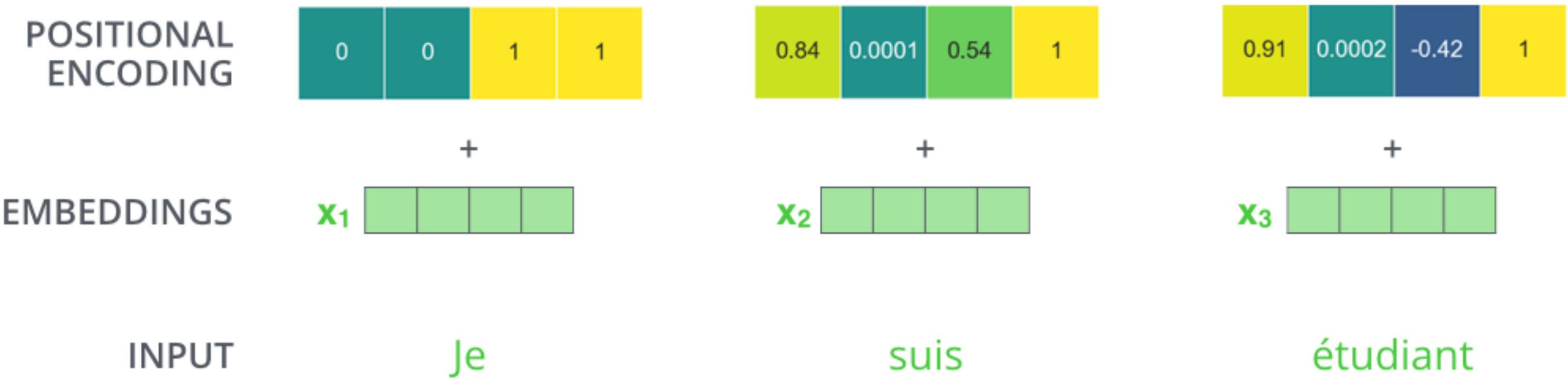


Transformers

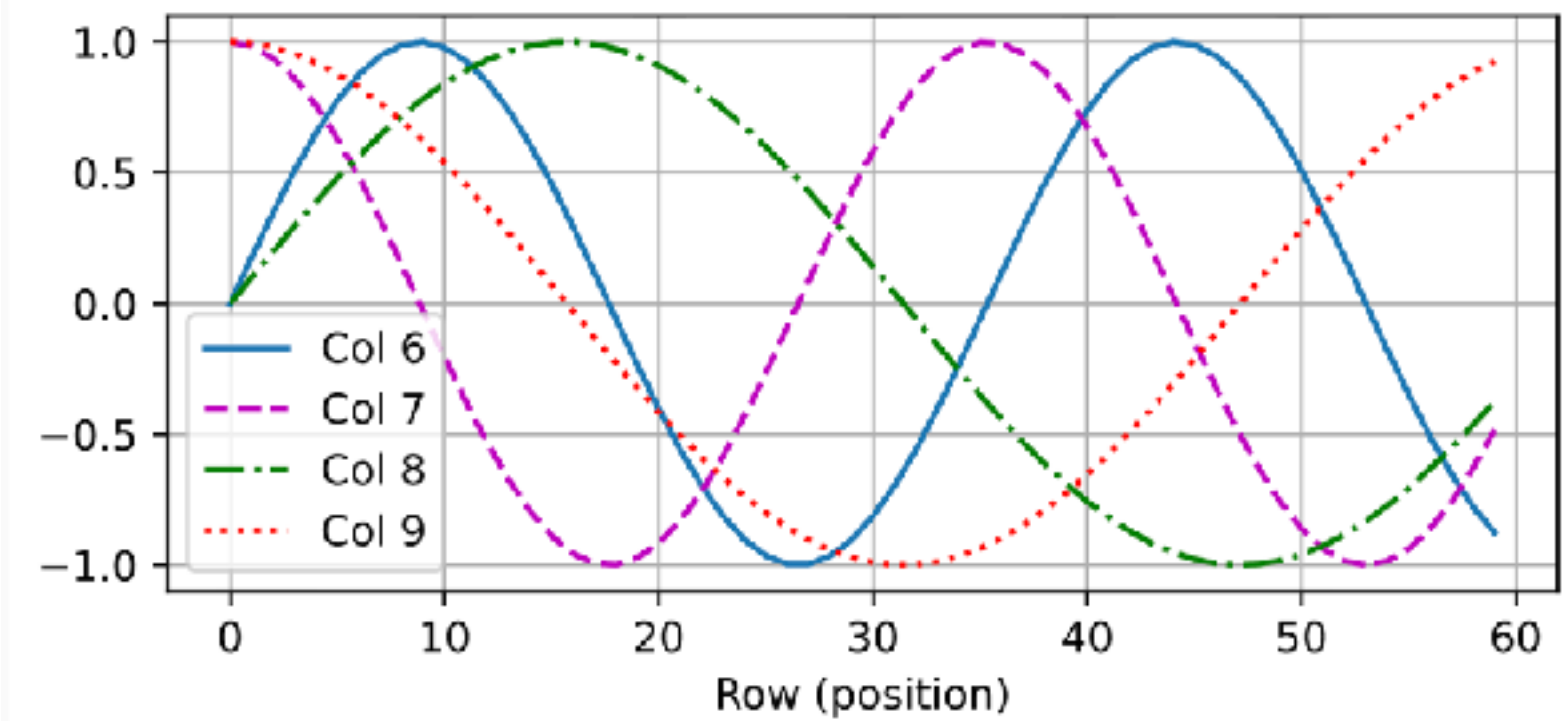
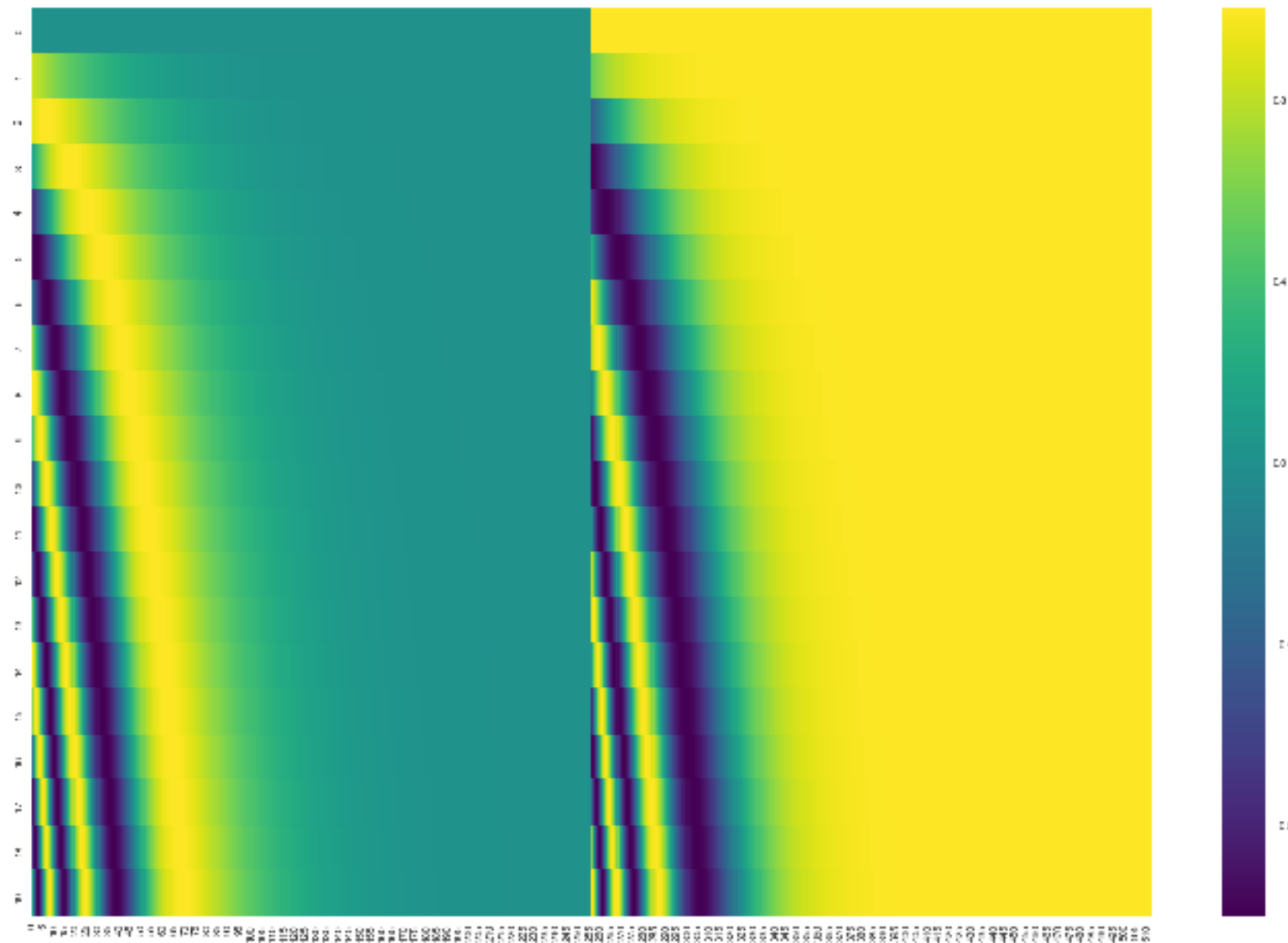


Transformers

Representing The Order of The Sequence Using Positional Encoding

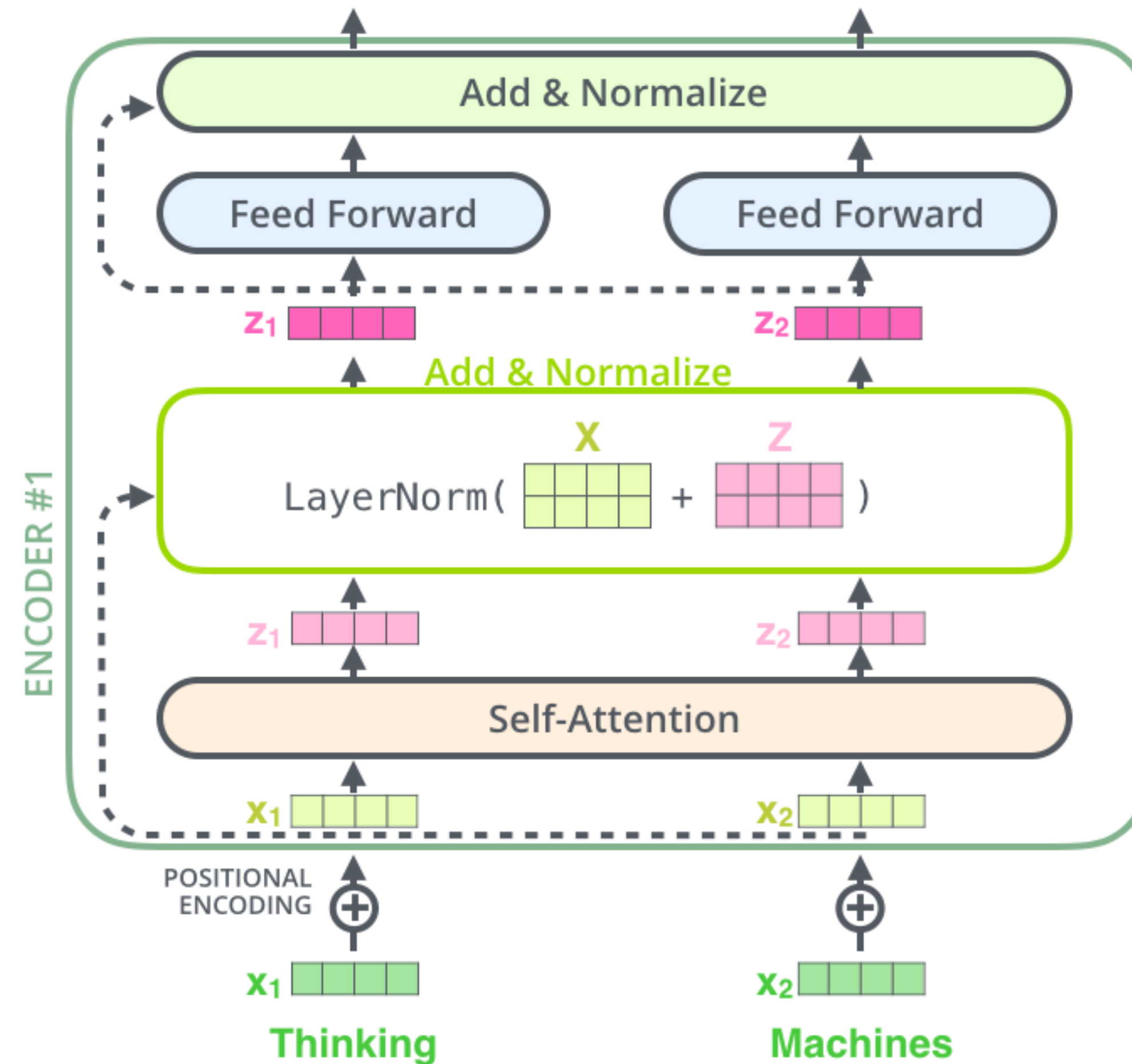


$$PE_{(pos,2i)} = \sin(pos/10000^{2i/d_{model}})$$
$$PE_{(pos,2i+1)} = \cos(pos/10000^{2i/d_{model}})$$



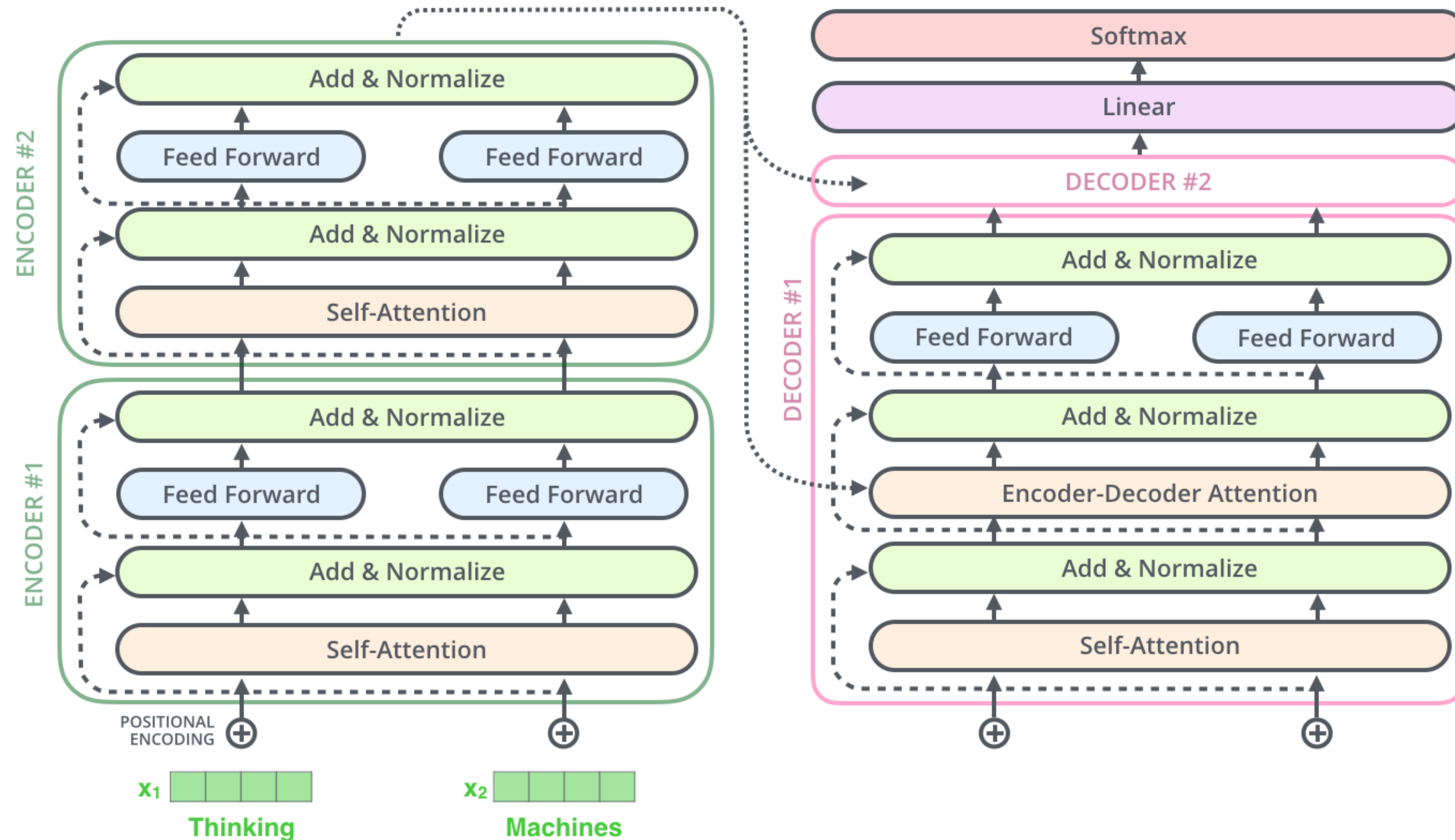
Transformers

The Residuals



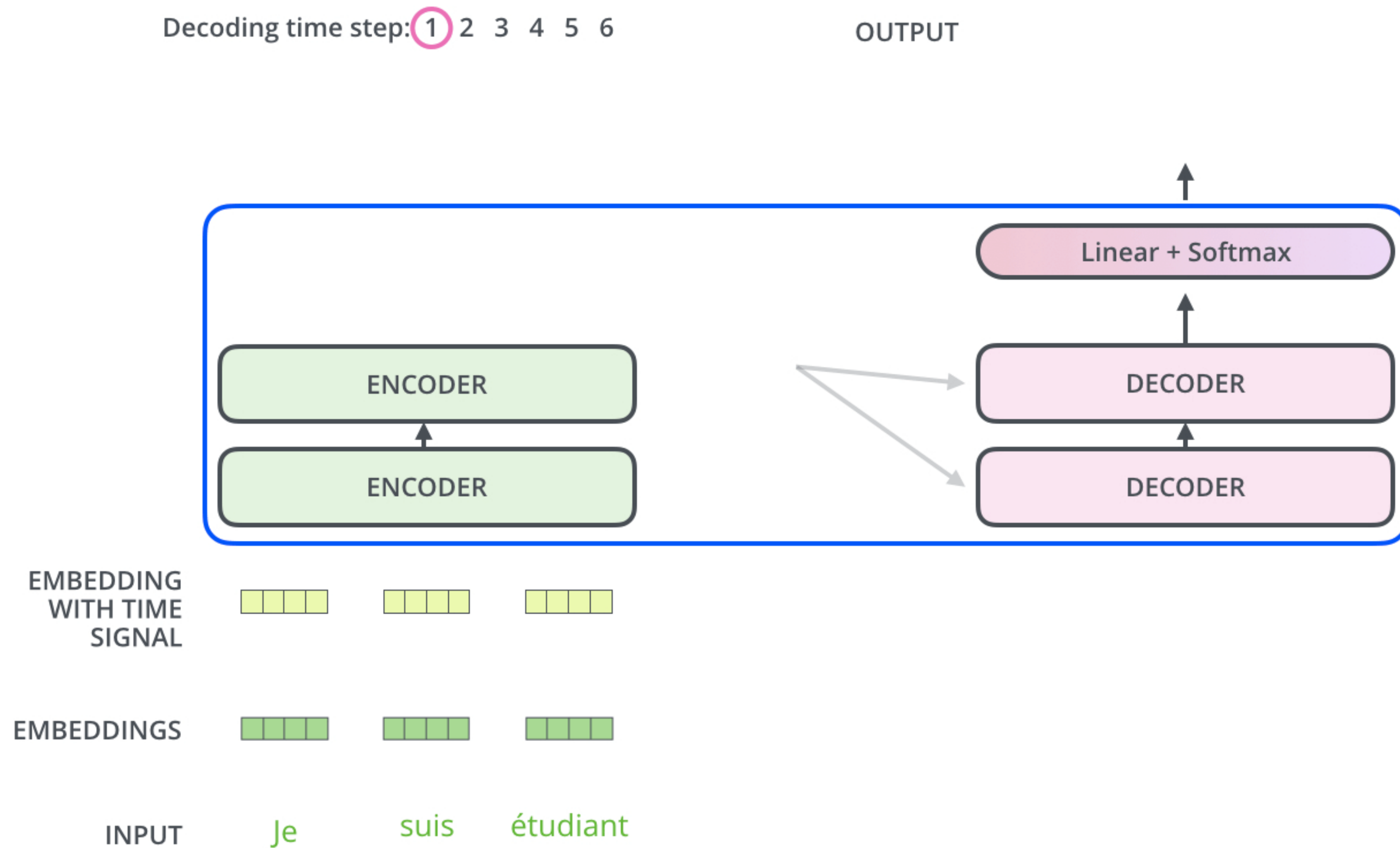
Transformers

The Residuals



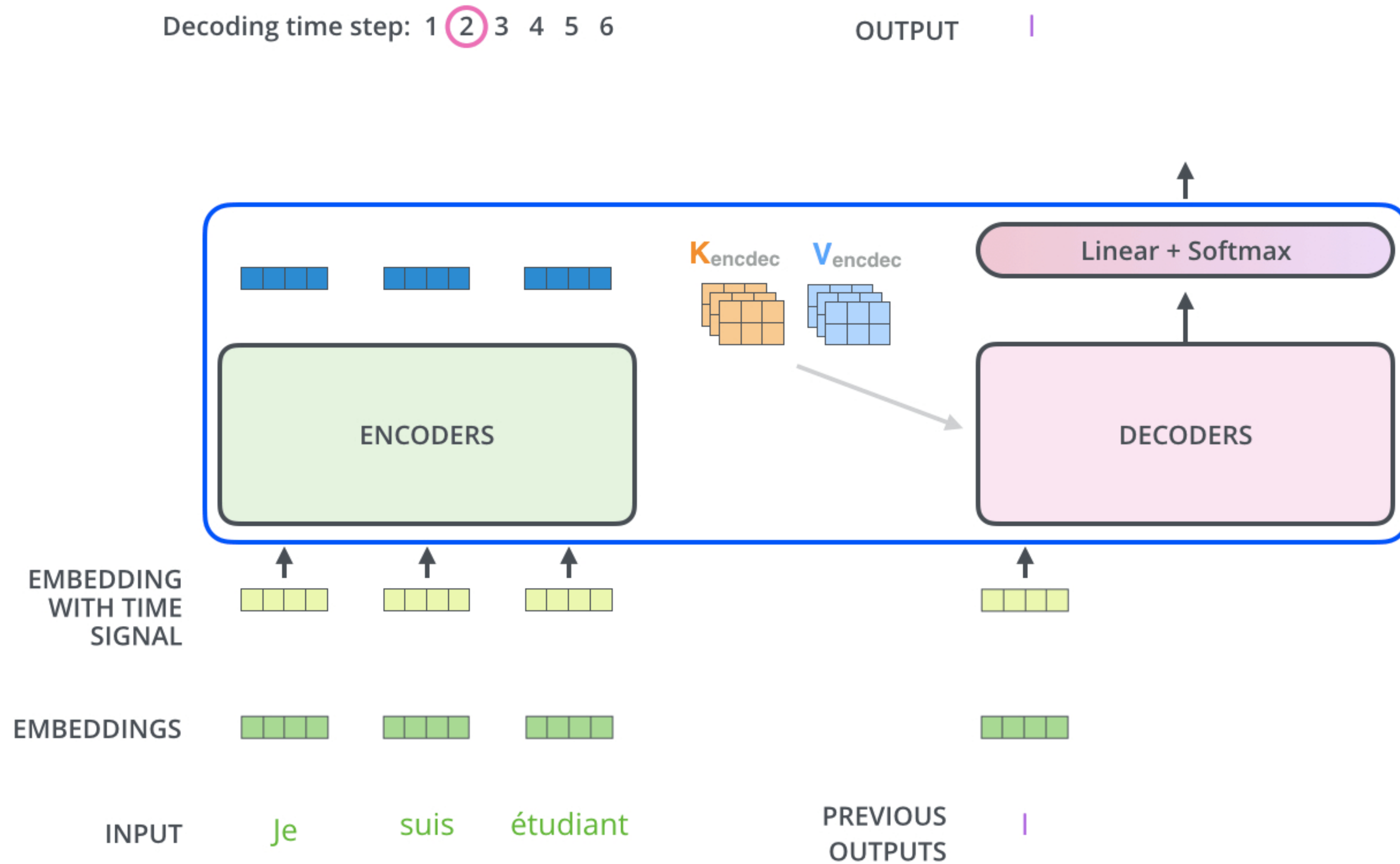
Transformers

The Decoder Side



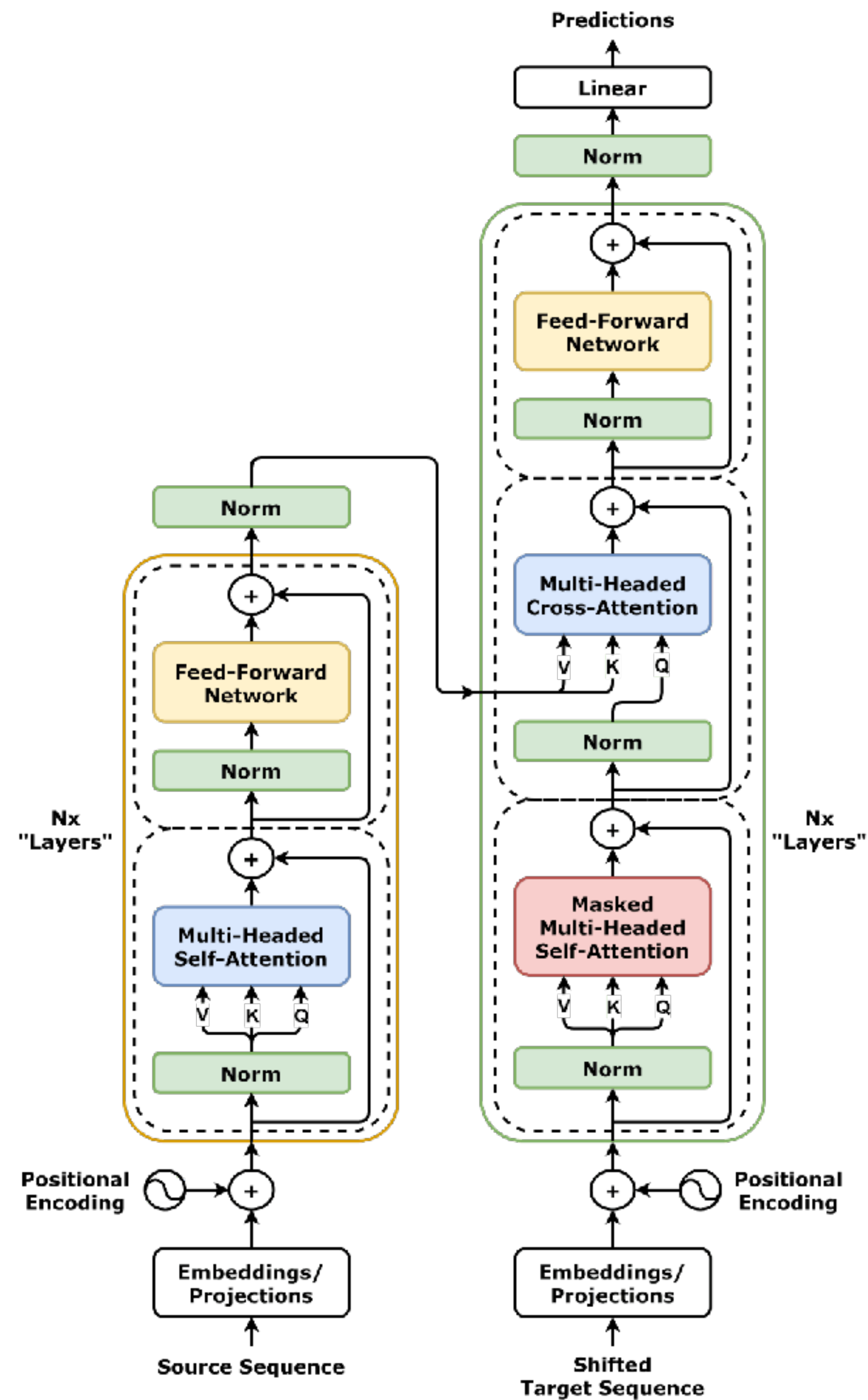
Transformers

The Decoder Side



$$\text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V + M$$

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[Llion Jones](#), [Aidan N. Gomez](#), [Lukasz Kaiser](#), [Illia Polosukhin](#)

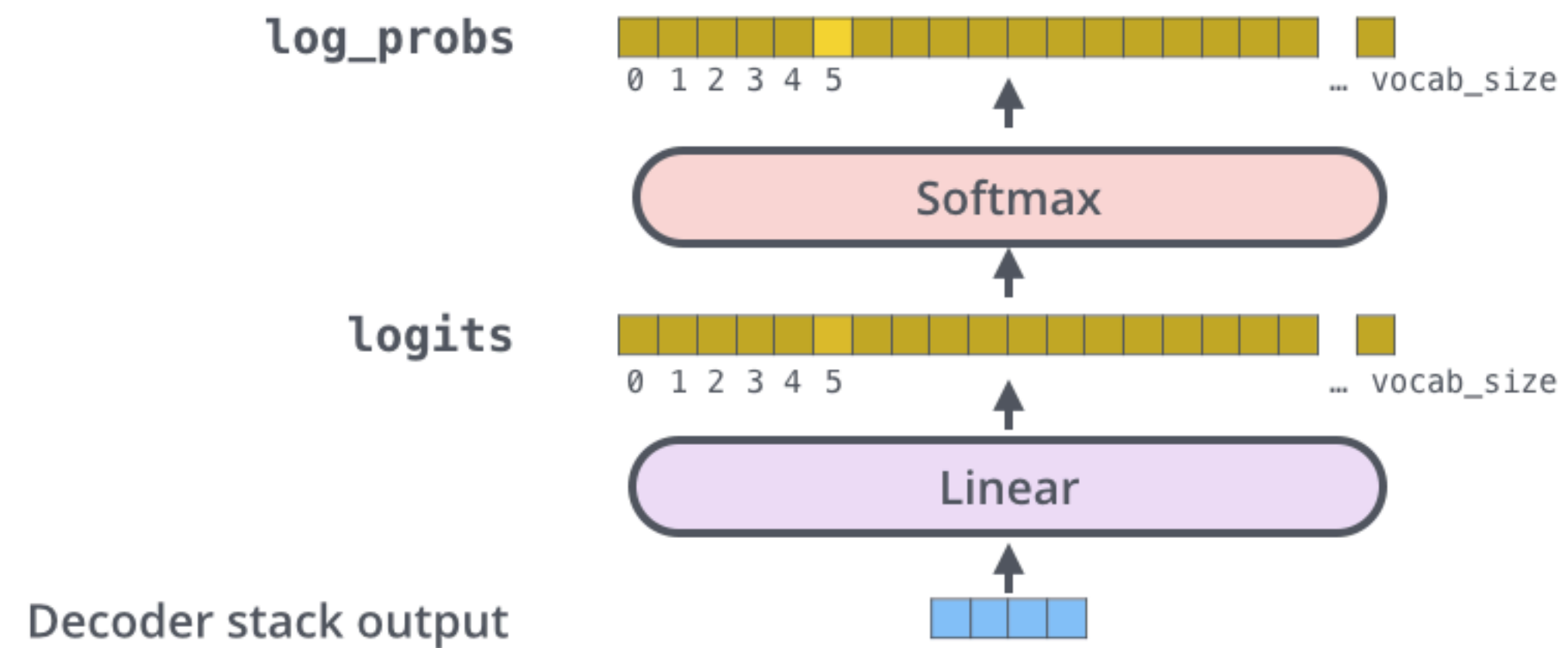
<https://arxiv.org/abs/1706.03762>

Transformers

The Final Linear and Softmax Layer

Which word in our vocabulary
is associated with this index?

Get the index of the cell
with the highest value
(**argmax**)



Output Vocabulary						
WORD	a	am	I	thanks	student	<eos>
INDEX	0	1	2	3	4	5

The output vocabulary of our model is created in the preprocessing phase before we even begin training.

Once we define our output vocabulary, we can use a vector of the same width to indicate each word in our vocabulary. This also known as one-hot encoding. So for example, we can indicate the word “am” using the following vector:

Output Vocabulary						
WORD	a	am	I	thanks	student	<eos>
INDEX	0	1	2	3	4	5

One-hot encoding of the word “am”

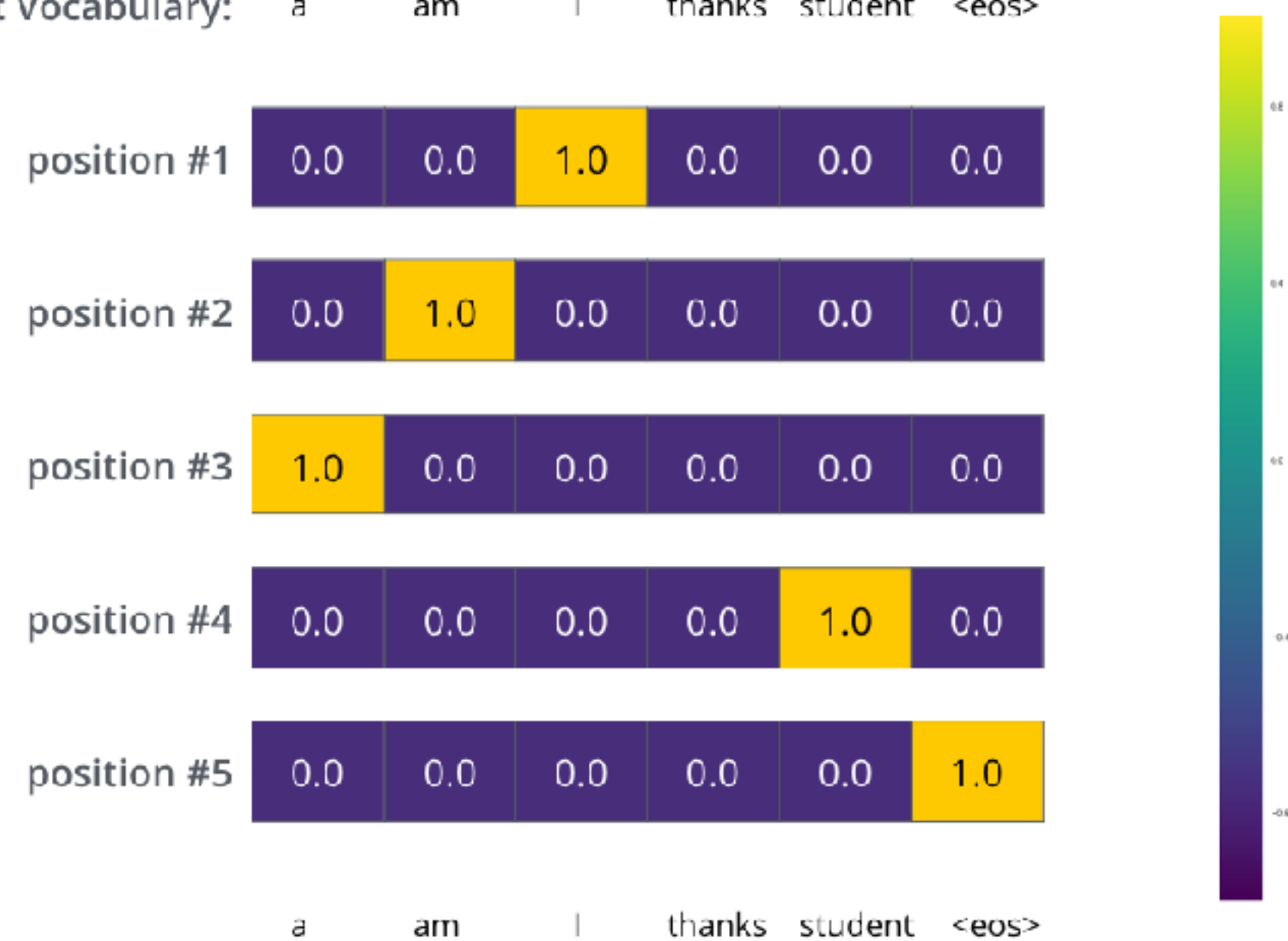
0.0	1.0	0.0	0.0	0.0	0.0
-----	-----	-----	-----	-----	-----

Example: one-hot encoding of our output vocabulary

Training probabilities

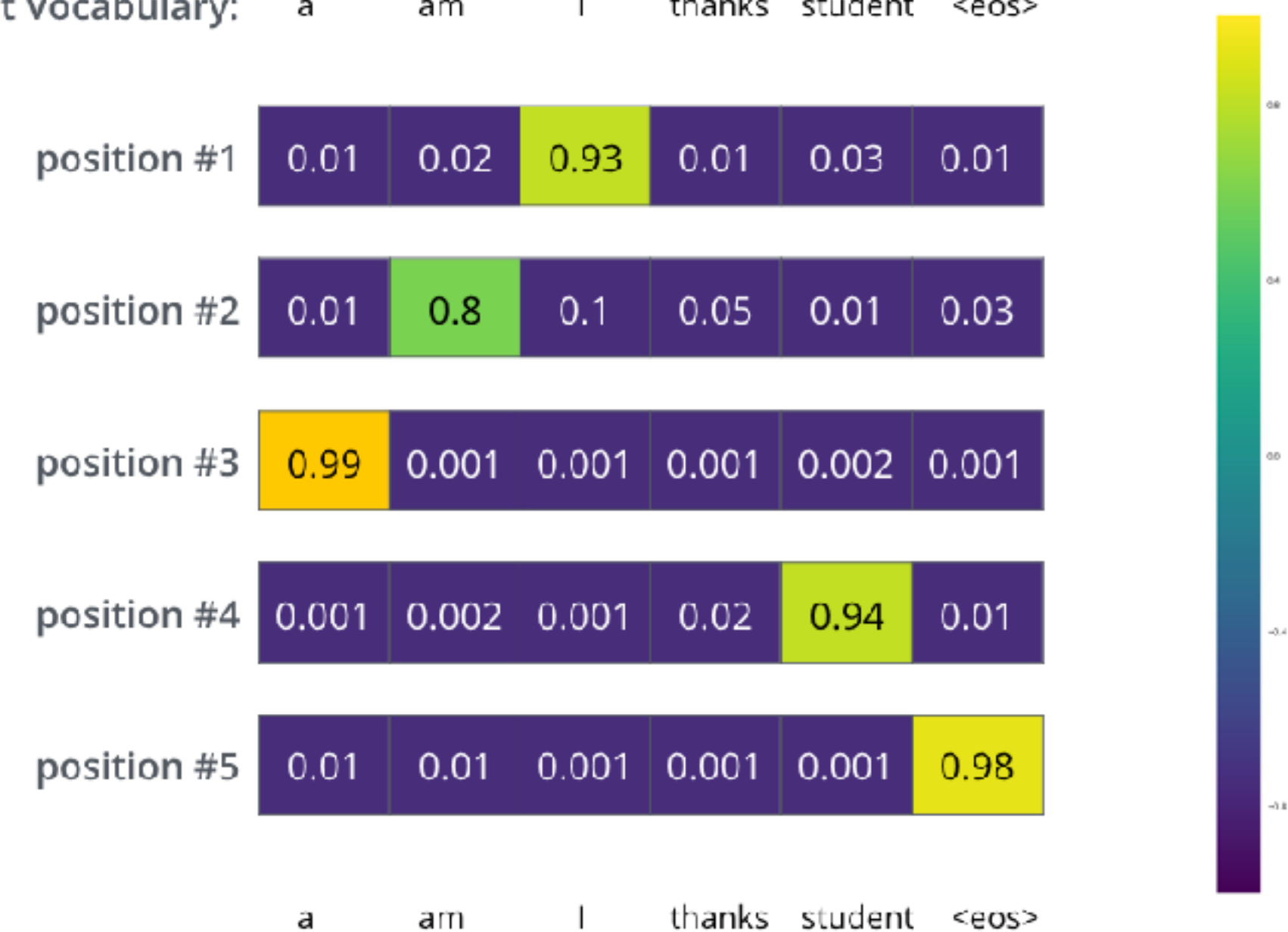
Target Model Outputs

Output Vocabulary: a am I thanks student <eos>

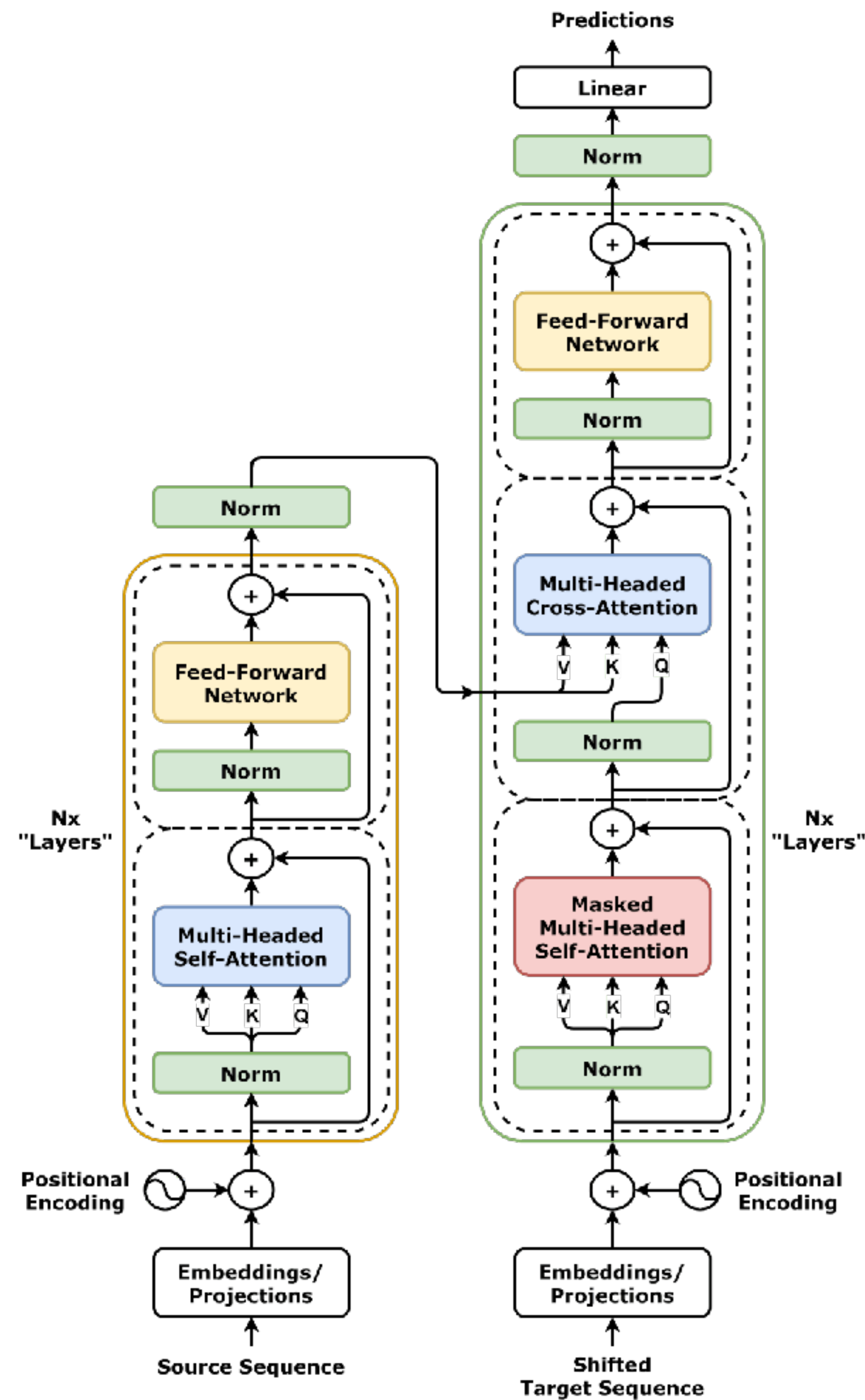


Trained Model Outputs

Output Vocabulary: a am I thanks student <eos>



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<https://arxiv.org/abs/1706.03762>